Module 1: Chapters

- **Fundamentals of Mathematics (with new exercise)**
- **Fundamentals of Algebra (Same as the previous chapter in some aspects)**

FUNDAMENTALS OF MATHEMATICS

1. SET THEORY

1.1 Sets : A set is a collection of well defined objects which are distinct from each other. Sets are generally denoted by capital letters A, B, C, ... etc. and the elements of the set by *a, b, c*,... etc.

If a is an element of a set A, then we write $a \in A$ (a belongs to A).

If a is **not** an element of a set A, then we write $a \notin A$ (a does not belong to A).

Ex. The collection of vowels in english alphabet is a set containing the elements *a, e, i, o, u*.

1.2 Methods to write a Set

(i) Roster Method : In this method a set is described by listing elements, separated by commas and enclosing them by curly brackets.

Ex. The set of vowels of English Alphabet may be described as $\{a, e, i, o, u\}$.

(ii) Set Builder Form : In this case we write down a property or rule or proposition, which gives us all the element of the set.

$A = \{x : P(x)\}$

Ex. $A = \{x : x \in \mathbb{N} \text{ and } x = 2n \text{ for } n \in \mathbb{N}\}$

i.e. $A = \{2, 4, 6, ...\}$

Ex. $B = \{x^2 : x \in \mathbb{N}\}$

i.e. $B = \{1, 4, 9, ...\}$

Illustration-1 : The set $A = \{x : x \in \mathbb{R}, x^2 = 16 \text{ and } 2x = 6\}$ is equal to

(A) ϕ (B) $\{14, 3, 4\}$ (C) $\{3\}$ (D) $\{4\}$

Solution : $x^2 = 16 \Rightarrow x = \pm 4$

$2x = 6 \Rightarrow x = 3$

There is no value of x which satisfies both the above equations.

Thus, $A = \phi$

Hence, (A) is the correct answer.

1.3 Cardinality of a Finite Set :

The number of elements in a finite set is called the cardinality of the set A and is denoted |A| or n(A). It is also called cardinal number of the set.

Ex. $A = \{a, b, c, d\} \Rightarrow n(A) = 4$

1.4 Types of Sets

(i) Null set or Empty set : A set having no element in it is called an Empty set or a Null set or Void set. It is denoted by ϕ or { }

Ex. $A = \{x \in \mathbb{N} : 5 < x < 6\} = \phi$

A set consisting of atleast one element is called a non-empty set or a non-void set.

(ii) Singleton set : A set consisting of a single element is called a singleton set.

Ex. set $\{0\}$, is a singleton set

(iii) Finite Set : A set which has only finite number of elements is called a finite set.

Ex. $A = \{a, b, c\}$

(iv) Infinite set : A set which has an infinite number of elements is called an infinite set.

Ex. $A = \{1, 2, 3, 4, ...\}$ is an infinite set

(v) Subset : Let A and B be two sets, if every element of A is an element of B, then A is called a subset of B. If A is a subset of B, we write $A \subseteq B$

Ex : $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6, 7\} \Rightarrow A \subseteq B$

The symbol "$\Rightarrow$" stands for "implies"

Note : $(x \in A \Rightarrow x \in B) \Leftrightarrow A \subseteq B$

(vi) Proper subset : If A is a subset of B and $A \neq B$ then A is a proper subset of B and we write $A \subset B$

Note :

- Every set is a subset of itself i.e. $A \subseteq A$ for all A
- Empty set ϕ is a subset of every set
- The total number of subsets of a finite set containing n elements is 2^n

(vii) Universal set : A set consisting of all possible elements which occur in the discussion is called a Universal set and is denoted by U

Note : All sets are contained in the universal set

Ex. If $A = \{1, 2, 3\}$, $B = \{2, 4, 5, 6\}$, $C = \{1, 3, 5, 7\}$ then $U = \{1, 2, 3, 4, 5, 6, 7\}$ can be taken as the Universal set.

(viii) Power set : Let A be any set. The set of all subsets of A is called power set of A and is denoted by P(A).

Note :

- If $A = \phi$ then P(A) has one element.
- Power set of a given set is always non empty

Illustration-2 : If $A = \{1, 2\}$ then find its power set.

Solution : $A = \{1, 2\}$ then $P(A) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

Illustration-3 : If $A = \{x, y\}$, then the power set of A is-

(A) $\{x^y, y^x\}$ (B) $\{\phi, x, y\}$

(C) $\{\phi, \{x\} \{2y\}\}$ (D) $\{\phi, \{x\}, \{y\}, \{x, y\}\}$

Solution : Clearly $P(A)$ = set of all subsets of A

$= \{\phi, \{x\}, \{y\}, \{x, y\}\}$

$\therefore$ (D) holds.

1.5 Some Operation on Sets

(i) Union of two sets : $A \cup B = \{x : x \in A \text{ or } x \in B\}$

Ex. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ then $A \cup B = \{1, 2, 3, 4\}$

Note : $x \in (A \cup B) \Leftrightarrow x \in A$ or $x \in B$

(ii) Intersection of two sets : $A \cap B = \{x : x \in A \text{ and } x \in B\}$

Ex. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ then $A \cap B = \{2, 3\}$

Note :

- $x \in (A \cap B) \Leftrightarrow x \in A$ and $x \in B$
- If $A \cap B = \phi$, then A, B are disjoint sets.

Ex. If $A = \{1, 2, 3\}$, $B = \{7, 8, 9\}$ then $A \cap B = \phi$

Illustration-4 : If $aN = \{ax : x \in N\}$, then the set $6N \cap 8N$ is equal to-

(A) 8N (B) 48N (C) 12N (D) 24N

Solution : $6N = \{6, 12, 18, 24, 30, ...\}$

$8N = \{8, 16, 24, 32, ...\}$

$\therefore$ $6N \cap 8N = \{24, 48, ...\} = 24N$

Short cut Method

$6N \cap 8N = 24N$ [24 is the L.C.M. of 6 and 8]

(iii) Difference of two sets : $A - B$ or $A\backslash B = \{x : x \in A \text{ and } x \notin B\}$

Ex. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$; $A - B = \{1\}$

Note : $x \in (A - B) \Leftrightarrow x \in A$ and $x \notin B$

(iv) Complement of a set : A' or A^c or $\overline{A} = \{x : x \notin A \text{ but } x \in U\} = U - A$

Ex. $U = \{1, 2, ..., 10\}$, $A = \{1, 2, 3, 4, 5\}$ then $A' = \{6, 7, 8, 9, 10\}$

Note :

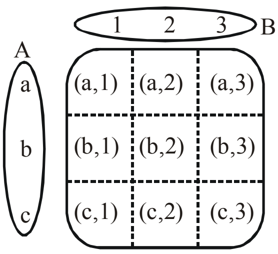
- $x \in A \Leftrightarrow x \notin A^c$
- $A \cap A' = \phi \quad \therefore$ A, A' are disjoint.
- $A \cup A' = U$
- $(A')' = A$.

(v) Cartesian Product of two sets

Cartersian product of two sets A and B, denoted as $A \times B$, is the set of all ordered pairs (a,b) where $a \in A$ and $b \in B$

$A \times B = \{(a,b) | \ a \in A \text{ and } b \in B\}$

Ex : Cartersian product $A \times B$ when $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$ is represented in the square grid



Note : $n(A\times B) = n(A) \times n(B)$

Illustration-5 : Find $A \times B$ when $A = \{x|x$ is prime number less than $5\}$ and $B = \{1,2,3\}$

Solution : $A = \{2,3\}$

$B = \{1,2,3\}$

$A \times B = \{(2,1), (2,2), (2,3), (3,1), (3,2),(3,3)\}$

1.6 Equality of two Sets :

Two sets A and B are said to be equal if every element of A is an element of B, and every element of B is an element of A.

If sets A and B are equal, we write $A = B$ and if A and B are not equal, then $A \neq B$

Ex. $A = \{1, 2, 6, 7\}$ and $B = \{6, 1, 2, 7\} \Rightarrow A = B$

Note : $A \subseteq B$ and $B \subseteq A \Leftrightarrow A = B$

Illustration-6 : Find the pairs of equal sets (if any), give reasons:

$A = \{0\}$, $\qquad B = \{x : x > 15 \text{ and } x < 5\}$,

$C = \{x : x - 5 = 0\}$, $D = \{x : x^2 = 25\}$,

$E = \{x : x$ is an integral positive root of the equation $x^2 - 2x - 15 = 0\}$.

Solution : Since $0 \in A$ and 0 does not belong to any of the sets B, C, D and E, it follows that,

$A \neq B$, $A \neq C$, $A \neq D$, $A \neq E$.

Since $B = \phi$ but none of the other sets are empty. Therefore $B \neq C$, $B \neq D$ and $B \neq E$.

Also $C = \{5\}$ but $-5 \in D$, hence $C \neq D$.

Since $E = \{5\}$,$C = E$. Further, $D = \{-5, 5\}$ and $E = \{5\}$, we find that, $D \neq E$.

Thus, the only pair of equal sets is C and E.

Illustration-7 : Show that $A \cup B = A \cap B$ implies $A = B$

Solution : Let $a \in A$. Then $a \in A \cup B$. Since $A \cup B = A \cap B$, $a \in A \cap B$. So $a \in B$.

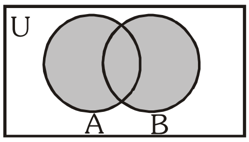
Therefore, $A \subseteq B$, Similarly, if $b \in B$, then $b \in A \cup B$. Since

$A \cup B = A \cap B$, $b \in A \cap B$. So, $b \in A$. Therefore, $B \subseteq A$. Thus, $A = B$

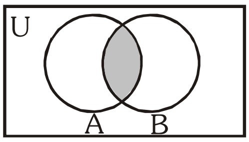
1.7 Venn Diagram

Venn diagram is a diagram representing sets pictorially as circles or closed curves within an enclosing rectangle/ a closed curve (the universal set), common elements of the sets being represented by intersections of the circles/closed curves.

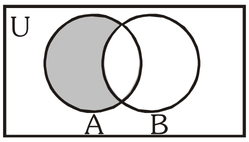
See the following Examples :



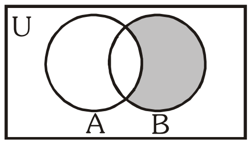
$A \cup B$



$A \cap B$

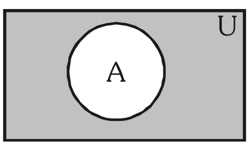


$A - B$

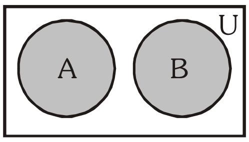


$B - A$

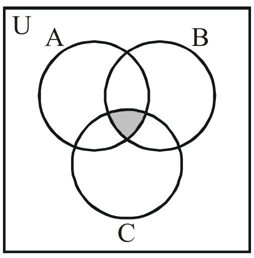
Clearly $(A - B) \cup (B - A) \cup (A \cap B) = A \cup B$



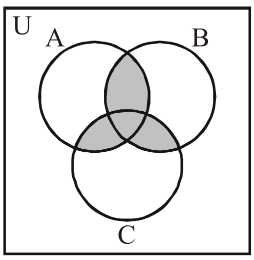
A'



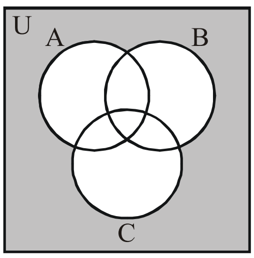
Disjoint Sets



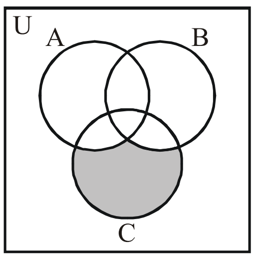
$(A \cap B \cap C)$



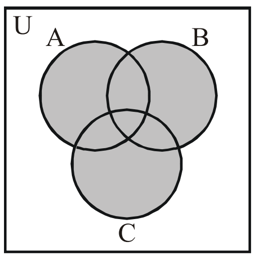
$(A \cap B) \cup (B \cap C) \cup (C \cap A)$



$(A \cup B \cup C)'$



$C-(A \cup B)$



$(A \cup B \cup C)$

Do yourself -1 :

1. Check whether the following statements are true or false :

(a) $A \cap \phi = \phi$ **(b)** $A \cap U = A$ **(c)** $A \cup \phi = A$ **(d)** $A \cup U = U$

(e) $A \cap B \subseteq A$ **(f)** $A \cap B \subseteq B$ **(g)** $A \subseteq A \cup B$ **(h)** $B \subseteq A \cup B$

(i) $A \subseteq B \Rightarrow A \cap B = A$ **(j)** $A \subseteq B \Rightarrow A \cup B = B$

2. $P(A) = P(B) \Rightarrow$

(A) $A \subseteq B$ (B) $B \subseteq A$ (C) $A = B$ (4D) none of these

3. The set of intelligent students in a class is-

(A) a null set (B) a singleton set

(C) a finite set (D) not a will defined collection

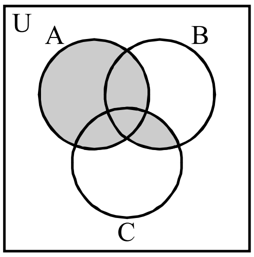
1.8 Some Important Laws

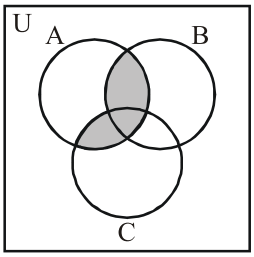
(i) **Commutative Law :** $A \cup B = B \cup A$; $A \cap B = B \cap A$

(ii) **Associative Law :** $(A \cup B) \cup C = A \cup (B \cup C)$; $(A \cap B) \cap C = A \cap (B \cap C)$

(iii) **Distributive Law :**

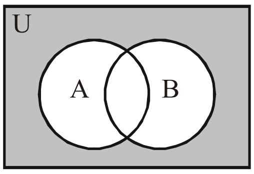
(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$; (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

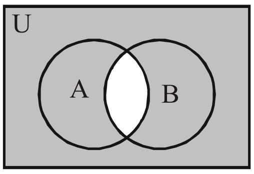




(iv) **De-Morgan's Law :**

(a) $(A \cup B)' = A' \cap B'$; (b) $(A \cap B)' = A' \cup B'$





Do yourself-2 :

1. If A and B are any two sets, then verify the following using venn diagram or otherwise :

(i) $A - B = A \cap B'$ **(ii)** $B - A = B \cap A'$

(iii) $A - B = A \Leftrightarrow A \cap B = \phi$ **(iv)** $(A - B) \cup B = A \cup B$

(v) $(A - B) \cap B = \phi$

(vi) $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

(vii) $A - (B \cup C) = (A - B) \cap (A - C)$

(viii) $A - (B \cap C) = (A - B) \cup (A - C)$

1.9 Some Important Results on Cardinality of Sets

If A, B and C are finite sets, and U be the finite universal set, then

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cup B) = n(A) + n(B) \Leftrightarrow$ A, B are disjoint sets.

(iii) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

Illustration-8 : In a group of 1000 people, there are 750 who can speak Hindi and 400 who can speak Bengali. How many can speak Hindi only ?How many can speak Bengali ? How many can speak both Hindi and Bengali?

Solution : Let A and B be the sets of persons who can speak Hindi and Bengali respectively.

then $n(A \cup B) = 1000$, $n(A) = 750$, $n(B) = 400$.

Number of persons whose can speak both Hindi and Bengali

$= n(A \cap B) = n(A) + n(B) - n(A \cup B)$

$= 750 + 400 - 1000 = 150$

Number of persons who can speak Hindi only

$= n(A - B) = n(A) - n(A \cap B) = 750 - 150 = 600$

Number of persons who can speak Bengali only

$= n(B - A) = n(B) - n(A \cap B) = 400 - 150 = 250$

Illustration-9 : Each person in a group of 80 can speak either Hindi or English or both. If 55 persons can speak English and 40 can speak both, find the number of persons who can speak Hindi.

Solution : Let E = set of persons who can speak English.

H = set of persons who can speak Hindi.

$n(E) = 55$, $n(H) = x$, $n(E \cap H) = 40$, $n(H \cup E) = 80$

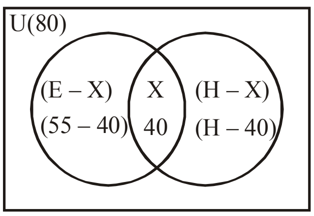
Using $n(H \cup E) = n(H) + n(E) - n(H \cap E)$

$\Rightarrow 80 = x + 55 - 40$

$\Rightarrow x = 80 - 55 + 40 = 65$

Alternate :

Using the Venn diagram



Let $X = E \cap H$

$n(U) = n(E - X) + n(X) + n(H - X)$

$80 = (55 - 40) + 40 + n(H) - 40$

$\Rightarrow n(H) = 80 - 15 = 65$

Illustration-10 : A group of members know at least one of the two languages, Hindi or Kannada. In the group, 150 members know Hindi and 80 members know Kannada and 70 of them know both. How many members are there in the group ?

Solution : Let H = set of persons who know Hindi.

K = set of persons who know Kannada.

$n(H \cap K)$ = the number of persons who know both Hindi and Kannada is 70.

$n(H \cup K) = n(H) + n(K) - n(H \cap K)$

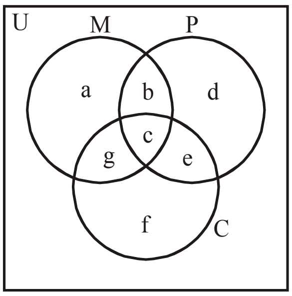
$= 150 + 80 - 70 = 160$

Illustration-11 : In a survey of 25 students, it was found that 15 had taken mathematics, 12 had taken physics, and 11 had taken chemistry, 5 had taken mathematics and chemistry, 9 had taken mathematics and physics, 4 had taken physics and chemistry, and 3 had taken all the three subjects. Find the number of students who had taken.

(A) Only mathematics (B) Mathematics and physics, but not chemistry

(C) At least one of the three subjects (D) Only one of the three subjects

Solution : Let M denote the set of students who had taken mathematics. P the set of students who had taken physics and C the set of students who had taken chemistry. In the Venn diagram, let a, b, c, d, e, f, g denote the number of students in the respective regions.



Now, $n(M \cap P \cap C) = c = 3$

$n(M \cap C) = g + c = 5 \Rightarrow g = 2$

$n(M \cap P) = b + c = 9 \Rightarrow b = 6$

$n(P \cap C) = e + c = 4 \Rightarrow e = 1$

$n(M) = a + b + g + c = 15 \Rightarrow a = 4$

$n(P) = b + c + d + e = 12 \Rightarrow d = 2$

$n(C) = e + f + g + c = 11 \Rightarrow f = 5$

(a) The number of students who had taken only mathematics = a = 4.

(b) The number of students who had taken mathematics and physics, but not chemistry = b=6

(c) The number of students who has taken at least one of the three subjects

$= a + b + c + d + e + f + g = 23.$

(d) The number of students who had taken only one of the three subjects = a + d + f = 11.

Do yourself-3 :

1. Verify the following using Venn diagram.

(i) $n(A - B) = n(A) - n(A \cap B)$ i.e. $n(A - B) + n(A \cap B) = n(A)$

(ii) $n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$

(iii) $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

2. If A and B are two sets, then $A \cap (A \cup B)'$ is equal to-

(A) A (B) B (C) ϕ (D) none of these

3. If A is any set, then-

(A) $A \cup A' = \phi$ (B) $A \cup A' = U$ (C) $A \cap A' = U$ (D) none of these

4. If A, B be any two sets, then $(A \cup B)'$ is equal to-

(A) $A' \cup B'$ (B) $A' \cap B'$ (C) $A \cap B$ (D) $A \cup B$

5. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 2, 5\}$, $B = \{6, 7\}$ then $A \cap B'$ is-

(A) B' (B) A (C) A' (D) B.

6. If A and B are two sets, then $A \cup B = A \cap B$ iff-

(A) $A \subseteq B$ (B) $B \subseteq A$ (C) $A = B$ (D) none of these

7. Two sets A, B are disjoint iff-

(A) $A \cup B = \phi$ (B) $A \cap B \neq \phi$ (C) $A \cap B = \phi$ (D) None of these

8. Which of the following is a null set ?

(A) $\{0\}$ (B) $\{x : x > 0 \text{ or } x < 0\}$

(C) $\{x : x^2 = 4 \text{ or } x = 3\}$ (D) $\{x : x^2 + 1 = 0, x \in \mathbb{R}\}$

9. If $A = \{2, 4, 5\}$, $B = \{7, 8, 9\}$ then $n(A \times B)$ is equal to-

(A) 6 (B) 9 (C) 3 (D) 0

10. Which set is the subset of all given sets ?

(A) $\{1, 2, 3, 4, ...\}$ (B) $\{1\}$ (C) $\{0\}$ (D) $\{\ \}$

11. If $Q = \left\{x : x = \frac{1}{y}, \text{ where } y \in \mathbb{N}\right\}$, then-

(A) $0 \in Q$ (B) $1 \in Q$ (C) $2 \in Q$ (D) $\frac{2}{3} \in Q$

12. $A = \{x : x \neq x\}$ represents-

(A) $\{0\}$ (B) $\{\ \}$ (C) $\{1\}$ (D) $\{x\}$

2. THEORY OF NUMBERS

2.1 Types of Numbers :

(i) Natural numbers : The counting numbers 1, 2, 3, 4, ... are called natural numbers. The set of natural numbers is denoted by $\mathbb{N}$. Thus $\mathbb{N} = \{1, 2, 3, 4, ...\}$

(ii) Whole numbers : Natural numbers including zero are called whole numbers. The set of whole numbers is denoted by **W** or $\mathbb{N}_0$. Thus $\mathbf{W} = \{0, 1, 2, 3, 4, ...\}$

(iii) Integers : The numbers ... –3, –2, –1, 0, 1, 2, 3, ... are called integers and the set is denoted by **I** or $\mathbb{Z}$. Thus ($\mathbf{I}$ or $\mathbb{Z}$) = $\{... -3, -2, -1, 0, 1, 2, 3, ...\}$

Note :

- Natural numbers are also called positive integers

  (some time denoted by $\mathbf{I}^+$ or $\mathbb{Z}^+$) = $\{1, 2, 3, ...\}$

- Whole numbers are also called non-negative integers

  (denoted by **W** or $\mathbf{I}_0^+$ or $\mathbb{Z}_0^+$) = $\{0, 1, 2, 3, ...\}$

- The set of negative integers, $\mathbf{I}^-$ or $\mathbb{Z}^- = \{..., -3, -2, -1\}$
- The set of non positive integers in $\mathbf{I}_0^-$ or $\mathbb{Z}_0^- = \{..., -3, -2, -1, 0\}$
- Zero is neither positive nor negative but 0 is a member of the set of non negative integers as well as of the set of non positive integers.

(iv) Even integers : Integers which are divisible by 2 are called even integers.

e.g. 0, ±2, ±4, ...

(v) Odd integers : Integers which are not divisible by 2 are called as odd integers.

e.g. ±1, ±3, ±5, ±7, ...

(vi) Prime number : Natural number having exactly two positive divisors i.e. 1 and itself are called prime numbers.

e.g. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ...

(vii) Composite number : Let 'a' be a natural number, 'a' is said to be composite if it has atleast three distinct positive divisors.

Note :

- 1 is the only natural number that is neither a prime number nor a composite number.
- 2 is the only prime number which is even.
- Numbers which are not prime are composite numbers (except 1).
- '4' is the smallest composite number.

(viii) Co-prime number : Two natural numbers (not necessarily prime) are coprime, if their H.C.F (Highest common factor) is 1.

e.g. (1, 2), (1, 3), (3, 4), (3, 10), (3, 8), (5, 6), (7, 8) (15, 16) etc.

These numbers are also called relatively prime numbers.

Note :

- Two prime numbers are always co-prime but converse need not be true.
- Two consecutive natural numbers are always co-prime numbers.
- Two consecutive odd natural numbers are always co-prime numbers.

(ix) Rational numbers : The numbers, which can be reduced in the form p/q where p, q $\in \mathbb{Z}$ and $q \neq 0$ and H.C.F (p, q) = 1, are called rational numbers. '$\mathbb{Q}$' represents their set.

Note :

- All integers are rational numbers with q = 1
- When numbers are expressed in reduced form of $\frac{p}{q}$, $q \neq 1$, the rational numbers are called fractions.
- Rational numbers when represented in decimal form are either 'terminating' or 'non-terminating but repeating'.

  e.g., 5/4 = 1.25 (terminating)

  5/3 = 1.6666 ... or $1.\overline{6}$ or $1.\dot{6}$ (non terminating but repeating)
- $0.\overline{9} = 0.9999... = 1$

Illustration-12 : Express the following rational numbers in the form of p/q, (where p, q $\in \mathbb{Z}$)

(i) $0.1\overline{2}$ (ii) $1.5\overline{23}$

Solution : **(i)** Let x = 0.1222...

$$10x = 1.\overline{2} \quad \text{... (i)}$$

$$100x = 12.\overline{2} \quad \text{... (ii)}$$

$$\Rightarrow \quad 90x = 11$$

$$\Rightarrow \quad x = \frac{11}{90} \quad \text{(so x is a rational number)}$$

(ii) Let $x = 1.5\overline{23}$

$$10x = 15.\overline{23}$$

$$1000x = 1523.\overline{23}$$

$$990x = 1508$$

$$\Rightarrow x = \frac{1508}{990} = \frac{754}{495} \quad \text{(so x is a rational number)}$$

(x) Irrational numbers : Numbers, which cannot be represented in p/q form as above. In decimal representation, they are neither terminating nor repeating.

e.g., $\sqrt{2},(15)^{1/3},\pi$ etc.

Note :

- $\pi \neq \frac{22}{7}$. Infact $\pi < \frac{22}{7}$

  $\left(\frac{22}{7} = 3.142857...\right)$ is only an approximate value of π in terms of rational numbers, taken for the sake of convenience.

  Actually π = 3.14159265359...

(xi) Real numbers : All rational and irrational numbers taken together form the set of real numbers, represented by $\mathbb{R}$. This is the largest set in the real world of numbers.

Note :

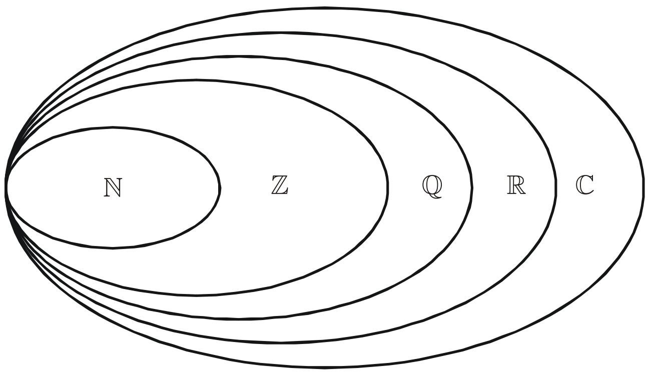
- Division by 0 is not defined.
- Integers are rational number, but converse need not be true.
- Sum of a rational number and an irrational number is always an irrational number.

  e.g. $5+\sqrt{6}$
- The product of a non zero rational number and an irrational number will always be an irrational number.
- If a $\in \mathbb{Q}$ and b $\notin \mathbb{Q}$ then ab = rational number, only if a = 0.
- Sum, difference, product and quotient of two irrational numbers need not be an irrational number or we can say, result may be a rational number also.
- Sum, difference, product and quotient of two rational numbers is always a rational number.

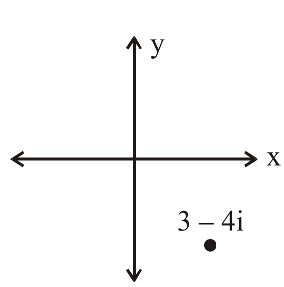
(xii) Complex number : A number z of the form a + ib is called complex number, where a, b $\in \mathbb{R}$ and i stands for $\sqrt{-1}$. Here 'a' is called real part of z denoted by Re(z) and 'b' is called imaginary part of z denoted by Im(z).

The set of complex number is represented by $\mathbb{C}$.

It may be noted that $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$



Complex numbers can be represented on the **complex plane,** in the same way as a point (x, y) is plotted on the Cartesian plane. We can plot the number x + iy by taking the real part 'x' as the horizontal coordinate and the imaginary part 'y' as the vertical coordinate. For example in the adjacent diagram, the point 3 – 4i is shown on the complex plane.



When graphing complex numbers, the horizontal axis is often referred as the **real axis** and the vertical axis as the **imaginary axis**.

The complex plane also gives us a way to visualize the magnitude of a complex number.

The magnitude of a complex number is its distance from the origin when plotted on the complex plane. We use |x + iy| to denote the distance between x + iy and the origin when plotted on the complex plane. So, for example, we have $|3-4i| = \sqrt{3^2+(-4)^2} = 5$. More generally, we have $|x+iy| = \sqrt{x^2+y^2}$

Conjugate of z :

If z = a + ib, a, b $\in \mathbb{R}$, then conjugate of z (denoted as $\bar{z}$), $\bar{z}$ = a – ib.

Illustration-13 : Find all complex numbers z when

(i) $\bar{z} = z$ (ii) $\bar{z} = -z$

Solution : (i) Let z = a + i b, $\forall$ a, b $\in \mathbb{R}$

$z = \bar{z} \Rightarrow a + ib = a - ib \Rightarrow 2ib = 0$

$\Rightarrow b = 0$, so $z = a$ where $a \in \mathbb{R}$

(ii) for $\bar{z} = -z \Rightarrow a - ib = -a - ib \Rightarrow 2a = 0$

$\Rightarrow a = 0$, so z = ib, $\forall$ b $\in \mathbb{R}$

Numbers to Remember :

Number	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Square	4	9	16	25	36	49	64	81	100	121	144	169	196	225	256	289	324	361	400
Cube	8	27	64	125	216	343	512	729	1000	1331	1728	2197	2744	3375	4096	4913	5832	6859	8000
Sq. Root	1.41	1.73	2	2.24	2.45	2.65	2.83	3	3.16										

← rounded upto two places of decimal

Note :

- Square of a real number is always non negative. (i.e. $x^2 \geq 0$)
- Square root of a positive number is always positive. e.g. $\sqrt{4}$ = 2

Do yourself-4 :

1. If $a\sqrt{2}+b=3\sqrt{2}+4$, find the integral value of a, b and justify your answer.

2. Express the following in form of p/q, where p, q $\in \mathbb{Z}$ and q $\neq$ 0

(i) $0.\overline{18}$ (ii) $0.\overline{16}$ (iii) $0.4\overline{23}$

3. Prove that $\sqrt{2}$ is an irrational number.

4. Identify rational, irrational among x + y, x – y, xy and x/y. When x and y $(xy \neq 0)$

(i) are both rational

(ii) are both irrational

(iii) one is rational other is irrational

5. $\sqrt{a}+\sqrt{b}$ and a – b are rational numbers. Prove that $\sqrt{a}, \sqrt{b}$, a and b all are rational.

6. Check x = 0.101001000100001... is rational or irrational, where the number of zeroes between units increased by 1.

2.2 Divisibility Test

Divisibility of	Test
2	The digit at the unit's place of the number is divisible by 2.
3	The sum of digits of the number is divisible by 3.
4	The last two digits of the number together are divisible by 4.
5	The digit at the unit's place is either 0 or 5.
6	The digit at the unit's place of the number is divisible by 2 & the sum of all digits of the number is divisible by 3.
8	The last 3 digits of the number all together are divisible by 8.
9	The sum of all digits is divisible by 9.
11	The difference between the sum of the digits at even places and the sum of digits at odd places is 0 or multiple of 11. e.g. 1298, 1221, 123321, 12344321, 1234554321, 123456654321

Illustration-14 : Consider a number N = 2 1 P 5 3 Q 4

(i) Number of ordered pairs (P, Q) so that the number 'N' is divisible by 9, is

(A) 11 (B) 12 (C) 10 (D) 8

(ii) Number of values of Q so that the number 'N' is divisible by 8, is

(A) 4 (B) 3 (C) 2 (D) 6

Solution : (i) Sum of digits = P + Q + 15

'N' is divisible by 9 if

P + Q + 15 = 18, 27

$\Rightarrow$ P + Q = 3 .... (i) or P + Q = 12 .... (ii)

From equation (i)

$$\left.\begin{array}{ll} P=0, & Q=3 \\ P=1, & Q=2 \\ P=2, & Q=1 \\ P=3, & Q=0 \end{array}\right\} \text{Number of ordered pairs is 4}$$

From equation (ii)

$$\left.\begin{array}{ll} P=3, & Q=9 \\ P=4, & Q=8 \\ \cdots\cdots\cdots & \cdots\cdots\cdots \\ P=8, & Q=4 \\ P=9, & Q=3 \end{array}\right\} \text{Number of ordered pairs is 7}$$

Total number of ordered pairs is 11

(ii) N is divisible by 8 if

Q = 0, 4, 8

Number of values of Q is 3

Do yourself-5 :

1. If $x, y \in \mathbb{N}$, and $x \cdot y = 20$ then find all possible ordered pairs (x, y).

2. Find all (x, y) where $x, y \in \mathbb{N}$ such that

(i) $\frac{1}{x}+\frac{1}{y}=1$ **(ii)** $xy + 5x = 4y + 38$

3. Let x and y be positive integers such that $\frac{1}{x}+\frac{1}{y}=\frac{1}{7}$. Find all (x,y).

4. How many integers in between 500 to 2020 (both inclusive) are multiple of 3 or 7?

5. How many integers in between 1000 to 2020 (both inclusive) are divisible by 5 or 7 but not divisible by 35 ?

Paragraph for Q.6 to Q.8

Consider the number N = 7 7 4 9 5 8 P 9 6 Q

6. If P = 2 and the number N is divisible by 3, then find the number of possible values of Q.

7. If N is divisible by 4, then find the number of possible ordered pairs (P, Q).

8. If N is divisible by 8 and 9 both, then find the number of possible ordered pairs (P,Q).

3. INTERVALS

Intervals are basically subsets of $\mathbb{R}$ and are very much important in mathematics as you will get to know shortly. If there are two numbers a, b $\in \mathbb{R}$ such that a < b, we can define four types of intervals as follows :

3.1 Closed Intervals

All numbers 'x' between a and b including both numbers is written in closed interval. It is denoted by [ ]. **i.e.** $a \le x \le b$ or $x \in [a,b]$ or $\{x : x \in \mathbb{R}$ and $a \le x \le b\}$

Graphical Representation :

3.2 Open Intervals

All numbers 'x' between a and b excluding both numbers is written in open interval. It is denoted by] [or ( ). **i.e.** $a < x < b$ or $x \in]a, b[$ or $x \in (a, b)$ or $\{x : x \in \mathbb{R}$ and $a < x < b\}$

Graphical Representation :

3.3 Open-Closed Intervals

All numbers 'x' between a and b including b and excluding a is written in open - closed interval. It is denoted by]a, b] or (a, b] or $a < x \le b$ or $\{x : x \in \mathbb{R}$ and $a < x \le b\}$

Graphical Representation :

3.4 Closed-Open Intervals

All numbers 'x' between a and b including a but excluding b is written in closed-open interval. It is denoted by [a, b[or [a, b) or $a \le x < b$ or $\{x : x \in \mathbb{R}$ and $a \le x < b\}$

Graphical Representation :

3.5 The Infinite Intervals are Defined as follows :

- $(a, \infty) = \{x : x > a\}$
- $[a, \infty) = \{x : x \ge a\}$
- $(-\infty, b] = \{x : x \le b\}$
- $(-\infty, b) = \{x : x < b\}$

Note : $x \in \{1, 2\}$ denotes some particular values of x, i.e. $x = 1, 2$

If there is no value of x, then we can say $x \in \phi$ (Null set)

Intervals are particularly important in solving inequalities.

Illustration-15 : Represent following sets on the number line

(i) $(-\infty, 3) \cup [5, \infty)$ (ii) $x \le 5$ or $x > 7$ (iii) $(-\infty, -2) \cup (-3, 5]$

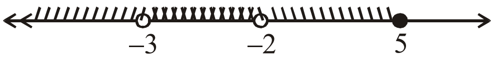
(iv) $(-2, 5] \cap [-3, 2)$ (v) $-1 \le x \le 3$ and $-2 < x$

(vi) $[-2, 10) - (1, 5]$ (vii) $(-1, 2) - (0, 3]$

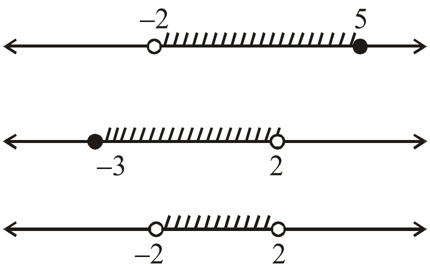
Solution :

(i) 

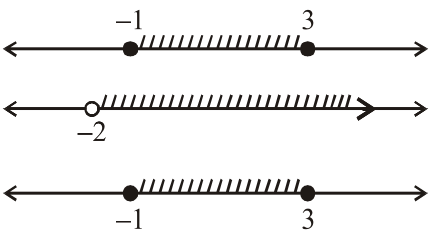
(ii) 

(iii) 

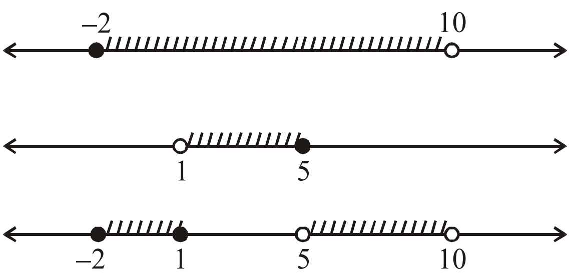
Union of given two sets is $(-\infty, 5]$

(iv) 

The intersection of given two sets is $(-2, 2)$.

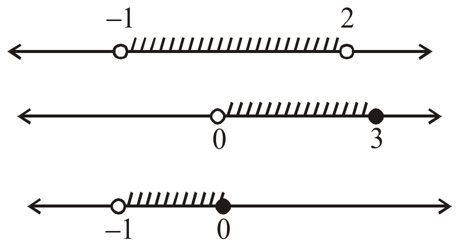
(v) 

The intersection of given two sets is $[-1, 3]$

(vi) 

Difference of given two sets is $[-2, 1] \cup (5, 10)$

(vii)



Difference of given two sets is (–1, 0].

Do yourself-6 :

Represent following sets on the number line :

(i) $(-3, 2] \cup [5, 10) \cup (-1, 4)$

(ii) $(A \cup B) \cap C$ when

$A = (-\infty, -2)$, $B = [4, 10]$ & $C = [-10, 5)$

(iii) $[-3, 5] \cap (-2, 6) \cap [-4, 4)$

(iv) $(A \cup B) - C$ when

$A = (-\infty, -5)$, $B = [4, 10]$ & $C = (-10, 10)$

4. INDICES AND SURDS

4.1 Indices

Definition of Indices :

If 'a' is any non zero real or imaginary number and 'm' is a positive integer, then $a^m = a.\ a.\ a....a$ (m times). Here a is called the base and m is the index, power or exponent.

Law of indices :

(i) $a^0 = 1, (a \neq 0)$

(ii) $a^{-m} = \dfrac{1}{a^m}, (a \neq 0)$

(iii) $a^m.a^n = a^{m+n}$

(iv) $\dfrac{a^m}{a^n} = a^{m-n}$, $a \neq 0$

(v) $(a^m)^n = a^{mn} = (a^n)^m$

(vi) $\sqrt[q]{a^p} = a^{p/q}$, $q \in \mathbb{N}$ and $q \geq 2$

(vii) If $x = y$, then $a^x = a^y$, but the converse may not be true. e.g. : $(1)^6 = (1)^8$, but $6 \neq 8$

For $a^x = a^y$ we have following possibilities

- If $a \neq \pm 1, 0$, then $x = y$
- If $a = 1$, then x, y may be any real number
- If $a = -1$, then x, y may be both even or both odd
- If $a = 0$, then x, y may be any positive real number

But if we have to solve the equations like $(f(x))^{g(x)} = (f(x))^{h(x)}$ (i.e same base, different indices) then we have to solve :

(a) $f(x) = 1$ (b) $f(x) = -1$ (c) $f(x) = 0$ (d) $g(x) = h(x)$

Verification should be done in (b) and (c) cases

Illustration-16 : Solve $(4-x)^{x^3-4x} = (4-x)^{(x^2-4)(x-1)}$

Solution : **Case-1 :** $4 - x = 1 \Rightarrow x = 3$

Case-2 : $4 - x = -1 \Rightarrow x = 5$

for $x = 5$, $x^3 - 4x$ is odd

and $(x^2 - 4)(x - 1)$ is even

so $x = 5$ does not satisfy the given equation

Case-3 : $4 - x = 0 \Rightarrow x = 4$

For $x = 4$, $x^3 - 4x > 0$ and $(x^2 - 4)(x - 1) > 0$

$\Rightarrow x = 4$ satisfies the given equation

Case-4 : $x^3 - 4x = (x^2 - 4)(x - 1)$

$\Rightarrow x^3 - 4x = x^3 - x^2 - 4x + 4$

$\Rightarrow x^2 - 4 = 0, \Rightarrow x = \pm 2$ which satisfies the given equation.

From case-1, case-2, case-3 and case-4, solutions set of the given equation is $\{-2, 2, 3, 4\}$

(viii) If $a^x = b^x$ then consider the following cases :

- If $a \neq \pm b$, then $x = 0$
- If $a = b \neq 0$, then x may have any real value for which a^x is well defined.
- If $a = -b \neq 0$, then x is even.
- If $a = b = 0$, then x can be any positive real.

If we have to solve the equation of the form $[f(x)]^{h(x)} = [g(x)]^{h(x)}$, i.e., same index, different bases, then we have to solve :

(a) $f(x) = g(x)$ (b) $f(x) = -g(x)$ (c) $h(x) = 0$

Verification should be done in (a), (b) and (c) cases.

Illustration-17 : Solve $(x^2-4)^{2x} = (x^2+2x)^{2x}$.

Solution : **Case-1 :** when $x = 0$

Base : $x^2 + 2x = 0$, so $x = 0$ does not satisfy given equation

Case-2 : $x^2 - 4 = x^2 + 2x \neq 0$

$\Rightarrow x \in \phi$

Case-3 : $x^2 - 4 = -x^2 - 2x \neq 0 \Rightarrow 2x^2 + 2x - 4 = 0$

$x^2 + x - 2 = 0 \Rightarrow x = 1$

satisfies the given equation

Case-4 : $x^2 - 4 = x^2 + 2x \Rightarrow x = -2$

$x = -2$ does not satisfy the given equation.

From all above cases, $x \in \{1\}$

Illustration-18 : If $a^x = b$, $b^y = c$, $c^z = a$, prove that $xyz = 1$, where a,b,c are distinct numbers.

Solution : We have, $a^{xyz} = (a^x)^{yz}$

$\Rightarrow a^{xyz} = (b)^{yz}$ $[\because a^x = b]$

$\Rightarrow a^{xyz} = (b^y)^z$

$\Rightarrow a^{xyz} = c^z$ $[\because b^y = c]$

$\Rightarrow a^{xyz} = a$ $[\because c^z = a]$

$\therefore a^{xyz} = a^1$

$\Rightarrow xyz = 1$

4.2 Surds

If 'x' is a rational number, which is not the n^{th} power ($n \in \mathbb{N}\backslash\{1\}$) of any rational number, then the number $x^{1/n}$ usually denoted by $\sqrt[n]{x}$ is called surd. The sign '$\sqrt[n]{\ }$' is called the radical sign. The number in the angular part of the sign, i.e., 'n' is called order of the surd. In case of n = 2 the expression $\sqrt[2]{x}$, simply written as $\sqrt{x}$.

Note :

- If $\sqrt[n]{x}$ is a surd then $-\left(\sqrt[n]{x}\right)$ is also a surd.
- Every surd is an irrational number (but every irrational number is not a surd).
- To rationalize the denominator of a fraction of the form $\frac{a}{\sqrt{b}}$, multiply the numerator and denominator

 of the fraction by $\sqrt{b} \Rightarrow \frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}}.\frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{\sqrt{b^2}} = \frac{a\sqrt{b}}{b}$.

Eg.

(a) $\sqrt{3}$ is a surd and $\sqrt{3}$ is an irrational number.

(b) $\sqrt[3]{5}$ is surd and $\sqrt[3]{5}$ is an irrational number.

(c) π is an irrational number, but it is not a surd.

4.2.1 Conjugate of a Surd

If two binomial surds (surds containing two terms such as $2+\sqrt{3}$, $2\sqrt{5}-\sqrt{7}$ etc) are such that only the sign connecting the individual terms are different, then they are said to be conjugate of each other. If these surds are quadratic, then their product would always be rational. So in case of a binomial quadratic surd, we use its conjugate as its rationalizing factor.

Ex. Conjugate of $3\sqrt{2}+\sqrt{5}$ is $3\sqrt{2}-\sqrt{5}$ or $-3\sqrt{2}+\sqrt{5}$

Illustration-19 : Rationalize the denominator of $\frac{1}{3\sqrt{2}+\sqrt{5}}$

Solution : A conjugate of $3\sqrt{2}+\sqrt{5}$ is $3\sqrt{2}-\sqrt{5}$

Therefore multiplying the conjugate in the numerator and denominator of the given fraction.

$$\frac{3\sqrt{2}-\sqrt{5}}{\left(3\sqrt{2}+\sqrt{5}\right)\left(3\sqrt{2}-\sqrt{5}\right)}$$

$$= \frac{3\sqrt{2}-\sqrt{5}}{\left(3\sqrt{2}\right)^2-\left(\sqrt{5}\right)^2}$$

$$= \frac{3\sqrt{2}-\sqrt{5}}{18-5}$$

$$= \frac{3\sqrt{2}-\sqrt{5}}{13}$$

Illustration-20 : Using the fact that $\left(\sqrt{x}+\sqrt{y}\right)^2 = x+y+\sqrt{xy}$ to find the square root of $7+2\sqrt{10}$. If this number can be expressed in the form $\sqrt{a}+\sqrt{b}$, where a ≤ b, find the value of b – a.

(A) 3 (B) 4 (C) 0 (D) 2

Solution : Let $\sqrt{7+2\sqrt{10}} = \sqrt{a}+\sqrt{b}$, where $a,b \in \mathbb{Q}$

$\Rightarrow 7+2\sqrt{10} = a+b+2\sqrt{ab}$

$\Rightarrow$ a + b = 7 and $\sqrt{ab} = \sqrt{10}$

$(b-a)^2 = (a+b)^2 - 4ab$

= 49 – 40 = 9

$\Rightarrow \sqrt{(b-a)^2} = 3 \;\Rightarrow\; b-a=3$ (since b > a)

OR

$$7+2\sqrt{10} = 7+2\cdot\sqrt{2}\cdot\sqrt{5} = \left(\sqrt{2}\right)^2+\left(\sqrt{5}\right)^2+2\cdot\sqrt{2}\cdot\sqrt{5} = \left(\sqrt{2}+\sqrt{5}\right)^2$$

So $\sqrt{7+2\sqrt{10}} = \sqrt{2}+\sqrt{5} \;\therefore\; b-a=3$

Illustration-21 : If $x = \frac{1}{2+\sqrt{3}}$, find the value of $x^3 - x^2 - 11x + 4$.

Solution : as $x = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{(2)^2-(\sqrt{3})^2}$

$x = \frac{2-\sqrt{3}}{4-3} = 2-\sqrt{3}$

$x - 2 = -\sqrt{3}$, squaring both sides, we get

$(x-2)^2 = \left(-\sqrt{3}\right)^2 \Rightarrow x^2+4-4x = 3$

$\Rightarrow x^2 - 4x + 1 = 0$

Now $x^3 - x^2 - 11x + 4$

$= x^3 - 4x^2 + x + 3x^2 - 12x + 4$

$= x(x^2-4x+1) + 3(x^2-4x+1) + 1 = x \times 0 + 3(0) + 1 = 0 + 0 = 0 = 1$

Illustration-22 : If $x = 3 - 2\sqrt{2}$, find $x^2 + \frac{1}{x^2}$

Solution : We have, $x = 3 - 2\sqrt{2}$.

$$\therefore \frac{1}{x} = \frac{1}{3-2\sqrt{2}} = \frac{1}{3-2\sqrt{2}} \times \frac{3+2\sqrt{2}}{3+2\sqrt{2}}$$

$$= \frac{3+2\sqrt{2}}{(3)^2 - (2\sqrt{2})^2} = \frac{3+2\sqrt{2}}{9-8} = 3 + 2\sqrt{2}$$

Thus, $x^2 + \frac{1}{x^2} = (3 - 2\sqrt{2})^2 + (3 + 2\sqrt{2})^2$

$= 2((3)^2 + (2\sqrt{2})^2) = 2(9 + 8) = 34$

Illustration-23 : Rationalise the denominator of $\frac{1}{\sqrt{3}-\sqrt{2}-1}$

Solution :
$$\frac{1}{\sqrt{3}-\sqrt{2}-1} = \frac{1}{\sqrt{3}-\sqrt{2}-1} \times \frac{\sqrt{3}+\sqrt{2}+1}{\sqrt{3}+\sqrt{2}+1}$$

$$= \frac{\sqrt{3}+\sqrt{2}+1}{(\sqrt{3}-\sqrt{2}-1)(\sqrt{3}+\sqrt{2}+1)} = \frac{\sqrt{3}+\sqrt{2}+1}{(\sqrt{3})^2-(\sqrt{2}+1)^2} = \frac{\sqrt{3}+\sqrt{2}+1}{-2\sqrt{2}} = -\left(\frac{\sqrt{6}+\sqrt{2}+2}{4}\right)$$

Illustration-24 : Evaluate the following

(i) $\left(\sqrt[3]{64}\right)^{\frac{-1}{2}}$ (ii) $\left(\frac{121}{169}\right)^{-3/2}$

Solution :
(i) $\left(\sqrt[3]{64}\right)^{\frac{-1}{2}} = \left[(64)^{\frac{1}{3}}\right]^{\frac{-1}{2}} = (64)^{\frac{1}{3}\times\frac{-1}{2}} = (64)^{\frac{-1}{6}}$

$= (2^6)^{\frac{-1}{6}} = 2^{6\times\left(\frac{-1}{6}\right)} = 2^{-1} = \frac{1}{2}$

(ii) $\left(\frac{11\times11}{13\times13}\right)^{-3/2} = \left(\frac{11^2}{13^2}\right)^{-3/2} = \left(\frac{11}{13}\right)^{2\times\frac{-3}{2}} = \left(\frac{11}{13}\right)^{-3} = \left(\frac{13}{11}\right)^{3} = \frac{2197}{1331}$

Illustration 25 : Simplify $\left[\sqrt[3]{\sqrt[6]{a^9}}\right]^4 \left[\sqrt[6]{\sqrt[3]{a^9}}\right]^4$

(A) a^{16} (B) a^{12} (C) a^8 (D) a^4

Solution. $a^{9(1/6)(1/3)4} \cdot a^{9(1/3)(1/6)4} = a^2 \cdot a^2 = a^4$.

Illustration 26 : Simplify $a\left(\frac{\sqrt{a}+\sqrt{b}}{2b\sqrt{a}}\right)^{-1}+b\left(\frac{\sqrt{a}+\sqrt{b}}{2a\sqrt{b}}\right)^{-1}$

Solution. The given expression is equal to

$$a\left(\frac{2b\sqrt{a}}{\sqrt{a}+\sqrt{b}}\right)+b\left(\frac{2a\sqrt{b}}{\sqrt{a}+\sqrt{b}}\right)$$

$$=\frac{2ab}{\sqrt{a}+\sqrt{b}}\left(\sqrt{a}+\sqrt{b}\right)=2ab$$

Illustration 27 : Evaluate $\sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{7+\sqrt{48}}}}$

Solution. $\sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{7+\sqrt{48}}}}$

$$=\sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{4+3+2\sqrt{12}}}}=\sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{4}+\sqrt{3}}}$$

$$=\sqrt{3+\sqrt{3}+\sqrt{3}+1}=\sqrt{4+2\sqrt{3}}=\sqrt{3}+1\ =\sqrt{3}+1$$

Illustration 28 : Find rational numbers a and b, such that $\frac{4+3\sqrt{5}}{4-3\sqrt{5}}=a+b\sqrt{5}$

Solution. $\frac{4+3\sqrt{5}}{4-3\sqrt{5}}\times\frac{4+3\sqrt{5}}{4+3\sqrt{5}}=a+b\sqrt{5}$

$$\frac{61+24\sqrt{5}}{-29}=a+b\sqrt{5}$$

$$a=-\frac{61}{29},b=-\frac{24}{29}$$

Do yourself-7 :
Only One Option is Correct for Q.1 to Q.5

1. $\left(\left((625)^{-1/2}\right)^{-1/4}\right)^{2}=$

(A) 4 (B) 5 (C) 2 (D) 3

2. $\left(5\left(8^{1/3}+27^{1/3}\right)^{3}\right)^{1/4}=$

(A) 3 (B) 6 (C) 5 (D) 4

3. $\left\{\sqrt[4]{\left(\frac{1}{x}\right)^{-12}}\right\}^{-2/3} =$

(A) $\frac{1}{x^2}$ (B) $\frac{1}{x^4}$ (C) $\frac{1}{x^3}$ (D) $\frac{1}{x}$

4. $\frac{\sqrt{x^3}\times\sqrt[3]{x^5}}{\sqrt[5]{x^3}}\times\sqrt[30]{x^{77}} =$

(A) $x^{76/15}$ (B) $x^{78/15}$ (C) $x^{79/15}$ (D) $x^{77/15}$

5. If $\sqrt[4]{\sqrt[3]{x^2}} = x^k$, then k =

(A) $\frac{2}{6}$ (B) 6 (C) $\frac{1}{6}$ (D) 7

6. Simplify :

(i) $\frac{\left(5\sqrt{3}+\sqrt{50}\right)\left(5-\sqrt{24}\right)}{\left(\sqrt{75}-5\sqrt{2}\right)}$ (ii) $\frac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}}-\frac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}}+\frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}}$

(iii) $\frac{\sqrt{\left[6+2\sqrt{3}+2\sqrt{2}+2\sqrt{6}\right]-1}}{\sqrt{5+2\sqrt{6}}}$ (iv) $\frac{2.3^{n+1}-7.3^{n-1}}{3^{n+1}+2\left(\frac{1}{3}\right)^{1-n}}$

(v) $\left(\frac{1}{3}\right)^{-10}.27^{-3}+\left(\frac{1}{5}\right)^{-4}.(25)^{-2}+\left(64^{-\frac{1}{9}}\right)^{-3}$

7. If $\frac{\left(2^{n+1}\right)^m\left(2^{2n}\right)2^n}{\left(2^{m+1}\right)^n\left(2^{2m}\right)} = 1$, then find the value of $\frac{m}{n}$

8. Find rational numbers a and b, such that $\frac{2+3\sqrt{5}}{1-3\sqrt{5}} = a+b\sqrt{5}$

9. The square root of $11+\sqrt{112}$ is $a+\sqrt{b}$, $a,b\in\mathbb{N}$ then $b-a$ is

10. $\sqrt{11+\sqrt{21}} = \sqrt{\frac{a}{b}}+\sqrt{\frac{c}{d}}$, where a, b, c and d are natural numbers with gcd(a, b) = gcd(c, d) = 1. Find a + b + c + d.

11. For $x \neq 0$ find the value of $\left(\frac{x^\ell}{x^m}\right)^{\ell^2+\ell m+m^2}.\left(\frac{x^m}{x^n}\right)^{m^2+mn+n^2}.\left(\frac{x^n}{x^\ell}\right)^{n^2+n\ell+\ell^2}$

12. For $a^x = (x + y + z)^y$, $a^y = (x + y + z)^z$, $a^z = (x + y + z)^x$, then find the values of x,y and z. Where $a > 0$ and $a \neq 1$.

5. FACTORIZATIONS

5.1 $a^2 - b^2 = (a - b)(a + b)$

Illustration-29 : $(3x - y)^2 - (2x - 3y)^2$

Solution : Use $a^2 - b^2 = (a - b)(a + b)$

$(3x - y)^2 - (2x - 3y)^2 = (3x - y + 2x - 3y)(3x - y - 2x + 3y) = (5x - 4y)(x + 2y)$

5.2 Factorising the Quadratic expression

Illustration-30 : $x^2 + 6x - 187$

Solution : $x^2 + 6x - 187 = x^2 + 17x - 11x - 187$

$= x(x + 17) - 11(x + 17)$

$= (x + 17)(x - 11)$

5.3 Factorisation by converting the given expression into a perfect square.

Illustration-31 : $9x^4 - 10x^2 + 1$

Solution : $9x^4 - 10x^2 + 1 = (3x^2)^2 - 2\cdot3x^2 + 1 - 4x^2$

$= (3x^2 - 1)^2 - (2x)^2$

$= (3x^2 - 1 - 2x)(3x^2 - 1 + 2x)$

$= (x - 1)(3x + 1)(x + 1)(3x - 1)$

5.4 $a^3 \pm b^3 \equiv (a \pm b)(a^2 \mp ab + b^2)$

Illustration-32 : $a^6 - b^6$

Solution : $a^6 - b^6 = (a^2)^3 - (b^2)^3$

$= (a^2 - b^2)(a^4 + a^2b^2 + b^4)$

$= (a - b)(a + b)(a^2 - ab + b^2)(a^2 + ab + b^2)$

5.5 Using Factor Theorem :

Illustration-33 : $x^3 - 13x - 12$

Solution : As $x = -1$ makes given expression 0, $x + 1$ is a factor

$$
\begin{array}{r|l|l}
x+1 & x^3 - 13x - 12 & x^2 - x - 12 \\
 & \underline{x^3 + x^2} & \\
 & -x^2 - 13x - 12 & \\
 & \underline{-x^2 - x} & \\
 & -12x - 12 & \\
 & \underline{-12x - 12} & \\
 & 0 &
\end{array}
$$

$\therefore\ x^3 - 13x - 12 = (x + 1)(x^2 - x - 12)$

$= (x + 1)(x - 4)(x + 3)$

5.6 $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$

Illustration-34 : $8x^3 + y^3 + 27z^3 - 18xyz$

Solution : $8x^3 + y^3 + 27z^3 - 18xyz = (2x)^3 + (y)^3 + (3z)^3 - 3(2x)(y)(3z)$

$= (2x + y + 3z)(4x^2 + y^2 + 9z^2 - 2xy - 6xz - 3yz)$

5.7 Cyclic Expression and its Factorization :

An expression is said to be cyclic with regard to the variables $x_1, x_2, ..., x_n$ arranged in this order, when it is unchanged by changing x_1 into x_2, x_2 into x_3, x_3 into x_4, ..., x_n into x_1.

For three variables

$E(x, y, z)$ is cyclic if $E(x, y, z) = E(y, z, x) = E(z, x, y)$

Ex. 1. $x + y + z$ is a cyclic expression

2. $x^2 + y^2 + z^2 + xy + yz + zx$ is a cyclic expression

3. $E(x, y, z) = x(y - z) + y(z - x) + z(x - y)$ is a cyclic expression because

$\Rightarrow E(y, z, x) = y(z - x) + z(x - y) + x(y - z)$

and $E(z, x, y) = z(x - y) + x(y - z) + y(z - x)$

$\Rightarrow E(x, y, z) = E(y, z, x) = E(z, x, y)$

Theorem :

If $E(x, y, z)$ is a cyclic expression and $x - y$ is a factor of $E(x, y, z)$ then $y - z$ and $z - x$ are also factors of $E(x, y, z)$.

Illustration-35 : Factorize $x^2(y - z) + y^2(z - x) + z^2(x - y)$

Solution : $x^2(y - z) + x(z^2 - y^2) + yz(y - z)$

$= (y - z)(x^2 - x(z + y) + yz)$

$= (y - z)(x^2 - xz - xy + yz)$

$= (y - z)(x(x - z) - y(x - y))$

$= (y - z)(x - z)(x - y)$

$= -(x - y)(y - z)(z - x)$

OR

Let $E(x, y, z) = x^2(y - z) + y^2(z - x) + z^2(x - y)$

when $y = x$, $E(x, y, z) = x^2(x - z) + x^2(z - x) + z^2(x - x) = x^3 - x^2z + x^2z - x^3 = 0$

$\Rightarrow x - y$ is factor of $E(x, y, z)$

So, $y - z$ and $z - x$ are also factors of $E(x, y, z)$

$E(x, y, z) = A(x - y)(y - z)(z - x)$ ...(i)

Since degree of $E(x, y, z)$ is 3 so A is constant. We can find A by substituting values of x, y & z in(i)

Let $x = 0, y = 1, z = 2$

$E(0, 1, 2) = A(-1)(-1)(2)$

$\Rightarrow 1(2 - 0) + 2^2(0 - 1) = 2A \Rightarrow A = -1$

$E(x, y, z) = -(x - y)(y - z)(z - x)$

5.8 Important Algebraic Identities

- $xy + ay + bx + ab = (x + a)(y + b)$
- $x^2 + 2xy + y^2 = (x + y)^2$
- $x^2 + y^2 + z^2 + 2xy + 2yz + 2zx = (x + y + z)^2$
- $x^2 - y^2 = (x - y)(x + y)$
- $x^4 + x^2 + 1 = (x^2 + 1)^2 - x^2 = (x^2 + x + 1)(x^2 - x + 1)$
- $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + ... + y^{n-1}),\ n \in \mathbb{N}$
- $x^{2n+1} + y^{2n+1} = (x + y)(x^{2n} - x^{2n-1}y + x^{2n-2}y^2 - ... + y^{2n}),\ n \in \mathbb{N}$
- $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$$= \frac{1}{2}(x+y+z)\left\{(x-y)^2 + (y-z)^2 + (z-x)^2\right\}$$

- $x^3 + y^3 + z^3 = 3xyz$ if $x + y + z = 0$ or $x = y = z$
- $x^3 + y^3 + z^3 + 3(x + y)(y + z)(z + x) = (x + y + z)^3$

Problem Solving Strategies :

- When facing a problem with gigantic numbers, try replacing them with smaller numbers and look for a pattern. You can often prove your pattern works and solve the problem by substituting variable expressions for the numbers.

Illustration-36 : Compute $\sqrt{2022 \times 2020 \times 2018 \times 2016 + 16}$ without using calculator.

Solution : Let $x = 2019$

$$\sqrt{2022 \times 2020 \times 2018 \times 2016 + 16}$$

$$= \sqrt{(x+3)(x+1)(x-1)(x-3) + 16}$$

$$= \sqrt{(x^2-9)(x^2-1) + 16}$$

$$= \sqrt{x^4 - 10x^2 + 25} = \sqrt{(x^2-5)^2} = x^2 - 5$$

$= 2019^2 - 5 = (2000)^2 + (19)^2 + 2 \times 2000 \times 19 - 5$

$= 4000000 + 361 + 76000 - 5$

$= 4,076,356$

- Focusing on what makes a problem tricky helps identify what strategies might be best to solve the problem.
- If you have a problem that involves expressions of the form $a + b$ and $a - b$, where a and/or b involve square roots, consider finding a way to multiply the expressions to get rid of the square roots.

Illustration-37 : Suppose $x = z - \sqrt{z^2 - 5}$ and $5y = z + \sqrt{z^2 - 5}$.

Find x when y = 2/3.

Solution : $x = z - \sqrt{z^2 - 5}$ and $5y = z + \sqrt{z^2 - 5}$, multiplying both equations

we get $5xy = z^2 - (z^2 - 5) = 5$

$\Rightarrow xy = 1 \Rightarrow x = \frac{3}{2}$.

- When making a substitution, take some time to look for the substitution that simplifies your work the most.
- If we have the product of two variables added to linear terms with both variables, such as $mn + 3m + 5n$, then there is a constant we can add that will allow us to factor. For example, adding 15 to $mn + 3m + 5n$ gives us $mn + 3m + 5n + 15 = (m + 5)(n + 3)$.

Illustration-38 : Find all integral solutions of x and y when $xy - y - 2x = 3$.

Solution : $xy - y - 2x = 3$

$\Rightarrow xy - y - 2x + 2 = 5$

$\Rightarrow y(x - 1) - 2(x - 1) = 5$

$\Rightarrow (x - 1)(y - 2) = 5$

$x-1$	$y-2$	(x,y)
5	1	(6,3)
1	5	(2,7)
–5	–1	(–4,1)
–1	–5	(0,–3)

All possible (x, y) and (6, 3), (2, 7), (0, –3) and (–4, 1)

- Guessing has a long and glorious history in mathematics and science. It is often a very important first step in many discoveries. Don't be afraid to guess! But make sure you test your guesses – a guess itself is only a first step.

Illustration-39 : Find all pairs of positive integers m and n such that m^2 is 105 greater than n^2.

Solution : Turning the words into math is easy :

$m^2 = n^2 + 105.$

$\Rightarrow (m - n)(m + n) = 105.$

$(m - n)(m + n) = 1.105 = 3.35 = 5.21 = 7.15$

Because m and n are positive, we know that m – n is smaller than m + n, so we only have these four cases to consider :

$m - n = 1$	$m - n = 3$	$m - n = 5$	$m - n = 7$
$m + n = 105$	$m + n = 35$	$m + n = 21$	$m + n = 15$

Each of these systems of equations gives us a solution (m, n). Adding the equations in the first case gives us 2m = 106, so m = 53. Substitution then gives n = 52. Similarly, we can work through each of the other three cases to find the four solutions (m, n) = (53, 52); (19, 16); (13, 8); (11, 4).

Illustration-40 : The number 7,999,999,999 has two prime factors. Find them.

Solution : Let 7,999,999,999 = 8,000,000,000 – 1

$= (2000)^3 - 1^3 = (2000 - 1)(2000^2 + 2000 + 1)$

$= (1999)(4002001)$

According to question, 7,999,999,999 has two prime factors, they must be 1999 and 4002001.

Illustration-41 : Factor $x^4 + 4y^4$.

Solution : $x^4 + 4y^4 = (x^2)^2 + (2y^2)^2 - 2(x^2)(2y^2) + (2xy)^2$

$= (x^2 + 2y^2)^2 - (2xy)^2$

$= (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)$

Do yourself-8 :

1. Factorize following expressions

(i) $x^4 - y^4$ **(ii)** $9a^2 - (2x - y)^2$ **(iii)** $4x^2 - 9y^2 - 6x - 9y$

2. Factorize following expressions

(i) $8x^3 - 27y^3$ **(ii)** $8x^3 - 125y^3 + 2x - 5y$

3. Factorize following expressions

(i) $x^2 + 3x - 40$ **(ii)** $x^2 - 3x - 40$ **(iii)** $x^2 + 5x - 14$

(iv) $x^2 - 3x - 4$ **(v)** $x^2 - 2x - 3$ **(vi)** $3x^2 - 10x + 8$

(vii) $12x^2 + x - 35$ **(viii)** $3x^2 - 5x + 2$ **(ix)** $3x^2 - 7x + 4$

(x) $7x^2 - 8x + 1$ **(xi)** $2x^2 - 17x + 26$ **(xii)** $3a^2 - 7a - 6$

(xiii) $14a^2 + a - 3$

4. Factorize following expressions

(i) $a^2 - 4a + 3 + 2b - b^2$ **(ii)** $x^4 + 324$ **(iii)** $x^4 - y^2 + 2x^2 + 1$

(iv) $4a^4 - 5a^2 + 1$ **(v)** $4x^4 + 81$ **(vi)** $1 + x^4 + x^8$

5. Factorize following expressions

(i) $x^3 - 6x^2 + 11x - 6$ **(ii)** $2x^3 + 9x^2 + 10x + 3$

(iii) $2x^3 - 9x^2 + 13x - 6$ **(iv)** $x^6 - 7x^2 - 6$

(v) $(x + y + z)^3 - x^3 - y^3 - z^3$

6. (i) Factorize the expressions $8a^6 + 5a^3 + 1$

(ii) Show that $(x - y)^3 + (y - z)^3 + (z - x)^3 = 3(x - y)(y - z)(z - x)$.

7. Factorize following expressions

(i) $(x + 1)(x + 2)(x + 3)(x + 4) - 15$

(ii) $4x(2x + 3)(2x - 1)(x + 1) - 54$

(iii) $(x - 3)(x + 2)(x + 3)(x + 8) + 56$

5.9 Miscellaneous Algebraic Manipulations

Illustration-42 : Suppose $x+\frac{1}{x}=5$ find **(i)** $x^2+\frac{1}{x^2}$ **(ii)** $x^4-\frac{1}{x^4}$

Solution : (i) Given : $x+\frac{1}{x}=5$

By squaring both sides $\left(x+\frac{1}{x}\right)^2=(5)^2 \Rightarrow x^2+\frac{1}{x^2}+2=25 \Rightarrow x^2+\frac{1}{x^2}=23$

(ii) $\left(x-\frac{1}{x}\right)^2=\left(x+\frac{1}{x}\right)^2-4=21 \Rightarrow x-\frac{1}{x}=\pm\sqrt{21}$

$x^2-\frac{1}{x^2}=\pm5\sqrt{21}$

from (i) we have $x^2+\frac{1}{x^2}=23$

So $x^4-\frac{1}{x^4}=\pm23\times5\sqrt{21}=\pm115\sqrt{21}$

Illustration-43 : Simplify $\sqrt{6+\sqrt{11}}+\sqrt{6-\sqrt{11}}$

Solution : Let $x=\sqrt{6+\sqrt{11}}+\sqrt{6-\sqrt{11}}$

By squaring both sides

$x^2=\left(\sqrt{6+\sqrt{11}}\right)^2+\left(\sqrt{6-\sqrt{11}}\right)^2+2\sqrt{36-11}$

$= 12 + 10 = 22$

Since x is positive, so $x=\sqrt{22}$.

OR

$$x=\sqrt{6+\sqrt{11}}+\sqrt{6-\sqrt{11}}=\frac{1}{\sqrt{2}}\left(\sqrt{12+2\sqrt{11}}+\sqrt{12-2\sqrt{11}}\right)$$

$$=\frac{1}{\sqrt{2}}\left(\sqrt{\left(\sqrt{11}+1\right)^2}+\sqrt{\left(\sqrt{11}-1\right)^2}\right)=\frac{1}{\sqrt{2}}\left(\sqrt{11}+1+\sqrt{11}-1\right)$$

$x=\sqrt{22}$

Illustration-44 : If $\left(a+\frac{1}{a}\right)^2 = 3$, then $a^3 + \frac{1}{a^3}$ equals :

(A) $6\sqrt{3}$ (B) $3\sqrt{3}$ (C) 0 (D) $6\sqrt{3}$

Solution : $a+\frac{1}{a}=\pm\sqrt{3}$

$$a^3+\frac{1}{a^3}=\left(a+\frac{1}{a}\right)^3-3\left(a+\frac{1}{a}\right)=\pm 3\sqrt{3}\mp 3\sqrt{3}=0$$

Illustration-45 : If $x=\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ and $y=\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$, then find $x^3 + y^3$.

Solution : $x=\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}=\left(\sqrt{3}-\sqrt{2}\right)^2=5-2\sqrt{6}$

$y=5+2\sqrt{6}$

$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

$= (x + y)[(x + y)^2 - 3xy] = 10 \times [100 - 3] = 970$

Do yourself-9 :

1. Suppose $a+\frac{1}{a}=3$:

(a) Find $a^2 + \frac{1}{a^2}$ **(b)** Find $a^4 + \frac{1}{a^4}$ **(c)** Find $a^3 + \frac{1}{a^3}$

2. If $x+\frac{1}{x}=2$, then prove that : $x^2+\frac{1}{x^2}=x^4+\frac{1}{x^4}=x^8+\frac{1}{x^8}$.

3. If $2x + 3y + 4z = 0$, then prove that $8x^3 + 27y^3 + 64z^3 = 72xyz$.

4. I'm thinking of two numbers. The sum of my numbers is 14 and the product of my numbers is 46. What is the sum of the squares of my numbers ?

5. Simplify $\sqrt{7-\sqrt{13}}-\sqrt{7+\sqrt{13}}$.

6. Simplify the expression $\sqrt[3]{2+\sqrt{5}}+\sqrt[3]{2-\sqrt{5}}$.

7. If x, y, z are all different real numbers, then prove that

$$\frac{1}{(x-y)^2}+\frac{1}{(y-z)^2}+\frac{1}{(z-x)^2}=\left(\frac{1}{x-y}+\frac{1}{y-z}+\frac{1}{z-x}\right)^2.$$

8. Solve for x :

(a) $\sqrt{2x+1}=\sqrt{x}+1$ **(b)** $\sqrt{x^2-1}=x-3$ **(c)** $\sqrt{x+1}+2\sqrt{2x-3}=-3$

6. POLYNOMIAL IN ONE VARIABLE

An algebraic expression of the form $p(x) = a_nx^n + a_{n-1}x^{n-1} + ... + a_1x + a_0$ is called a polynomial function in 'x' where $a_i(i = 0, 1, 2, ..., n)$ is a constant which belongs to the set of real numbers and sometimes to the set of complex numbers and n is a whole number.

- a_i is the coefficient of x^i for i = 1, 2, 3, ..., n and a_0 is constant term of p(x).
- If $a_n \neq 0$, then a_nx^n is called leading term and a_n is called **leading co-efficient.**
- If $a_n = 1$, then polynomial is called **monic polynomial.**
- If $a_n \neq 0$, then **degree of the polynomial** is n.
- $f(x) = a_0$ is called **constant polynomial.** Its degree is 0, if $a_0 \neq 0$. If $a_0 = 0$ the polynomial $f(x)$ is called **ZERO polynomial.** Its degree is defined as $-\infty$ to preserve following two properties listed below. Some people prefer not to define degree of zero polynomial.

If $f(x)$ is a polynomial of degree n and g(x) is a polynomial of degree m then

1. $f(x) \pm g(x)$ is a polynomial of degree $\leq \max\{n, m\}$
2. $f(x) \times g(x)$ is a polynomial of degree m + n.

6.1 Types of Polynomials (w.r.t. Degree)

Degree of the polynomial in one variable is the largest exponent of the variable. For example, the degree of the polynomial $3x^7 - 4x^6 + x + 9$ is 7 and the degree of the polynomial $5x^6 - 4x^2 - 6$ is 6.

Polynomials classified by degree

Degree	Name	General form	Example
undefined or $-\infty$	Zero polynomial	0	0
0	(Non-zero) constant polynomial	a; $(a \neq 0)$	4
1	Linear polynomial	$ax + b$; $(a \neq 0)$	$x + 1$
2	Quadratic polynomial	$ax^2 + bx + c$; $(a \neq 0)$	$x^2 + 1$
3	Cubic polynomial	$ax^3 + bx^2 + cx + d$; $(a \neq 0)$	$x^3 + 1$

Usually, a polynomial of degree n, for n greater than 3, is called a polynomial of degree n, although the phrases quartic polynomial for degree 4 and quintic polynomial for degree 5 are sometimes used.

Note :

Polynomials having only one term are called monomials. E.g. 2, 2x, $7y^5$, $12t^7$ etc. Polynomials having exactly two dissimilar terms are called binomials. E.g. $p(x) = 2x + 1$, $r(y) = 2y^7 + 5y^6$ etc. Polynomials having exactly three distinct terms are called trinomials. E.g. $p(x) = 2x^2 + x + 6$, $q(y) = 9y^6 + 4y^2 + 1$ etc.

6.2 Division in Polynomials

Consider two polynomials P(x) & d(x) with d(x) being not identically zero and degree of d(x) $\le$ degree of P(x) then there exists unique polynomials Q(x) and r(x) such that

$P(x) = Q(x) \cdot d(x) + r(x)$

Here P(x) is called as dividend,

d(x) is called as divisor,

Q(x) is called as quotient,

& r(x) is called as remainder with degree of r(x) < degree of d(x)

Note : If d(x) is a divisor of P(x) then kd(x) will also be a divisor of P(x); $k \in \mathbb{R} - \{0\}$ and d(–x) will be a divisor of P(–x).

6.3 Remainder Theorem

Statement : Let p(x) be a polynomial of degree ≥ 1 and 'a' is any real number. If p(x) is divided by (x – a), then the remainder is p(a).

Illustration-46 : Let p(x) be $x^3 - 7x^2 + 6x + 4$. Divide p(x) with (x – 6) and find the remainder

Solution : Put x = 6 in p(x) i.e. p(6) will be the remainder.

$\therefore$ required remainder be

$p(6) = (6)^3 - 7\cdot 6^2 + 6\cdot 6 + 4 = 216 - 252 + 36 + 4 = 256 - 252 = 4$

$$\begin{array}{r|l|l}
x - 6 & x^3 - 7x^2 + 6x + 4 & x^2 - x \\
 & \underset{-\quad +}{x^3 - 6x^2} & \\
\hline
 & -x^2 + 6x + 4 & \\
 & \underset{+\quad -}{-x^2 + 6x} & \\
\hline
 & \text{Remainder} = 4 &
\end{array}$$

Thus, p(a) is remainder on dividing p(x) by (x – a).

Remark : (i) p(–a) is remainder on dividing p(x) by (x + a)

$[\because x + a = 0 \Rightarrow x = -a]$

(ii) $p\left(\frac{b}{a}\right)$ is remainder on dividing p(x) by (ax – b)

$[\because ax - b = 0 \Rightarrow x = b/a]$

(iii) $p\left(\frac{-b}{a}\right)$ is remainder on dividing p(x) by (ax + b)

$[\because ax + b = 0 \Rightarrow x = -b/a]$

(iv) $p\left(\frac{b}{a}\right)$ is remainder on dividing p(x) by (b – ax)

$[\because b - ax = 0 \Rightarrow x = b/a]$

Illustration-47 : Find the remainder when

$x^3 - ax^2 + 6x - a$ is divided by $x - a$

Solution : Let $p(x) = x^3 - ax^2 + 6x - a$

$p(a) = a^3 - a(a)^2 + 6(a) - a$

$= a^3 - a^3 + 6a - a = 5a$

So, by the Remainder theorem, remainder = 5a

6.4 Factor Theorem

Statement : Let f(x) be a polynomial of degree ≥ 1 and a be any real constant such that f(a) = 0, then (x – a) is a factor of f(x). Conversely, if (x – a) is a factor of f(x), then f(a) = 0.

Proof : By Remainder theorem, if f(x) is divided by (x – a), the remainder will be f(a). Let q(x) be the quotient. Then, we can write,

$f(x) = (x - a) \times q(x) + f(a)$ ($\because$ Dividend = Divisor × Quotient + Remainder)

If f(a) = 0, then $f(x) = (x - a) \times q(x)$

Thus, (x – a) is a factor of f(x).

Converse Let (x – a) is a factor of f(x).

Then we have a polynomial q(x) such that $f(x) = (x - a) \times q(x)$

Replacing x by a, we get f(a) = 0.

Hence, proved.

Illustration-48 : Use the factor theorem to determine whether (x – 1) is a factor of $f(x) = 2\sqrt{2}x^3 + 5\sqrt{2}x^2 - 7\sqrt{2}$

Solution : By using factor theorem, (x – 1) is a factor of f(x), only when f(1) = 0

$f(1) = 2\sqrt{2}(1)^3 + 5\sqrt{2}(1)^2 - 7\sqrt{2} = 2\sqrt{2} + 5\sqrt{2} - 7\sqrt{2} = 0$

Hence, (x – 1) is a factor of f(x).

6.5 Fundamental Theorem of Algebra

Every polynomial function of degree ≥ 1 has atleast one zero in the complex numbers.

In other words, if we have

$f(x) = a_nx^n + a_{n-1}x^{n-1} + ... + a_1x + a_0$, $a_i \in$ complex number $\forall$ i = 0, 1, 2, ..., n

with n ≥ 1, then there exists atleast one $h \in \mathbb{C}$, such that

$a_nh^n + a_{n-1}h^{n-1} + ... + a_1h + a_0 = 0.$

From this, it is easy to deduce that a polynomial function of degree 'n' (≥ 1) has exactly n zeroes.

i.e., $f(x) = a(x - r_1)(x - r_2)...(x - r_n)$

Illustration-49: Find the constants a, b, c such that $(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

Solution : **Method : 1**

$(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

By comparing coefficient of x^4 from both sides

$$2a = 2 \Rightarrow a = 1$$

By comparing coefficient of x^3 from both sides

$$2b + 3a = 11 \Rightarrow b = 4$$

By comparing coefficient of x^2 from both sides

$$2c + 3b + 7a = 9 \Rightarrow c = -5$$

By comparing coefficient of x from both sides

$$3c + 7b = 13 \qquad \text{(b \& c satisfy)}$$

By comparing constant

$$7c = -35 \quad \Rightarrow c = -5$$

So $\quad a = 1,\ b = 4,\ c = -5$

Method : 2

Given that $(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

$$\Rightarrow \qquad ax^2 + bx + c = \frac{2x^4 + 11x^3 + 9x^2 + 13x - 35}{2x^2 + 3x + 7}$$

$$= x^2 + 4x - 5$$

By comparing

$$a = 1,\ b = 4 \text{ and } c = -5.$$

Illustration 50 : Show that $(x - 3)$ is a factor of the polynomial $x^3 - 3x^2 + 4x - 12$.

Solution. Let $p(x) = x^3 - 3x^2 + 4x - 12$ be the given polynomial. By factor theorem, $(x - a)$ is a factor of a polynomial p(x) iff $p(a) = 0$. Therefore, in order to prove that $x - 3$ is a factor of p(x), it is sufficient to show that $p(3) = 0$.

Now, $p(x) = x^3 - 3x^2 + 4x - 12$

$\Rightarrow p(3) = 3^3 - 3 \times 3^2 + 4 \times 3 - 12 = 27 - 27 + 12 - 12 = 0$

Hence, $(x - 3)$ is a factor of $p(x) = x^3 - 3x^2 + 4x - 12$.

Illustration 51 : Without actual division prove that $2x^4 - 6x^3 + 3x^2 + 3x - 2$ is exactly divisible by $x^2 - 3x + 2$.

Solution. Let $f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$ and $g(x) = x^2 - 3x + 2$ be the given polynomials.

Then $g(x) = x^2 - 3x + 2 = x^2 - 2x - x + 2 = x(x - 2) - 1(x - 2) = (x - l)(x - 2)$

In order to prove that f(x) is exactly divisible by g(x), it is sufficient to prove that $x - 1$ and $x - 2$ are factors of f(x). For this it is sufficient to prove that $f(l) = 0$ and $f(2) = 0$.

Now, $f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$

$\Rightarrow \quad f(l) = 2 \times l^4 - 6 \times 1^3 + 3 \times l^2 + 3 \times 1 - 2 \Rightarrow f(l) = 0$

and, $f(2) = 2 \times 2^4 - 6 \times 2^3 \times 2^2 + 3 \times 2 - 2 \Rightarrow f(2) = 0$

Hence, $(x - 1)$ and $(x - 2)$ are factors of f(x).

$\Rightarrow$ $g(x) = (x - 1)(x - 2)$ is a factors of f(x).

Hence, f(x) is exactly divisible by g(x).

Illustration 52 : The polynomials $P(x) = kx^3 + 3x^2 - 3$ and $Q(x) = 2x^3 - 5x + k$, when divided by $(x - 4)$ leave the same remainder. The value of k is

(A) 2 (B) 1 (C) 0 (D) –1

Solution : $P(4) = 64k + 48 - 3 = 64k + 45$

$Q(4) = 128 - 20 + k = k + 108$

given $P(4) = Q(4)$

$64k + 45 = k + 108$

$\Rightarrow 63k = 63$ $\Rightarrow k = 1$ $\Rightarrow$ Option **(B)** is correct

Illustration-53 : Let P(x) be a polynomial such that when P(x) is divided by $x - 19$, the remainder is 99, and when P(x) is divided by $x - 99$, the remainder is 19. What is the remainder when P(x) is divided by $(x - 19)(x - 99)$?

Solution : Because we are dividing by a quadratic, the degree of the remainder is not greater than 1. So, the remainder is $ax + b$, for some constants a and b. Therefore, we have

$$P(x) = (x - 19)(x - 99)Q(x) + ax + b,$$

where Q(x) is the quotient when P(x) is divided by $(x - 19)(x - 99)$. We eliminate the Q(x) term by letting $x = 19$ or by letting $x = 99$. Doing each in turn gives us the system of equations

$$P(19) = 19a + b = 99,$$

$$P(99) = 99a + b = 19.$$

Solving this system of equations gives us $a = -1$ and $b = 118$.

So, the remainder is $-x + 118$.

Illustration-54 : The polynomial $f(x) = x^4 + ax^3 + bx^2 + cx + d$ has roots 1, 3, 5 and 7. Determine all the coefficients of $f(x)$.

Solution : By applying factor theorem

$f(x) = (x - 1)(x - 3)(x - 5)(x - 7)$

$= (x^2 - 4x + 3)(x^2 - 12x + 35)$

$= x^2(x^2 - 12x + 35) - 4x(x^2 - 12x + 35) + 3(x^2 - 12x + 35)$

$= x^4 - 16x^3 + 86x^2 - 176x + 105$

So $a = -16$, $b = 86$, $c = -176$ and $c = 105$

Illustration-55 : If $f(x)$ is monic polynomial of degree 6 such that $f(0) = 0$, $f(1) = -1$, $f(2) = -2$, $f(3) = -3$, $f(4) = -4$ and $f(5) = -5$, then find $f(x)$.

Solution : According to question

$f(0) = 0,\ f(1) = -1,\ f(2) = -2,\ ...,\ f(5) = -5$

$\Rightarrow f(x) + x = 0$ has the roots $x = 0, 1, 2, ..., 5$

$\Rightarrow f(x) + x = x(x - 1)(x - 2)(x - 3)(x - 4)(x - 5)$ (By factor theorem)

$\Rightarrow f(x) = x(x - 1)(x - 2)(x - 3)(x - 4)(x - 5) - x.$

Illustration-56 : If $P(x) = 2x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{Z}$. If $P\left(\sqrt{3}\right) = 10 - 2\sqrt{3}$.

Find (i) $P\left(-\sqrt{3}\right)$ (ii) $3a + c$

Solution : $P(x) = 2x^3 + ax^2 + bx + c$

$P\left(\sqrt{3}\right) = 10 - 2\sqrt{3}$

$\Rightarrow 6\sqrt{3} + 3a + b\sqrt{3} + c = 10 - 2\sqrt{3}$

$\Rightarrow (3a + c) + \sqrt{3}(6 + b) = 10 - 2\sqrt{3}$

$\Rightarrow 3a + c = 10$ and $6 + b = -2$ $(a, b, c \in I)$

$3a + c = 10$ and $b = -8$

$P\left(-\sqrt{3}\right) = -6\sqrt{3} + 3a - \sqrt{3}b + c$

$= -6\sqrt{3} + (3a + c) + 8\sqrt{3}$

$P\left(-\sqrt{3}\right) = 2\sqrt{3} + 10$

Illustration-57 : Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where a, b, c, d are constants. If $P(1) = 10$, $P(2) = 20$, $P(3) = 30$, then compute $P(4) + P(0)$.

Solution : $P(1) = 10$, $P(2) = 20$ and $P(3) = 30$

we can write these information as

$P(x) = 10x$ for $x = 1, 2, 3$

$\Rightarrow P(x) - 10x = 0$ has roots $x = 1, 2$ and 3

By factor theorem

$P(x) - 10x = (x - \alpha)(x - 1)(x - 2)(x - 3)$

$\Rightarrow P(x) = (x - \alpha)(x - 1)(x - 2)(x - 3) + 10x$

$P(4) + P(0) = (4 - \alpha)(3)(2)(1) + 40 + (-\alpha)(-1)(-2)(-3) + 0$

$= 24 - 6\alpha + 40 + 6\alpha$

$= 64$

Illustration-58 : Let a, b and c are roots of $2x^3 + x^2 + x + 1 = 0$

find (i) $a + b + c$ (ii) abc

Solution : By factor theorem

$2x^3 + x^2 + x + 1 = 2(x - a)(x - b)(x - c)$

$\Rightarrow 2x^3 + x^2 + x + 1 = 2(x^3 - (a + b + c)x^2 + (ab + bc + ca)x - abc)$

$\Rightarrow 2x^3 + x^2 + x + 1 = 2x^3 - 2(a + b + c)x^2 + 2(ab + bc + ca)x - 2abc$

By comparing coefficient of x^2 and constant term, we have

$-2(a + b + c) = 1$ and $-2abc = 1$

$\Rightarrow a + b + c = \frac{-1}{2}$ and $abc = \frac{-1}{2}$.

Do yourself-10 :

1. Determine the remainder when the polynomial $P(x) = x^4 - 3x^2 + 2x + 1$ is divided by $(x - 1)$

2. Find the remainder when $f(x) = 3x^3 + 6x^2 - 4x - 5$ is divided by $(x + 3)$.

3. Determine the value of k for which $x^3 - 6x + k$ may be divisible by $(x - 2)$.

4. Find the value of a, if $(x - a)$ is a factor of $x^3 - a^2x + x + 2$.

5. Find remainder when $f(x) = x^5 - x^3 + 3x^2 + 3x + 1$ is divided by $(x^2 - 1)$.

6. Find the value of ℓ and m if $8x^3 + \ell x^2 - 27x + m$ is divisible by $2x^2 - x - 6$.

7. Find ℓ and m if $2x^3 - (2\ell + 1)x^2 + (\ell + m)x + m$ may be exactly divisible by $2x^2 - x - 3$.

8. f(x) when divided by $x^2 - 3x + 2$ leaves the remainder $ax + b$. If $f(1) = 4$ and $f(2) = 7$, determine a and b.

9. A polynomial in x of the third degree which will vanish when $x = 1$ & $x = -2$ and will have the values 4 & 28 when $x = -1$ and $x = 2$ respectively. Find the polynomial

10. If $f(x)$ is polynomial of degree 4 such that $f(1) = 1$, $f(2) = 2$, $f(3) = 3$, $f(4) = 4$ & $f(0) = 1$ find $f(5)$.

7. EQUATIONS REDUCIABLE TO QUADRATIC EQUATIONS

There are certain equations which can be reduced to $ax^2 + bx + c = 0$ by some proper substitution.

7.1 $a(f(x))^2 + b(f(x)) + c = 0$, where $f(x)$ is expression of x

Method of solving : Put $f(x) = y$

Illustration-59 : **(a)** Solve $2^{x+3} + 2^{-x} - 6 = 0$ **(b)** Solve $\left(\frac{x}{x+1}\right)^2 + 6 - 5\left(\frac{x}{x+1}\right) = 0$

Solution : **(a)** Put $2^x = y$

$$8y + \frac{1}{y} - 6 = 0$$

$$8y^2 - 6y + 1 = 0$$

$$(4y - 1)(2y - 1) = 0$$

$$y = \frac{1}{4}, \frac{1}{2}$$

$\therefore\ 2^x = 2^{-2}$ and $2^x = 2^{-1}$ $\therefore\ x = -2, x = -1$

(b) Put $\frac{x}{x+1} = y$

$$y^2 - 5y + 6 = 0$$

$$(y - 2)(y - 3) = 0 \Rightarrow \frac{x}{x+1} = 2 \text{ and } \frac{x}{x+1} = 3$$

7.2 $(x - a)^4 + (x - b)^4 = c$,

Method of Solving : Put $\frac{x-a+x-b}{2} = t \Rightarrow x = \frac{(a+b)}{2} + t$

Illustration-60 : $(x - 1)^4 + (x - 7)^4 = 272$

Solution : Put $x = \frac{7+1}{2} + t$

$$\Rightarrow x = t + 4$$

$$\Rightarrow (t + 3)^4 + (t - 3)^4 = 272$$

$$\Rightarrow 2(t^4 + 6.9t^2 + 81) = 272$$

$$\Rightarrow t^4 + 54t^2 + 81 = 136$$

$$\Rightarrow t^4 + 54t^2 - 55 = 0$$

$$\Rightarrow (t^2 - 1)(t^2 + 55) = 0$$

$$\Rightarrow t^2 = 1$$

$$\Rightarrow t = \pm 1$$

$$\Rightarrow x = 5,3$$

7.3 $ma^{2x} + n(ab)^x + rb^{2x} = 0$ (a & b > 0)

Method of Solving : Divide the equation by b^{2x} and put $\left(\frac{a}{b}\right)^x = t$ for $t > 0$

Illustration-61 : $3^{2x+2} + 5.6^x - 4^{x+1} = 0$

Solution : $3^{2x+2} + 5.6^x - 4^{x+1} = 0$

$\Rightarrow 9 \times (3)^{2x} + 5 \times (2 \times 3)^x - 4(2)^{2x} = 0$

$$\Rightarrow 9\left(\frac{3}{2}\right)^{2x} + 5\left(\frac{3}{2}\right)^x - 4 = 0 \quad ...(1)$$

Let $\left(\frac{3}{2}\right)^x = t$ for $t > 0$

Equation 1 becomes

$9t^2 + 5t - 4 = 0$

$$\Rightarrow t = \frac{-5 \pm \sqrt{25+144}}{18} = -1 \text{ or } \frac{4}{9}$$

$t = -1$ (rejected)

$$t = \frac{4}{9} \Rightarrow \left(\frac{3}{2}\right)^x = \left(\frac{2}{3}\right)^2 = \left(\frac{3}{2}\right)^{-2}$$

$\Rightarrow x = -2$

Solution of the given equation is $x = -2$

7.4 $m \cdot a^{f(x)} + n \cdot b^{f(x)} + r = 0$, where ab = 1, a & b > 0 and $f(x)$ is expression of x

Method of Solving : put $a^{f(x)} = t$, then $b^{f(x)} = \frac{1}{t}$

Illustration-62 : Solve $\left(\sqrt{5+2\sqrt{6}}\right)^x+\left(\sqrt{5-2\sqrt{6}}\right)^x=10$

Solution : Let $a=\sqrt{5+2\sqrt{6}}$ and $b=\sqrt{5-2\sqrt{6}}$

$ab = 1$

Let $a^x = t$

Given equation become

$t+\frac{1}{t}=10 \Rightarrow t^2-10t+1=0$

$\Rightarrow t=\frac{10\pm\sqrt{96}}{2}=5\pm2\sqrt{6}$

$\Rightarrow \left(5+2\sqrt{6}\right)^{x/2}=\left(5+2\sqrt{6}\right) \Rightarrow \frac{x}{2}=1 \Rightarrow x=2$

or $\left(5+2\sqrt{6}\right)^{x/2}=\left(5-2\sqrt{6}\right)=\left(5+2\sqrt{6}\right)^{-1}$

$\Rightarrow \frac{x}{2}=-1 \Rightarrow x=-2$

Solutions of the given equation is $x = 2$ or -2.

7.5 (x + a)(x + b)(x + c)(x + d) + e = 0 when b + c = a + d

Illustration-63 : Solve $x(x + 1)(x + 2)(x + 3) - 8 = 0$

Solution : $x(x + 1)(x + 2)(x + 3) - 8 = 0$

$\Rightarrow x(x + 3)(x + 1)(x + 2) - 8 = 0$

$\Rightarrow (x^2 + 3x)(x^2 + 3x + 2) - 8 = 0$

$\Rightarrow (x^2 + 3x)^2 + 2(x^2 + 3x) - 8 = 0$

$\Rightarrow (x^2 + 3x)^2 + 4(x^2 + 3x) - 2(x^2 + 3x) - 8 = 0$

$\Rightarrow (x^2 + 3x)(x^2 + 3x + 4) - 2(x^2 + 3x + 4) = 0$

$\Rightarrow (x^2 + 3x - 2)(x^2 + 3x + 4) = 0$

$\Rightarrow (x^2 + 3x - 2) = 0$ or $(x^2 + 3x + 4) = 0$

$\Rightarrow x=\frac{-3\pm\sqrt{17}}{2}$ or $x=\frac{-3\pm\sqrt{7}\,i}{2}$

7.6 $(x + a)(x + b)(x + c)(x + d) + ex^2 = 0$, where $ad = bc$.

Method of solving : Divide given equation by x^2 and put $x+\frac{ad}{x}=t$

Illustration-64 : $(x + 1)(x + 2)(x + 3)(x + 6) = 3x^2$

Solution : $(x + 1)(x + 6)\,(x + 2)(x + 3) - 3x^2 = 0$

$\Rightarrow (x^2 + 7x + 6)(x^2 + 5x + 6) - 3x^2 = 0$

$$\Rightarrow \left(x+\frac{6}{x}+7\right)\left(x+\frac{6}{x}+5\right)-3=0$$

Let $x+\frac{6}{x}=t$

$(t + 7)(t + 5) - 3 = 0$

$\Rightarrow t^2 + 12t + 32 = 0$

$t = -8$ or -4

when $x+\frac{6}{x}=-8 \Rightarrow x^2 + 8x + 6 = 0 \Rightarrow x=-4\pm\sqrt{10}$

$x+\frac{6}{x}=-4 \Rightarrow x^2 + 4x + 6 = 0 \Rightarrow x=-2\pm\sqrt{2}\,i$

Illustration-65 : Solve $(x^2 - 3x)(x^2 - 3x + 2) + 1 = 0$

Solution : Let $x^2 - 3x = y$

$\Rightarrow y(y + 2) + 1 = 0$

$\Rightarrow y^2 + 2y + 1 = 0$

$\Rightarrow (y + 1)^2 = 0$

$\Rightarrow y = -1$

Putting $y = 1$ in $x^2 - 3x = y$

we have $x^2 - 3x = -1$

$\Rightarrow x^2 - 3x + 1 = 0$

$$\Rightarrow x=\frac{3\pm\sqrt{5}}{2}$$

7.7 $ax^4 + bx^3 + cx^2 + dx + e = 0$, where $a = e$ & $b = \pm d$

Method of solving : Divide given equation by x^2 and put $x + \frac{1}{x} = t$ or $x - \frac{1}{x} = t$ whichever is applicable.

Illustration-66 : Solve $x^4 - 2x^3 + 3x^2 - 2x + 1 = 0$

Solution : $x^4 - 2x^3 + 3x^2 - 2x + 1 = 0$

By dividing x^2 both sides we have

$$x^2 - 2x + 3 - \frac{2}{x} + \frac{1}{x^2} = 0$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2\left(x + \frac{1}{x}\right) + 3 = 0$$

Let $x + \frac{1}{x} = t$

Above equation become

$t^2 - 2 - 2t + 3 = 0$

$\Rightarrow t^2 - 2t + 1 = 0 \Rightarrow (t - 1)^2 = 0$

$$\Rightarrow t = 1 \quad \Rightarrow x + \frac{1}{x} = 1 \Rightarrow x^2 - x + 1 = 0 \Rightarrow x = \frac{1 \pm \sqrt{3}\, i}{2}$$

Roots are $\frac{1 \pm \sqrt{3}\, i}{2}$

7.8 By Guessing Rational Roots of Polynomial.

Illustration-67 : Solve : $x^4 + x^3 - 2x^2 - x + 1 = 0$

Solution : Let $P(x) = x^4 + x^3 - 2x^2 - x + 1$

$P(1) = 0$ and $P(-1) = 0$

$\Rightarrow (x - 1)(x + 1)$ factor of $P(x)$

We can find other factor of $P(x)$ by dividing $x^2 - 1$ from $P(x)$.

$P(x) = (x^2 - 1)(x^2 + x - 1) = 0$

$\Rightarrow x = \pm 1$ or $x^2 + x - 1 = 0$

$$\Rightarrow x = \pm 1 \text{ or } x = \frac{-1 \pm \sqrt{5}}{2}$$

solution of $P(x) = 0$ are $x = \pm 1$ or $\frac{-1 \pm \sqrt{5}}{2}$.

Do yourself-11 :

Solve the following equations for x :

1. $x^{2/3} + x^{1/3} - 2 = 0$

2. $x^{\frac{2}{5}} - 3x^{\frac{1}{5}} + 2 = 0$

3. $3x^4 - 8x^2 + 4 = 0$

4. $4^x - 3.2^{x+3} + 128 = 0$

5. $x^2 + \frac{1}{x^2} - 5\left(x + \frac{1}{x}\right) + 8 = 0$

6. $2\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 4 = 0$

7. $2^{2x+1} - 7 \times 10^x + 5^{2x+1} = 0$

8. $(x-1)^4 + (x-5)^4 = 82$

9. $\left(\sqrt{3}+\sqrt{2}\right)^x + \left(\sqrt{3}-\sqrt{2}\right)^x - 2\sqrt{3} = 0$

10. $\left(3+2\sqrt{2}\right)^{x/2} + \left(3-2\sqrt{2}\right)^{x/2} - 34 = 0$

11. $x(x+1)(x+2)(x+3) = 24$

12. $(x+1)(x+2)(x+3)(x+4) = 120$

13. $(x+1)(x+2)^2(x+4) - 2x^2 = 0$

14. $x^4 - 2x^3 - 2x^2 + 2x + 1 = 0$

15. Match the values of x given in **Column-II** satisfying the exponential equation in Column-I (Do not verify). Remember that for a > 0, the term a^x is always greater than zero $\forall\, x \in R$.

Column-I	**Column-II**
(A) $5^x - 24 = \frac{25}{5^x}$	**(P)** –3
(B) $(2^{x+1})(5^x) = 200$	**(Q)** –2
(C) $4^{2/x} - 5(4^{1/x}) + 4 = 0$	**(R)** –1
(D) $\frac{2^{x-1}.4^{x+1}}{8^{x-1}} = 16$	**(S)** 0
(E) $4^{x^2+2} - 9\left(2^{x^2+2}\right) + 8 = 0$	**(T)** 1
(F) $5^{2x} - 7^x - 5^{2x}(35) + 7^x(35) = 0$	**(U)** 2
	(V) 3
	(X) None

8. SYSTEM OF EQUATIONS

Observe the following Illustrations :

Illustrations-68 : If $x - y = 2$ and $xy = 24$, find the value of $\frac{1}{x}+\frac{1}{y}$.

Solution : $(x + y)^2 = (x - y)^2 + 4xy = 4 + 4(24)$

$\Rightarrow (x + y)^2 = 100$

$\Rightarrow x + y = 10, -10$

$\therefore \frac{x+y}{xy}=\frac{10}{24}=\frac{5}{12};\frac{x+y}{xy}=-\frac{10}{24}=-\frac{5}{12}$

Illustrations-69 : If $2x - 3y - z = 0$ and $x + 3y - 14z = 0$, then find $\frac{x^2+3xy}{y^2+z^2}$.

Solution : $\frac{2x}{z}-\frac{3y}{z}=1$ & $\frac{x}{z}+\frac{3y}{z}=14$

Solving $\frac{x}{z}=5;\frac{y}{z}=3$

$$\Rightarrow \frac{\left(\frac{x}{z}\right)^2+\frac{3x}{z}.\frac{y}{z}}{\left(\frac{y}{z}\right)^2+1}=\frac{25+3(5)(3)}{(3)^2+1}=\frac{70}{10}=7$$

Illustration-70 : Given $a + b = 20$ and $a^3 + b^3 = 800$, find $a^2 + b^2$.

Solution : $a + b = 20$

$\Rightarrow a^2 + b^2 + 2ab = 400$ .... (1)

$a^3 + b^3 = 800$

$\Rightarrow (a + b)(a^2 - ab + b^2) = 800$

$\Rightarrow a^2 + b^2 - ab = 40$ .... (2)

By adding twice of second equation with first equation.

$3(a^2 + b^2) = 480$

$\Rightarrow a^2 + b^2 = 160.$

Illustrations-71 : $x(y + z) = 29$, $y(z + x) = 26$; $z(x + y) = 51$ find x,y,z.

Solution : $xy + zx = 29$ ...(1)

$yz + xy = 26$ ...(2)

$xz + yz = 51$ ...(3)

$\Rightarrow 2[xy + yz + zx] = 106$

$\Rightarrow xy + yz + zx = 53$

Now : $xy = 2$, $zx = 27$; $yz = 24$

$\Rightarrow x^2y^2z^2 = 24 \times 2 \times 27 = (36)^2$

$\Rightarrow xyz = \pm 36 \therefore (x, y, z) \equiv \left(\frac{3}{2}, \frac{4}{3}, 18\right)$ or $\left(-\frac{3}{2}, -\frac{4}{3}, -18\right)$

Illustrations-72 : If $x^3 + y^3 = 35$; and $x^2y + xy^2 = 30$, then find (x,y).

Solution : $\frac{x^3 + y^3}{x^2y + xy^2} = \frac{(x+y)(x^2 - xy + y^2)}{xy(x+y)} = \frac{35}{30} = \frac{7}{6}$

$\Rightarrow 6x^2 - 13xy + 6y^2 = 0 \Rightarrow (3x - 2y)(2x - 3y) = 0 \Rightarrow 3x = 2y$ or $2x = 3y$

Case -I : $3x = 2y$, $x^3 + \left(\frac{3x}{2}\right)^3 = 35 \Rightarrow 35x^3 = 8 \times 35 \Rightarrow x = 2, y = 3$

Case -II : $2x = 3y$, $x^3 + \left(\frac{2x}{3}\right)^3 = 35 \Rightarrow 35x^3 = 27 \times 35 \Rightarrow x = 3, y = 2$

Illustrations-73 : If x,y and z are real numbers such that $x^2 + y^2 + 2z^2 - 4x + 2z - 2yz + 5 = 0$, find the value of $(x - y - z)$.

Solution : $x^2 + y^2 + 2z^2 - 4x + 2z - 2yz + 5 = 0$

$\Rightarrow x^2 - 4x + 4 + y^2 + z^2 - 2yz + z^2 + 2z + 1 = 0$

$\Rightarrow (x - 2)^2 + (y - z)^2 + (z + 1)^2 = 0$

$\Rightarrow x = 2, y = z, z = -1$

$\therefore x - y - z = 2 + 1 + 1 = 4$

Illustrations-74 : $xy = 12$, $yz = 15$, $zx = 20$ find $x + y + z$; $x,y,z \in \mathbb{R}^+$

Solution : $(xyz)^2 = (3 \times 4 \times 5)^2$

$\Rightarrow xy\,z = 3 \times 4 \times 5$

$\Rightarrow z = 5, y = 3, x = 4$

$\Rightarrow x + y + z = 5 + 3 + 4 = 12$

Do yourself-12 :

Solve the following systems of equations :

1. $\begin{cases} x^2 - y^2 = 16, \\ x + y = 8 \end{cases}$

2. $\begin{cases} x - y = 1, \\ x^3 - y^3 = 7 \end{cases}$

3. $x + y = 2$ and $x^3 + y^3 = 56$; $x, y \in \mathbb{R}$, then find x and y.

4. $\begin{cases} x^2 + y^2 + 6x + 2y = 0, \\ x + y + 8 = 0 \end{cases}$

5. $\begin{cases} \dfrac{x}{y} - \dfrac{y}{x} = \dfrac{5}{6}, \\ x^2 - y^2 = 5 \end{cases}$

6. $\begin{cases} \dfrac{x+y}{x-y} + \dfrac{x-y}{x+y} = \dfrac{13}{6}, \\ xy = 5 \end{cases}$

7. $\begin{cases} \dfrac{1}{x+1} + \dfrac{1}{y} = \dfrac{1}{3}, \\ \dfrac{1}{(x+1)^2} - \dfrac{1}{y^2} = \dfrac{1}{4} \end{cases}$

8. $\begin{cases} x^2 + y^2 = 25 - 2xy \\ y(x+y) = 10 \end{cases}$

9. $\begin{cases} \dfrac{1}{y-1} - \dfrac{1}{y+1} = \dfrac{1}{x}, \\ y^2 - x - 5 = 0 \end{cases}$

10. $\begin{cases} 2xy + y^2 - 4x - 3y + 2 = 0, \\ xy + 3y^2 - 2x - 14y + 16 = 0 \end{cases}$

11. $\begin{cases} x^3 + y^3 = 7, \\ xy(x+y) = -2 \end{cases}$

12. $\begin{cases} x^4 + y^4 = 82, \\ xy = 3 \end{cases}$

9. INEQUALITIES

9.1 Basic Rules :

- If a > b and b > c, then a > c.
- If x > y, then x + c > y + c for any real number c. Additionally, if a > b, then x + a > y + b.
- If x > y and a > 0, then xa > ya.
- If we multiply or divide an inequality by a negative number, we must reverse the sign. For example, if x > y and a < 0, then xa < ya.
- If x > y > 0 and a > b > 0, then xa > yb.

- If x > y and x and y have the same sign (positive or negative), then $\frac{1}{x} < \frac{1}{y}$.
- If $x > y \geq 0$, then for any positive real number a, we have $x^a > y^a$.

  In particular, if $0 < a < b$, then $\sqrt[n]{a} < \sqrt[n]{b}$ for all positive integral values of $n > 1$.

  E.g. $\sqrt[4]{2} < \sqrt[4]{7}$, $\sqrt[3]{3} < \sqrt[3]{5}$, $\sqrt[5]{10} < \sqrt[5]{13}$ etc.

  If two simple surds of different orders viz. $\sqrt[n]{a}$ and $\sqrt[m]{b}$ have to be compared, they have to be expressed as surds of the same order i.e. LCM of n and m .

  Ex. compare $\sqrt[4]{6}$ and $\sqrt[3]{5}$,

  we express both as the surds of 12th order.

  $\therefore \sqrt[4]{6} = \sqrt[12]{6^3}$ and $\sqrt[3]{5} = \sqrt[12]{5^4}$. As $6^3 < 5^4 \Rightarrow \sqrt[4]{6} < \sqrt[3]{5}$

Illustration-75 : Solve :

(i) $7 - 3x \geq 8 + 2x$ (ii) $\frac{8-x}{7} \geq 4$ (iii) $\frac{1}{x} \geq 5$ (iv) $\frac{4}{x+1} \leq 2$

Solution : (i) $7 - 3x \geq 8 + 2x \Rightarrow 7 - 8 \geq 2x + 3x$

$\Rightarrow \frac{-1}{5} \geq x \Rightarrow x \in \left(-\infty, -\frac{1}{5}\right]$

(ii) $\frac{8-x}{7} \geq 4 \Rightarrow 8 - x \geq 28 \Rightarrow -x \geq 20$

$\Rightarrow x \leq -20 \Rightarrow x \in (-\infty, -20]$

(iii) $\frac{1}{x} \geq 5$

$\Rightarrow \infty > \frac{1}{x} \geq 5 \qquad \Rightarrow 0 < x \leq \frac{1}{5} \qquad \Rightarrow x \in \left(0, \frac{1}{5}\right]$

(iv) $\frac{4}{x+1} \leq 2$

$\Rightarrow -\infty < \frac{4}{x+1} < 0$ or $0 < \frac{4}{x+1} \leq 2$ ($\frac{4}{x+1} = 0$ is not possible)

$\Rightarrow 0 > \frac{x+1}{4} > -\infty$ or $\infty > \frac{x+1}{4} \geq \frac{1}{2}$

$\Rightarrow x + 1 < 0$ or $2 \leq x + 1$

$\Rightarrow x < -1$ or $1 \leq x \quad \Rightarrow x \in (-\infty, -1) \cup [1, \infty)$

Do yourself : 13

1. Determine which of the following statements is true or false. If it is false, provide an example that shows the statement is false. Assume *a, b, c, x* and *y* are real numbers.

(a) If $a \le b$ and $b \le c$, then $a < c$.
(b) If $a \ge b \ge a$, then $a = b$.
(c) If $a > b$, then $ac > bc$.
(d) If $a > b$ and $c \le 0$, then $ac \le bc$.
(e) If $x + a \ge y + a$, then $x \ge y$.
(f) If $x + a \ge y + b$, then $x \ge y$ and $a \ge b$.

2. Which fraction is larger ?

(a) $\frac{13}{17}$ or $\frac{17}{21}$ (b) $\frac{31}{35}$ or $\frac{37}{41}$

3. Which of the following numbers is largest : $2^{36}, 3^{30}, 5^{18}, 6^{12}, 7^{8}, 8^{4}$? (No calculators!)

4. What values of x satisfy the inequality, $7 - 3x < x - 1 \le 2x + 9$?

5. Which of the following is greater $5+\sqrt{3}, 3+\sqrt{14}$? (Without calculating the values of $\sqrt{3}, \sqrt{14}$)

9.2 Trivial and Sum of squares (SOS) Inequality

The square of any real number is non-negative. So if x is real, then $x^2 \ge 0$. This is known as Trivial inequality. Equality holds only if x = 0.

Sum of squares (SOS) of real numbers is non negative. That is $\Sigma x_i^2 \ge 0$. This is know as SOS inequality. Equality holds if $x_i = 0 \ \forall \ i$

Ex. $x, y, z \in \mathbb{R}$ and $x^2 + y^2 + z^2 = 0 \Rightarrow x = y = z = 0$.

Note :

- $f(x) = [g(x)]^{2n}$ where $n \in \mathbb{N} \Rightarrow f(x) \ge 0$
- $f(x) = [g(x)]^{1/2n}$, $n \in \mathbb{N}$, $g(x) \ge 0 \Rightarrow f(x) \ge 0$

Illustration-76 : If $a, b, c \in \mathbb{R}$ and $a^2 + b^2 + c^2 - ab - bc - ca = 0$, prove that $a = b = c$.

Solution : $a^2 + b^2 + c^2 - ab - bc - ca = 0$

$\Rightarrow 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 0$

$\Rightarrow (a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ac + a^2) = 0$

$\Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 = 0$

$\Rightarrow a - b = b - c = c - a = 0$

$\Rightarrow a = b = c$.

Illustration-77 : For $x,y \in \mathbb{R}$, find the all possible values (range) of expression $4x^2 + 9y^2 - 12x + 6y$.

Solution : If $E(x,y) = 4x^2 + 9y^2 - 12x + 6y$

$= (2x)^2 - 2(2x) \times 3 + (3)^2 + (3y)^2 + 2(3y) + (1)^2 - 10$

$= (2x - 3)^2 + (3y + 1)^2 - 10$

By sum of square (SOS), $(2x - 3)^2 + (3y + 1)^2 \ge 0$

$\Rightarrow E(x,y) = (2x - 3)^2 + (3y + 1)^2 - 10 \ge 0 - 10$

So Range of $E(x,y) = [-10, \infty)$

Illustration-78 : Find all ordered pairs of real numbers (x,y) such that $(4x^2 + 4x + 3)(y^2 - 6y + 13) = 8$

Solution : Rewriting the equation

$((2x + 1)^2 + 2)\ ((y - 3)^2 + 4) = 8$

from the SOS, $(2x + 1)^2 \geq 0$ and $(y - 3)^2 \geq 0$,

so we have

$(2x + 1)^2 + 2 \geq 2$ and $(y - 3)^2 + 4 \geq 4$

$\Rightarrow ((2x + 1)^2 + 2)\ ((y - 3)^2 + 4) \geq 8$

Equality holds only if

$(2x + 1)^2 = (y - 3)^2 = 0 \Rightarrow (x, y) = \left(-\frac{1}{2}, 3\right)$

Illustration-79 : If $b^2 - 4ac < 0$ and $a > 0$, then show that $ax^2 + bx + c > 0\ \forall\ x \in \mathbb{R}$

Solution : $ax^2 + bx + c = a\left[x^2 + \frac{b}{a}x\right] + c$

$$= a\left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2}\right] + c$$

$$= a\left[x + \frac{b}{2a}\right]^2 - \left(\frac{b^2 - 4ac}{4a}\right) > 0\ \ \forall\ x \in \mathbb{R}.$$ Hence proved

9.3 Mean :

In any collection of data a specific value between two extremes (minimum/maximum) is called a mean of the data.

For Two Variables :

Let x, y be two **positive** real numbers with $x \leq y$.

- The **Arithmetic Mean (A)** of x and y is $\frac{x + y}{2}$

  Observe $x \leq \frac{x + y}{2} \leq y$

- The **Geometric Mean (G)** of x and y is $\sqrt{xy}$

  Observe $x \leq \sqrt{xy} \leq y$

We have $x \leq G \leq A \leq y$

Equality holds only when $x = y$.

Proof For two positive numbers x and y, the Trivial inequality gives us $\left(\sqrt{x}-\sqrt{y}\right)^2 \geq 0$

$$\Rightarrow x+y-2\sqrt{xy} \geq 0$$

$$\Rightarrow \frac{x+y}{2} \geq \sqrt{xy} \Rightarrow A \geq G$$

Note :

- $x+\frac{1}{x} \geq 2 \ \ \forall\ x>0$ and $x+\frac{1}{x} \leq -2 \ \ \forall\ x<0$

Illustration : 80 Find all possible values (range) of the expression $x+\frac{4}{x}$, when $x \in \mathbb{R}-\{0\}$.

Sol. When x > 0, $\frac{x+\frac{4}{x}}{2} \geq \sqrt{x \times \frac{4}{x}} = 2$ (By A ≥ G)

$$\Rightarrow x+\frac{4}{x} \geq 4,$$

So, when x > 0, range = $[4,\infty)$ ...(1)

for x < 0, $\frac{x+\frac{4}{x}}{2} \leq -\sqrt{x \times \frac{4}{x}}$

$$\Rightarrow x+\frac{4}{x} \leq -4 \quad ...(2)$$

From (1) and (2), range = $(-\infty,-4]\cup[4,\infty)$

Do yourself : 14

1. Find the minimum value of $\frac{x^4+8}{x^2}$.

2. For x < 0, find the maximum value of $\frac{3x^2+12}{x}$.

3. If a, b $\in \mathbb{R}^+$, find minimum possible value of $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right)$.

10. RATIO AND PROPORTION

If a and b be two quantities of the same kind, then their ratio is a : b; which may be denoted by the fraction $\frac{a}{b}$ (This may be an integer or fraction)

In the ratio a : b, a is the first term (Antecedent) and b is the second term (Consequent)

A ratio may represented in a number of ways e.g. $\frac{a}{b}=\frac{ma}{mb}=\frac{na}{nb}=...$ where m, n, ... are non-zero numbers.

Let a, b, c, d be positive integers now to compare two ratios a : b and c : d we use following :

- $(a : b) > (c : d)$ if $ad > bc$
- $(a : b) = (c : d)$ if $ad = bc$
- $(a : b) < (c : d)$ if $ad < bc$

To compare two or more ratio, reduce them to common denominator.

Note :

- If $a > b > 0$ and $x > 0$, then $\frac{a}{b} > \frac{a+x}{b+x}$, e.g. $\frac{41}{40} > \frac{45}{44}$
- If $0 < a < b$ and $x > 0$, then $\frac{a}{b} < \frac{a+x}{b+x}$

Illustration-81 : What term must be added to each term of the ratio 5 : 37 to make it equal to 1 : 3 ?

Solution : Let x be added to each term of the ratio 5 : 37.

Then $\frac{x+5}{x+37}=\frac{1}{3}$ $\Rightarrow 3x + 15 = x + 37$ i.e. $x = 11$

Illustration-82 : If x : y = 3 : 4; find the ratio of 7x – 4y : 3x + y

Solution : $\frac{x}{y}=\frac{3}{4} \Rightarrow x=\frac{3}{4}y$

Now $\frac{7x-4y}{3x+y}=\frac{7.\frac{3}{4}y-4y}{3.\frac{3}{4}y+y}$ (putting the value of x)

$=\frac{5y}{13y}=\frac{5}{13}$ i.e. 5 : 13

10.1 Proportion :

When two ratios are equal, then the four quantities compositing them are said to be proportional.

so, if $\frac{a}{b}=\frac{c}{d}$, then it is written as a : b = c : d or a : b :: c : d

Where 'a' and 'd' are known as extremes and 'b and c' are known as means.

(i) An important property of proportion : Product of extremes = product of means.

(ii) If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}$ then each is equal to $\frac{a+c+e}{b+d+f}$

(iii) If a : b = c : d, then b : a = d : c (Invertando)

i.e. $\frac{a}{b}=\frac{c}{d} \quad \Rightarrow \quad \frac{b}{a}=\frac{d}{c}$

(iv) If a : b = c : d, then a : c = b : d (Alternando)

i.e. $\frac{a}{b}=\frac{c}{d} \quad \Rightarrow \quad \frac{a}{c}=\frac{b}{d}$

(v) If a : b = c : d, then $\frac{a+b}{b}=\frac{c+d}{d}$ (Componendo)

i.e. $\frac{a}{b}=\frac{c}{d} \quad \Rightarrow \quad \frac{a}{b}+1=\frac{c}{d}+1$

(vi) If a : b = c : d, then $\frac{a-b}{b}=\frac{c-d}{d}$ (Dividendo)

i.e. $\frac{a}{b}=\frac{c}{d} \quad \Rightarrow \quad \frac{a}{b}-1=\frac{c}{d}-1$

(vii) If a : b = c : d, then $\frac{a+b}{a-b}=\frac{c+d}{c-d}$ (Componendo and dividendo)

i.e. $\frac{a}{b}=\frac{c}{d} \quad \Rightarrow \quad \frac{a}{b}+1=\frac{c}{d}+1 \quad \Rightarrow \quad \frac{a+b}{b}=\frac{c+d}{d}$ ... (1)

$\frac{a}{b}-1=\frac{c}{d}-1 \quad \Rightarrow \quad \frac{a-b}{b}=\frac{c-d}{d}$ ... (2)

Dividing equation (1) by (2) we obtain

$$\frac{a+b}{a-b}=\frac{c+d}{c-d}$$

Illustration-83 : If $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}$ show that $\frac{x^3+a^3}{x^2+a^2}+\frac{y^3+b^3}{y^2+b^2}+\frac{z^3+c^3}{z^2+c^2}=\frac{(x+y+z)^3+(a+b+c)^3}{(x+y+z)^2+(a+b+c)^2}$

Solution : $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}=k$ (constant)

x = ak; y = bk; z = ck

substituting these values of x, y, z in the given expression, we obtain

$$\text{L.H.S.} = \frac{a^3k^3+a^3}{a^2k^2+a^2}+\frac{b^3k^3+b^3}{b^2k^2+b^2}+\frac{c^3k^3+c^3}{c^2k^2+c^2}$$

$$=\frac{a^3(k^3+1)}{a^2(k^2+1)}+\frac{b^3(k^3+1)}{b^2(k^2+1)}+\frac{c^3(k^3+1)}{c^2(k^2+1)}=\frac{(k^3+1)}{(k^2+1)}.(a+b+c)$$

$$\text{Now R.H.S} =\frac{(ak+bk+ck)^3+(a+b+c)^3}{(ak+bk+ck)^2+(a+b+c)^2}$$

$$=\frac{k^3(a+b+c)^3+(a+b+c)^3}{k^2(a+b+c)^2+(a+b+c)^2}$$

$$=\frac{(k^3+1)(a+b+c)^3}{(k^2+1)(a+b+c)^2}=\frac{(k^3+1)}{(k^2+1)}.(a+b+c) = \text{L.H.S}$$

Illustration-84 : If $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}$, prove that

$(ab + bc + cd + de)^2 = (a^2 + b^2 + c^2 + d^2)(b^2 + c^2 + d^2 + e^2)$

Solution : $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}$, then we have

$$\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}=\frac{\sqrt{(a^2+b^2+c^2+d^2)}}{\sqrt{(b^2+c^2+d^2+e^2)}}=k \quad \text{(say)}$$

i.e. $a = bk$ $\quad\therefore ab = b^2k$

$b = ck$ $\quad\therefore bc = c^2k$

$c = dk$ $\quad\therefore cd = d^2k$

$d = ek$ $\quad\therefore de = e^2k$

so, $(a^2 + b^2 + c^2 + d^2) = k^2 (b^2 + c^2 + d^2 + e^2)$ ... (i)

Now L.H.S. $= (ab + bc + cd + de)^2$

$= (kb^2 + kc^2 + kd^2 + ke^2)^2$

$= k^2(b^2 + c^2 + d^2 + e^2)^2$

$= k^2(b^2 + c^2 + d^2 + e^2)(b^2 + c^2 + d^2 + e^2)$

$= (a^2 + b^2 + c^2 + d^2)(b^2 + c^2 + d^2 + e^2) =$ R.H.S (by use of (i))

Illustration-85 : Solve the equation $\dfrac{3x^4+x^2-2x-3}{3x^4-x^2+2x+3} = \dfrac{5x^4+2x^2-7x+3}{5x^4-2x^2+7x-3}$

Solution : $\dfrac{3x^4+x^2-2x-3}{3x^4-x^2+2x+3} = \dfrac{5x^4+2x^2-7x+3}{5x^4-2x^2+7x-3}$

Applying componendo and dividendo, we have

$$\frac{3x^4}{x^2-2x-3} = \frac{5x^4}{2x^2-7x+3}$$

or $3x^4 (2x^2 - 7x + 3) - 5x^4(x^2 - 2x - 3) = 0$

or $x^4 [6x^2 - 21x + 9 - 5x^2 + 10x + 15] = 0$

or $x^4 (x^2 - 11x + 24) = 0$

$\therefore\ x = 0$ or $x^2 - 11x + 24 = 0$

$\therefore\ x = 0$ or $(x - 8)(x - 3) = 0$

$\therefore\ x = 0, 8, 3$

Do yourself : 15

1. If $\dfrac{a}{b} = \dfrac{2}{3}$ and $\dfrac{b}{c} = \dfrac{4}{5}$, then find value of $\dfrac{a+b}{b+c}$

2. If $\dfrac{a}{b} = \dfrac{3}{5}$ and $\dfrac{b}{c} = \dfrac{7}{13}$, then find the value of a : b : c

3. If sum of two numbers is s and their quotient is $\dfrac{p}{q}$. Find number.

4. If $\dfrac{a}{b} = \dfrac{c}{d} = \dfrac{e}{f}$, then find the value of $\dfrac{2a^4b^2+3a^2c^2-5e^4f}{2b^6+3b^2d^2-5f^5}$ in terms of a and b.

5. If x : a = y : b = z : c, then show that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

11. SIGN-SCHEME (WAVY CURVE) METHOD

Given $f(x)$ and g(x) are polynomials.

To solve the inequalities of the type $\frac{f(x)}{g(x)} * 0$, where '*' can be >, ≥, < or ≤, we take the following steps :

(i) Find all the roots of f(x) = 0 and g(x) = 0

(ii) Write all these roots on the real line in increasing order of values.

(iii) Check the sign of the expression $\frac{f(x)}{g(x)}$ at some x greater than the largest root. If it is positive, put + sign in rightmost interval. In case of negative, put -ve sign in rightmost interval and while moving from right to left change sign in accordance with step (iv).

(iv) If a root occurs even number of times, then sign of expression will be same on both sides of the root and if a root occurs odd number of times, then sign of the expression will be different on both sides of the root.

(v) Write the answer according to need of the question.

Note :

- We don't give equality sign on '±∞' in the solution as they are two improper points of number line.
- We can't take zeroes of denominator in the final answer as at these points expression is not defined (because division by '0' is not defined).
- In case of ≥ 0 or ≤ 0, zeroes of numerator will be part of the answer provided they are not appearing in denominator also.
- Do not cross multiply the terms in the inequalities

Illustration-86 : Find the solution of $-x^2 + 6x + 7 \geq 0$

Solution : $-x^2 + 6x + 7 \geq 0$

$\Rightarrow x^2 - 6x - 7 \leq 0$

$\Rightarrow (x + 1)(x - 7) \leq 0$

$\Rightarrow x \in [-1,7]$

Illustration-87 : Find the solution of $2x^2 - 7x + 3 \geq 0$

Solution : $2x^2 - 7x + 3 \geq 0$

$\Rightarrow (2x - 1)(x - 3) \geq 0$

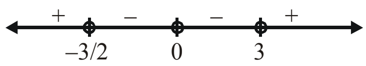
$x \in \left(-\infty, \frac{1}{2}\right] \cup [3, \infty)$

Illustration-88 : Solve $2x^4 > 3x^3 + 9x^2$

Solution : $2x^4 - 3x^3 - 9x^2 > 0$

$x^2(2x^2 - 3x - 9) > 0$

$x^2(2x + 3)(x - 3) > 0$



$$x \in \left(-\infty, -\frac{3}{2}\right) \cup (3, \infty)$$

Illustration-89 : Find the solution of the inequalities $\dfrac{(x-1)(x-2)}{(x-3)} \geq 0$

Solution : $x - 1 = 0,\ x - 2 = 0,\ x - 3 = 0$

$\Rightarrow x = 1, 2, 3$

Since $x - 3 \neq 0,\ x \neq 3$

so, $x \in [1, 2] \cup (3, \infty)$

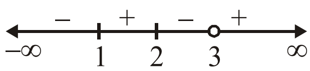
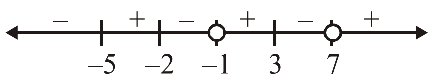


Illustration-90 : If $f(x) = \dfrac{x^3 + 4x^2 - 11x - 30}{x^2 - 6x - 7}$ then find x such that

(i) $f(x) > 0$ **(ii)** $f(x) < 0$.

Solution : Given $f(x) = \dfrac{(x-3)(x+2)(x+5)}{(x+1)(x-7)}$



(i) $f(x) > 0 \Rightarrow x \in (-5, -2) \cup (-1, 3) \cup (7, \infty)$

(ii) $f(x) < 0 \Rightarrow x \in (-\infty, -5) \cup (-2, -1) \cup (3, 7)$

Illustration 91 *:* Solve for real x : $\dfrac{1}{x+1} + \dfrac{2}{x+2} > \dfrac{3}{x+3}$

Solution : $\dfrac{3x+4}{(x+1)(x+2)} > \dfrac{3}{x+3}$

$$\Rightarrow \frac{3x+4}{(x+1)(x+2)} - \frac{3}{x+3} > 0 \qquad \Rightarrow \frac{4x+6}{(x+1)(x+2)(x+3)} > 0$$

so, $x \in (-\infty, -3) \cup \left(-2, -\frac{3}{2}\right) \cup (-1, \infty)$

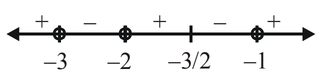
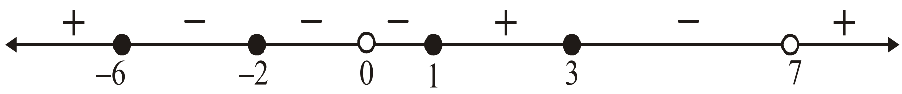


Illustration 92: Let $f(x)=\dfrac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$. Solve the following inequality

(i) $f(x)>0$ (ii) $f(x)\geq 0$ (iii) $f(x)<0$ (iv) $f(x)\leq 0$

Solution : We mark on the number line zeroes of numerator of expression : 1, –2, 3 and –6 (with black circles) and the zeroes of denominator 0 and 7 (with white circles), isolate the double points : –2 and 0 and draw the wavy curve :



From graph, we get

(i) $x \in (-\infty, -6) \cup (1, 3) \cup (7, \infty)$

(ii) $x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$

(iii) $x \in (-6, -2) \cup (-2, 0) \cup (0, 1) \cup (3, 7)$

(iv) $x \in [-6, 0) \cup (0, 1] \cup [3, 7)$

Do yourself-16 :

Solve following inequalities over the set of real numbers :

1. $(x-1)^2(x+1)^3(x-4)<0$

2. $\dfrac{6x-5}{4x+1}<0$

3. $\dfrac{(x-1)(x+2)^2}{-1-x}<0$

4. $\dfrac{(2x-1)(x-1)^2(x-2)^3}{(x-4)^4}>0$

5. $\dfrac{(x-1)^2(x+1)^3}{x^4(x-2)}\leq 0$

6. $\dfrac{(x-2)^2(1-x)(x-3)^3(x-4)^2}{(x+1)}\leq 0$

7. $\dfrac{x^2+4x+4}{2x^2-x-1}<0$

8. $\dfrac{x^3(x-2)(5-x)}{(x^2-4)(x+1)}>0$

9. $\dfrac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)}\geq 0$

10. $x^4-5x^2+4<0$

11. $\dfrac{15-4x}{x^2-x-12}<4$

12. $\dfrac{x^2+1}{4x-3}>2$

13. $\dfrac{1}{x-2}+\dfrac{1}{x-1}>\dfrac{1}{x}$

14. $(x-2)(x+3)\geq 0$

15 $\dfrac{x}{x+1}>2$

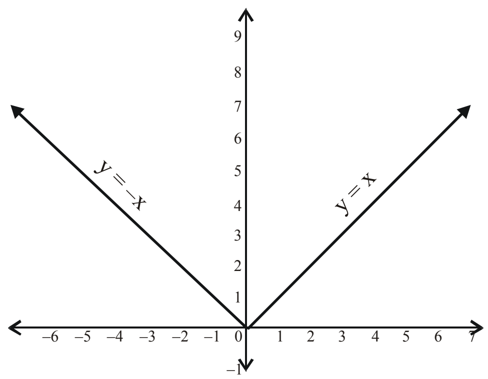
16. $\dfrac{3x-1}{4x+1}\leq 0$

12. MODULUS AND MODULUS EQUATIONS

For any real number x, modulus or absolute value of x is denoted by |x| and is defined as

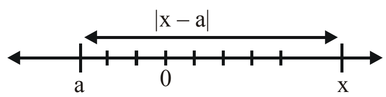
$$|x| = \begin{cases} x, & \text{if} \quad x \geq 0 \\ -x, & \text{if} \quad x < 0 \end{cases}$$

- Graph of y = |x|



Note :

- $|x| = |-x| \geq 0$
- Geometrically |x| is distance of real number x from zero along the real number line
- More generally |x – a| is distance between 'x' and 'a' on the number line.



- $|x| = \sqrt{x^2}$
- $|xy| = |x|\,|y|$

Illustration-93 : Sketch the graph of following equation and also find all possible values (Range) of y

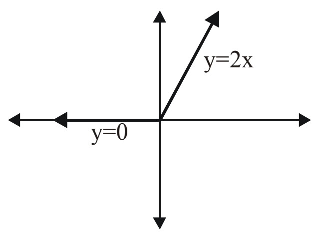
(i) y = |x| + x (ii) y = |x – 2| + x + 1

(iii) y = |x – 3| – |x| (iv) y =2|x – 2| – |x + 1|

Soluiton :

(i) $y = |x| + x = \begin{cases} x + x & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$

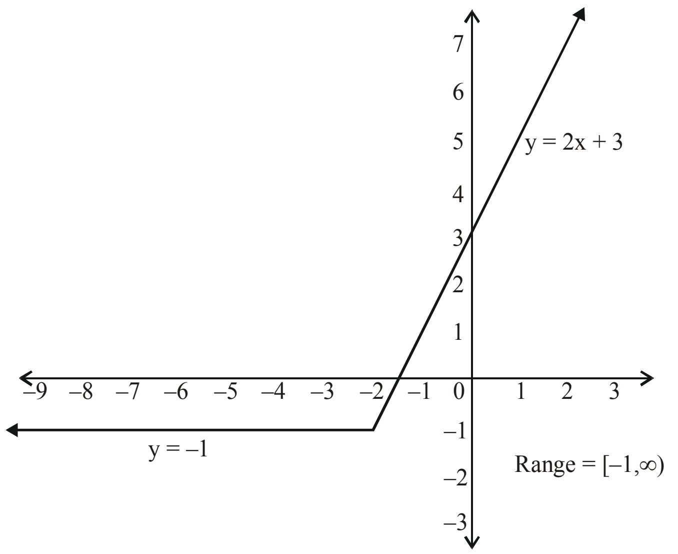
$$= \begin{cases} 2x & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$$



From graph we can find all possible values (range) of y which is $[0,\infty)$

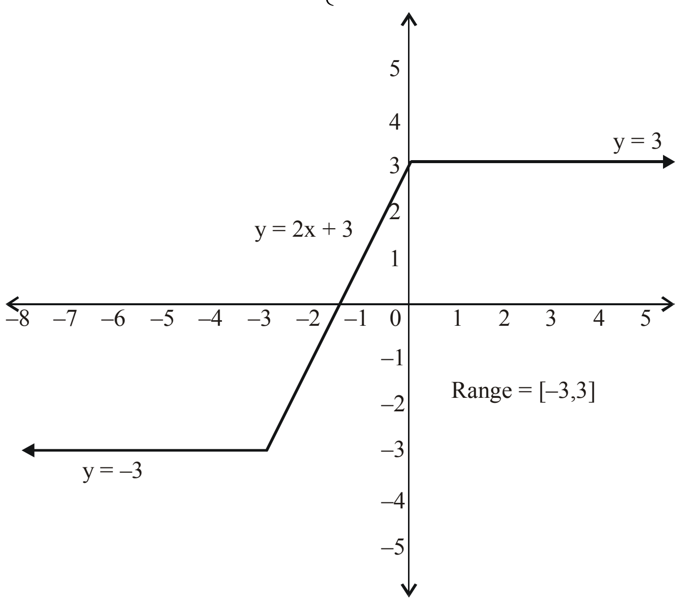
(ii) $y = |x+2| + x + 1 = \begin{cases} x+2+x+1 & , \quad x \geq -2 \\ -x-2+x+1 & , \quad x < -2 \end{cases}$

$$= \begin{cases} 2x+3 & , \quad x \geq -2 \\ -1 & , \quad x < -2 \end{cases}$$

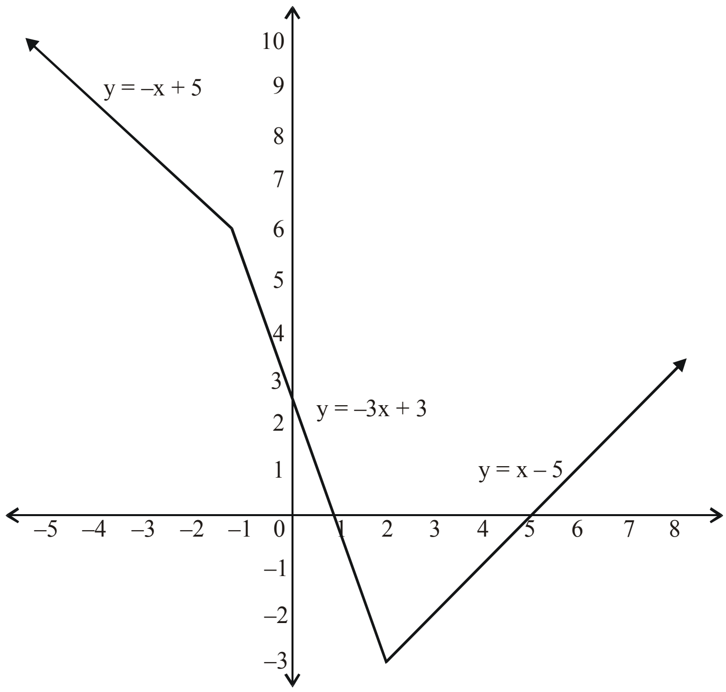


(iii) $y = |x+3| - |x| = \begin{cases} x+3-x & , \quad x \geq 0 \\ x+3-(-x) & , \quad x \in [-3,0) \\ -x-3-(-x) & , \quad x < -3 \end{cases}$

$$= \begin{cases} 3 & , \quad x \geq 0 \\ 2x+3 & , \quad x \in [-3,0) \\ -3 & , \quad x < -3 \end{cases}$$



(iv) $y = 2|x-2| - |x+1| = \begin{cases} 2(x-2)-(x+1) & , \quad x \ge 2 \\ 2(-x+2)-(x+1) & , \quad x \in [-1,2] \\ 2(-x+2)+(x+1) & , \quad x < -1 \end{cases}$



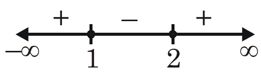
Range = $[-3,\infty)$

Illustration-94 : If $|x-1||x-2| = -(x^2-3x+2)$, then find the interval in which x lies ?

Solution : $|(x-1)(x-2)| = -(x-2)(x-1)$

$\Rightarrow (x-1)(x-2) \le 0$

$\Rightarrow 1 \le x \le 2$



Do yourself-17

(1) Sketch the graph of following

(i) $y = |x-2|$ (ii) $y = |x| - 2$ (iii) $y = 5 - |x|$

Solve for x

(2) $|2x+5| = 2$ **(3)** $|2x-5| = 7$ **(4)** $|x-3| = -1$

(5) $|2x-3| + 4 = 2$ **(6)** $\left|\frac{3x+4}{3}\right| = 7$ **(7)** $|x^2-3x+2| = 2$

(8) $|2x-3| = |3x+5|$ **(9)** $2|x+3| = 3|x-4|$ **(10)** $|x^2+x+1| = |x^2+x+2|$

(11) $|x^2-4x+3| = |x^2-5x+4|$ **(12)** $|x-6| + |x-3| = 1$ **(13)** $|2x-1| + |2x+3| = 6$

(14) $|3x+5| + |4x+7| = 12$ **(15)** $|x| + |x+1| + |x+2| = 3$

(16) $|2x-3| + |2x+1| + |2x+5| = 12$ **(17)** $|x| - 2|x+1| + 3|x+2| = 0$

(18) $|x| - x = 0$ **(19)** $|x^2+3x+2| + x + 1 = 0$

(20) $x^2 + 3|x| + 2 = 0$ **(21)** $|x^2+1| - x^2 - 1 = 0$

Useful Mathematical Symbols for Reference

Symbol	How it is read
$\forall$	For all...
$\exists$	There exists...
$\wedge$	and
$\vee$	or
$<$	is less than
$>$	is greater than
$\leq$	is less than or equal to
$\geq$	is greater than or equal to
$\nless$	is not less than
$\ngtr$	is not greater than
$\in$	belongs to
$\notin$	does not belong to
$\mid$, : , s.t.	such that
$\Rightarrow$	implies (If... then...)
$\Leftrightarrow$	implies and implied by (if and only if / iff)
!	factorial
$\sqrt{\ }$	The square root of
$\sqrt[n]{\ }$	n^{th} root, $n \in N$, $n \geq 2$
Σ	The summation of
Π	The product of

Greek Letters (Capital, Small) and there pronounciations

Symbol (Capital, Small)	How it is read	Symbol (Capital, Small)	How it is read
A,α	alpha	N,ν	nu
B,β	beta	Ξ,ξ	xi
Γ,γ	gamma	O,o	omicron
Δ,δ	delta	Π,π	pi
E,ε	epsilon	P,ρ	rho
Z,ζ	zeta	Σ,σ	sigma
H,η	eta	T,τ	tau
Θ,θ	theta	Y,υ	upsilon
I,ι	iota	Φ,ϕ	phi
K,κ	kappa	X,χ	chi
Λ,λ	lambda	Ψ,ψ	psi
M,μ	mu	Ω,ω	omega

EXERCISE (O-1)

Straight Objective Type

1. If A and B be any two sets, then $(A \cap B)'$ is equal to-

(A) $A' \cap B'$ (B) $A' \cup B'$ (C) $A \cap B$ (D) $A \cup B$

FM0001

2. Let A and B be two sets in the universal set. Then $A - B$ equals-

(A) $A \cap B'$ (B) $A' \cap B$ (C) $A \cap B$ (D) none of these

FM0002

3. If $A \subseteq B$, then $A \cap B$ is equal to-

(A) A (B) B (C) A' (D) B'

FM0003

4. If A and B are any two sets, then $A \cup (A \cap B)$ is equal to-

(A) A (B) B (C) A' (D) B'

FM0004

5. Which of the following statements is true ?

(A) $3 \subseteq \{1, 3, 5\}$ (B) $3 \in \{1, 3, 5\}$ (C) $\{3\} \in \{1, 3, 5\}$ (D) $\{3, 5\} \in \{1, 3, 5\}$

FM0005

6. Which of the following is a null set ?

(A) $A = \{x : x > 1 \text{ and } x < 1]$ (B) $B = \{x : x + 3 = 3\}$
(C) $C = \{\phi\}$ (D) $D = \{x : x \geq 1 \text{ and } x \leq 1\}$

FM0006

7. If A and B are not disjoint, then $n(A \cup B)$ is equal to-

(A) $n(A) + n(B)$ (B) $n(A) + n(B) - n(A \cap B)$
(C) $n(A) + n(B) + n(A \cap B)$ (D) $n(A).n(B)$

FM0007

8. $\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6+\ldots}}}} =$

(A) 3 (B) 2 (C) 1 (D) ± 3

FM0008

9. If $x = 8-\sqrt{60}$, then $\frac{1}{2}\left[\sqrt{x}+\frac{2}{\sqrt{x}}\right] =$

(A) $\sqrt{5}$ (B) $\sqrt{3}$ (C) $2\sqrt{5}$ (D) $2\sqrt{3}$

FM0009

10. If $\frac{4+3\sqrt{5}}{4-3\sqrt{5}} = a+b\sqrt{5}$, a, b are rational numbers, then (a, b) =

(A) $\left(\frac{61}{29}, \frac{-24}{29}\right)$ (B) $\left(\frac{-61}{29}, \frac{24}{29}\right)$ (C) $\left(\frac{61}{29}, \frac{24}{29}\right)$ (D) $\left(\frac{-61}{29}, \frac{-24}{29}\right)$

FM0010

11. The square root $5+2\sqrt{6}$ is :

(A) $\sqrt{3}+2$ (B) $\sqrt{3}-\sqrt{2}$ (C) $\sqrt{2}-\sqrt{3}$ (D) $\sqrt{3}+\sqrt{2}$

FM0011

12. If $\frac{4}{2+\sqrt{3}+\sqrt{7}} = \sqrt{a}+\sqrt{b}-\sqrt{c}$, then which of the following can be true -

(A) a = 1, b = 4/3, c = 7/3 (B) a = 1, b = 2/3, c = 7/9
(C) a = 2/3, b = 1, c = 7/3 (D) a = 7/9, b = 4/3, c = 1

FM0012

13. The numerical value of $\left(x^{1/(a-b)}\right)^{1/(a-c)} \times \left(x^{1/(b-c)}\right)^{1/(b-a)} \times \left(x^{1/(c-a)}\right)^{1/(c-b)}$ is (a, b, c are distinct real numbers)

(A) 1 (B) 8 (C) 0 (D) None

FM0013

14. $(1^3 + 2^3 + 3^3 + 4^3)^{-3/2} =$
(A) 10^{-3} (B) 10^{-2} (C) 10^{-4} (D) 10^{-1}

FM0014

15. $\left(7^{\left(-\frac{1}{2}\right)} \times 5^2\right)^2 \div \sqrt{25^3} =$

(A) $\frac{5}{7}$ (B) $\frac{7}{5}$ (C) 35 (D) $-\frac{5}{7}$

FM0015

16. $(2d^2e^{-1})^3 \times \left(\frac{d^3}{e}\right)^{-2} =$

(A) $8e^{-2}$ (B) $8e^{-3}$ (C) $8e^{-1}$ (D) $8e^{-4}$

FM0016

17. If $x^y = y^x$ and x = 2y, then the values of x and y are (x, y > 0)
(A) x = 4, y = 2 (B) x = 3, y = 2 (C) x = 1, y = 1 (D) None of these

FM0017

18. If $(a^m)^n = a^{m^n}$, then express 'm' in the terms of n is ($a > 0, a \neq 1, m > 1, n > 1$)

(A) $n^{\left(\frac{1}{n-1}\right)}$ (B) $n^{\left(\frac{1}{n+1}\right)}$ (C) $n^{\left(\frac{1}{n}\right)}$ (D) None

FM0018

19. If $a = x + \frac{1}{x}$, then $x^3 + x^{-3} =$

(A) $a^3 + 3a$ (B) $a^3 - 3a$ (C) $a^3 + 3$ (D) $a^3 - 3$

FM0019

20. If $\left(\sqrt[3]{4}\right)^{2x+\frac{1}{2}} = \frac{1}{32}$, then x =

(A) –2 (B) 4 (C) –6 (D) –4

FM0020

21. If $(5+2\sqrt{6})^{x^2-3}+(5-2\sqrt{6})^{x^2-3}=10$, then all possible values of x are

(A) –2, 2 (B) $\sqrt{2}$, $-\sqrt{2}$ (C) 2, $+\sqrt{2}$ (D) 2, –2, $\sqrt{2}$, $-\sqrt{2}$

FM0021

22. If $3^{2x^2}-2.3^{x^2+x+6}+3^{2(x+6)}=0$ then the value of x is

(A) – 2 (B) 3 (C) Both (A) and (B) (D) None of these

FM0022

23. How many integers in between 100 to 1500 (both inclusive) are multiples of 5 or 11 ?

(A) 408 (B) 26 (C) 382 (D) 380

FM0023

24. Square root of $4+\sqrt{15}$ is equal to

(A) $\frac{\sqrt{3}+\sqrt{5}}{2}$ (B) $\sqrt{\frac{3}{2}}+\sqrt{\frac{5}{2}}$ (C) $\sqrt{\frac{5}{2}}-\sqrt{\frac{3}{2}}$ (D) None of these

FM0024

25. If A = {(x, y) | xy = 8 and x, y $\in \mathbb{Z}$}, then n(A) =

(A) 4 (B) 8 (C) 12 (D) 16

FM0025

26. The set of all real numbers x that satisfy $\frac{x-2}{x+2}>\frac{2x-3}{4x-1}$

(A) $(-\infty,-2)\cup\left(\frac{1}{4},\frac{3}{2}\right)\cup(2,\infty)$ (B) $\left(-2,\frac{1}{4}\right)\cup\left(\frac{3}{2},2\right)$

(C) $(-\infty,-2)\cup\left(\frac{1}{4},1\right)\cup(4,\infty)$ (D) $\left(-2,\frac{1}{4}\right)\cup(1,4)$

FM0026

27. The set of all real numbers x that satisfy $\frac{x^2-4}{x^2-5x+6}\le 0$

(A) [–2, 3] (B) [–2, 3)

(C) (–∞, –2] ∪ (3, ∞) (D) None of these

FM0027

28. If $\frac{(x+3)^2(x-1)^9(x+1)^5}{(x-3)(x-5)^4(x-6)^5}\le 0$, then number of possible integral values of x is -

(A) 6 (B) 3 (C) 4 (D) 5

FM0028

29. If $\left(\sqrt{2}+1\right)^x+\left(\sqrt{2}-1\right)^x-2\sqrt{2}=0$, then sum of all possible values of x is

(A) 0 (B) 1 (C) 2 (D) 3

FM0029

30. If $x^2 + y^2 + 4z^2 - 6x - 2y - 4z + 11 = 0$, then xyz is equal to

(A) 3/2 (B) 4 (C) 6 (D) 3

FM0030

EXERCISE (O-2)

Straight objective Type

1. In a college of 300 students every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is

(A) at least 30 (B) at most 20 (C) exactly 25 (D) none of these

FM0031

2. If $x + y = 1$ and $x^2 + y^2 = 2$ then the value of $x^4 + y^4$ equals

(A) 7 (B) 6 (C) $\frac{7}{2}$ (D) $\frac{19}{4}$

FM0032

3. If $x = 2^{1/3} - 2^{-1/3}$ then the value of $2x^3 + 6x$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

FM0033

4. If $\frac{\sqrt{3}+4\sqrt{2}}{4\sqrt{2}-\sqrt{3}} = \frac{a+b\sqrt{6}}{c}$, then the value of $a + b + c$ (where $a, b, c \in N$ and are relatively prime)

(A) 70 (B) 72 (C) 50 (D) 40

FM0034

5. If $x^2 + 4y^2 + z^2 - 2xy - 2yz - zx = 0$ then $x : y : z$ equals

(A) 1 : 2 : 1 (B) 2 : 1 : 2 (C) 1 : 2 : 3 (D) 1 : 1 : 2

FM0035

More than one correct

6. If $A \subseteq B$ then which of the following option(s) is/are correct ?

(A) $A' \subseteq B'$ (B) $B' \subseteq A'$ (C) $A \cap B' = \phi$ (D) $A' \cap B = \phi$

FM0036

7. If $x = \sqrt{7\sqrt{7\sqrt{7\sqrt{7...}}}}$ where $x, y > 0$

$y = \sqrt{20+\sqrt{20+\sqrt{20+...}}}$ then which of the following is/are correct.

(A) $x + y = 12$ (B) $x - y = 3$ (C) $x^2 + y^2 = 74$ (D) $x^2 - y^2 = 24$

FM0037

8. If complex number $-3 + ix^2y$ and $x^2 + y + 4i$ are conjugate of each other. Then real value of x & y can be

(A) $x = 1, y = -4$ (B) $x = -1, y = -4$ (C) $x = 1, y = 4$ (D) $x = -2, y = -3$

FM0038

9. If $8x^3 + 27x^2 - 9x - 50$ is divided by $(x + 2)$ then remainder is λ then

(A) $\frac{3\lambda+4}{10}$ is equal to 4

(B) If $\lambda = \frac{p}{q}$ then $(p + q)$ is divisible by 13 (where p & q are coprime)

(C) λ is a natural number

(D) $(\lambda - 2)$ is divisible by 3

FM0039

10. If $f(x) = ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \mathbb{R}$, $a \neq 0$ then which of the following option(s) is(are) correct ?
(A) if $a + b + c + d = 0$ then $x - 1$ is factor of $f(x)$.
(B) if $a + b = c + d$ then $x + 1$ is factor of $f(x)$.
(C) if $a + c = b + d$ then $x + 1$ is factor of $f(x)$.
(D) none of these

FM0040

11. If $x = a$ and $x = b$ are the two roots of the equation $9^x - 4 \times 3^{x+1} + 27 = 0$ then

(A) $a + b = 3$ (B) $(a - b)^2 = 1$ (C) $\frac{a}{b} + \frac{b}{a} = \frac{5}{2}$ (D) $a + b = 4$

FM0041

12. The real values of 'x' satisfying the equation $\left(4+\sqrt{15}\right)^{x^2-x-1} + \left(4-\sqrt{15}\right)^{x^2-x-1} = 8$ is/are :
(A) 0 (B) 1 (C) –1 (D) –2

FM0042

Linked Comprehension Type

Paragraph for question no. 13 to 15

If $(5+2\sqrt{6})^{x^2-8} + (5-2\sqrt{6})^{x^2-8} = 10, x \in \mathbb{R}$
On the basis of above information, answer the following questions :

13. Number of solution(s) of the given equation is/are-
(A) 1 (B) 2 (C) 4 (D) infinite

FM0043

14. Sum of positive solutions is
(A) 3 (B) $3+\sqrt{7}$ (C) $2+\sqrt{5}$ (D) 2

FM0043

15. If $x \in (-3,5]$, then number of possible values of x, is-
(A) 1 (B) 2 (C) 3 (D) 4

FM0043

Paragraph for question no. 16 & 17

A polynomial p(x) when divided by $(x - 1)$, $(x + 1)$ and $(x + 2)$ gives remainder 5, 7 and 2 respectively. If p(x) is divided by $(x^2 - 1)$ and $(x - 1)(x + 2)$ gives remainder as another polynomial R(x) and r(x) respectively. Then

16. The value of R(50) is :
(A) 34 (B) –44 (C) 44 (D) 104

FM0044

17. The value of r(100) is :
(A) 34 (B) 44 (C) 54 (D) 104

FM0044

Matching list type

18. Match the list

List-I		List-II	
(P)	The units digit of $2^7 \times 3^{10}$ is	(1)	1
(Q)	The number of prime factors of $6^{100} \times 15^{200}$ is	(2)	2
(R)	Number of solutions of $xy = 35$ in natural numbers with $x > y > 1$ is	(3)	3
(S)	Remainder when $x^{100} - 3x^{99} + 2x^{98} + 3x - 2$ is divided by $x - 2$ is equal to	(4)	4

FM0045 **FM0046** **FM0047** **FM0048**

Codes :

(A) P-1, Q-2, R-3, S-4 (B) P-2, Q-3, R-1, S-4

(C) P-2, Q-3, R-2, S-3 (D) P-1, Q-4, R-1, S-2

19. Match the list

List-I		List-II	
(P)	If $x = \sqrt{5} - 2$ then value of $2x^3 + 11x^2 + 10x + 4$ is equal to	(1)	8
(Q)	If $x = \sqrt{5} + 2$ then value of $2x^3 - 7x^2 - 6x + 7$	(2)	7
(R)	If $\left(\left(\sqrt{11+2\sqrt{30}}\right)-\left(\sqrt{10+4\sqrt{6}}\right)\right)^2 = p + \sqrt{q} + \sqrt{r} + \sqrt{s}$ where p, q, r, s are integers & q, r, s are not perfect squares the $p + q + r + s$ is equal to	(3)	161
(S)	If $\left(\left(\sqrt{11+2\sqrt{30}}\right)-\left(\sqrt{10+4\sqrt{6}}\right)\right)^2 = \alpha - \sqrt{\beta}$ where α, β are integers & β is not a perfect square then $\alpha^2 + \beta$ is equal to	(4)	977

FM0049 **FM0050** **FM0051** **FM0052**

Codes :

(A) P-1, Q-2, R-3, S-4 (B) P-2, Q-3, R-1, S-4

(C) P-2, Q-1, R-3, S-4 (D) P-2, Q-1, R-4, S-3

Answer the question question 20 and 21 by appropriately matching the lists based on the information given

	List-I		List-II
(I)	If $2\sqrt{x+5} = x+2$, then integral value of x	(P)	3
(II)	The number of real solution of the equation $x^2 + 5\|x\| + 4 = 0$, is	(Q)	–4
(III)	The real solution of the equation $(x-2)^2 + \|x-2\| - 2 = 0$, is	(R)	–1
(IV)	The number of real solution(s) $\|x+2\| = 2(3-x)$, is	(S)	0
		(T)	4
		(U)	1

20. Which of the following is only **CORRECT** option ?
(A) I $\rightarrow$ R, U (B) II $\rightarrow$ S, T (C) I $\rightarrow$ Q, S (D) II $\rightarrow$ S

FM0053

21. Which of the following is only **INCORRECT** option ?
(A) III $\rightarrow$ P, U (B) IV $\rightarrow$ U (C) III $\rightarrow$ Q, T (D) III $\rightarrow$ P

FM0053

Numerical Grid Type

22. If $a - \frac{1}{a} = \frac{1}{2}$ then $\left(4a^2 + \frac{4}{a^2}\right)$ is equal to

FM0054

23. If polynomial $Ax^3 + 4x^2 + Bx + 5$ leaves same remainder, when divided by $x-1$ and $x+2$ respectively then value of $3A + B$ is equal to

FM0055

24. If $9^x + 6^x = 2.4^x$ then the value of $\frac{x^3+64}{5}$

FM0056

25. If $a + b + c = 6$ & $a^2 + b^2 + c^2 = 14$ and $a^3 + b^3 + c^3 = 36$ then the value of $3\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

FM0057

EXERCISE (S-1)

1. An investigator interviewed 100 students to determine their preferences for the three drinks : milk (M), coffee (C) and tea (T). He reported the following : 10 students had all the three drinks M, C and T; 20 had M and C; 30 had C and T; 25 had M and T; 12 had M only; 5 had C only; and 8 had T only. Using a Venn diagram find how many did not take any of the three drinks.

FM0058

2. Suppose $a + \frac{1}{a} = 3$. Find the values of

(a) $a^2 + \frac{1}{a^2}$ (b) $a^4 + \frac{1}{a^4}$ (c) $a^3 + \frac{1}{a^3}$

FM0059

3. Which of the following equation(s) has (have) only unity as the solution.
(A) $2(3^{x+1}) - 6(3^{x-1}) - 3^x = 9$ (B) $7(3^{x+1}) - 5^{x+2} = 3^{x+4} - 5^{x+3}$

FM0060

4. Which of the following equation (s) has (have) only natural solution(s)

(A) $6.9^{1/x} - 13.6^{1/x} + 6.\ 4^{1/x} = 0$ (B) $4^x.\sqrt{8^{x-1}} = 4$ **FM0061**

5. If $\frac{29}{12}$ can be expressed as $a + \cfrac{1}{b + \cfrac{1}{c + \cfrac{1}{d}}}$, $(a, b, c, d \in \mathbb{N})$ then find the value of $a^3 + b^3 + c^3 + d^3$ is equal to

FM0062

6. **Factorize the following expressions :**

(i) $(x - y)(x - y - 1) - 20$ **FM0063**

(ii) $(x + 2y)(x + 2y + 2) - 8$ **FM0064**

(iii) $3(4x + 5)^2 - 2(4x + 5) - 1$ **FM0065**

(iv) $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2$ **FM0066**

(v) $x^6 - 7x^3 - 8$ **FM0067**

(vi) $x^4 + 15x^2 - 16$ **FM0068**

(vii) $(x^2 - x - 3)(x^2 - x - 5) - 3$ **FM0069**

(viii) $(x^2 + 5x + 6)(x^2 + 5x + 4) - 120$ **FM0070**

(ix) $(a^2 + 1)^2 + (a^2 + 5)^2 - 4(a^2 + 3)^2$ **FM0071**

(x) $(x - 1)^2 + (y - 1)^2 + (z - 1)^2 + 2(x - 1)(y - 1) + 2(y - 1)(z - 1) + 2(z - 1)(x - 1)$ **FM0072**

(xi) $2x^4 - x^3 - 6x^2 - x + 2$ **FM0073**

(xii) $(x^4 + x^2 - 4)(x^4 + x^2 + 3) + 10$ **FM0074**

7. The value of $(4 + 1)(4^2 + 1)(4^4 + 1)(4^8 + 1) + \frac{1}{3} = \frac{2^\lambda}{3}$ where $\lambda \in \mathbb{N}$ then find the value of 'λ' is equal to :

FM0075

8. Simplify the following expressions :

(a) $\sqrt[3]{2+\sqrt{5}} + \sqrt[3]{2-\sqrt{5}}$.

FM0076

(b) $\sqrt[3]{18+5\sqrt{13}} + \sqrt[3]{18-5\sqrt{13}}$.

FM0077

(c) $\sqrt{\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}} + \ldots \text{ upto 99 terms}}$

FM0078

9. Suppose a and b are constants such that
$(x^3 + bx^2 - 7x + 9)(x^2 + ax + 5) = x^5 + 13x^4 + 38x^3 - 22x^2 + 37x + 45\ \forall\ x \in \mathbb{R}$. Find a and b.

FM0079

10. If P(x) is a cubic polynomial such that P(1) = 1; P(2) = 2; P(3) = 3 with leading coefficient 3 then find the value of P(4).

FM0080

11. Find all conditions on a and b $(a, b \in \mathbb{R})$ when $ax + b = 4x + 10$ has

(i) exactly one solution in 'x'. (ii) no solution in 'x'.

(iii) has exactly two solutions in 'x' (iv) has infinite solutions in x

FM0081

12. Let x, y, z are real numbers satisfying following equations :

$3x - 2y + 4z = 3$

$x - y + 2z = 10$ and

$2x - 3y + 3z = 5$ then the value of $(2x - 2y + 3z)^2$ is equal to

FM0082

13. How may ordered pairs of natural numbers (a, x) satisfy $ax = a + 4x$? **FM0083**

14. Let $x = \sqrt{3-\sqrt{5}}$ & $y = \sqrt{3+\sqrt{5}}$. If the value of the expression $x - y + 2x^2y + 2xy^2 - x^4y + xy^4$ can be expressed is the form $\sqrt{p} + \sqrt{q}$ (where $p, q \in \mathbb{N}$), then find the value of $(p + q)$

FM0084

15. Solve following Inequalities over the set of real numbers -

(i) $x^2 + 2x - 3 \le 0$ **FM0085**

(ii) $x^2 + 6x - 7 \le 0$ **FM0086**

(iii) $x^4 - 2x^2 - 63 \le 0$ **FM0087**

(iv) $\dfrac{x+1}{(x-1)^2} < 1$ **FM0088**

(v) $\dfrac{x^2 - 7x + 12}{2x^2 + 4x + 5} > 0$ **FM0089**

(vi) $\dfrac{(x-1)(x+2)^2}{-1-x} < 0$ **FM0090**

(vii) $\dfrac{x^4 + x^2 + 1}{x^2 - 4x - 5} < 0$ **FM0091**

(viii) $\dfrac{x+7}{x-5} + \dfrac{3x+1}{2} \ge 0$ **FM0092**

(ix) $\dfrac{1}{x+2} < \dfrac{3}{x-3}$ **FM0093**

(x) $\dfrac{14x}{x+1} - \dfrac{9x-30}{x-4} < 0$ **FM0094**

(xi) $\dfrac{x^2 + 2}{x^2 - 1} < -2$ **FM0095**

(xii) $\dfrac{5-4x}{3x^2 - x - 4} < 4$ **FM0096**

(xiii) $\dfrac{(x+2)(x^2 - 2x + 1)}{4 + 3x - x^2} \ge 0$ **FM0097**

(xiv) $\dfrac{x^4 - 3x^3 + 2x^2}{x^2 - x - 30} > 0$ **FM0098**

(xv) $\dfrac{2x}{x^2 - 9} \le \dfrac{1}{x+2}$ **FM0099**

(xvi) $\dfrac{20}{(x-3)(x-4)} + \dfrac{10}{x-4} + 1 > 0$ **FM0100**

EXERCISE (S-2)

1. A survey shows that 63% of the Americans like cheese whereas 76% like apples. If x% of the Americans like both cheese and apples, find the value of x.

FM0101

2. How many ordered pairs of integers which satisfy the equation $\frac{1}{m}+\frac{2}{n}=\frac{1}{2}$?

FM0102

3. Find the polynomial of 4^{th} degree with integer co-efficients such that $x=\sqrt{3}+\sqrt{2}$ is root of polynomial.

FM0103

4. If $x + y + z = 12$ & $x^2 + y^2 + z^2 = 96$ and $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=36$. Find the value of $\frac{x^3+y^3+z^3}{4}$

FM0104

5. If $x=\sqrt{4-2\sqrt{3}}$ and $y=\sqrt{9-4\sqrt{5}}$ then the value of $\left(\sqrt{5}x-\sqrt{3}y\right)^2$ is equal to $\left(a-b\sqrt{c}\right)$ where a, b, c are coprime numbers then a + b + c is equal to (where 'c' is an odd integer)

FM0105

6. If $x, y, z \in R$ and $121x^2 + 4y^2 + 9z^2 - 22x + 4y + 6z + 3 = 0$ then value of $x^{-1} - y^{-1} - z^{-1}$ is equal to

FM0106

7. If $\frac{x}{a}=\frac{y}{b}=\frac{z}{c}$ prove that $\frac{x^2-yz}{a^2-bc}=\frac{y^2-zx}{b^2-ca}=\frac{z^2-xy}{c^2-ab}$; $\forall$ a,b,c $\in R^*$ and $a^2 \neq bc$, $b^2 \neq ca$, $c^2 \neq ab$.

FM0107

8. If $\frac{p}{q}=\frac{x+2}{x-2}$, then evaluate $\frac{p^2-q^2}{p^2+q^2}$

FM0108

9. If $(a^2 + b^2)(x^2 + y^2) = (ax + by)^2$ prove that $\frac{a}{x}=\frac{b}{y}$ (for x,y $\in R^*$)

FM0109

10. $\frac{3x+\sqrt{9x^2-5}}{3x-\sqrt{9x^2-5}}=5$ find 'x'

FM0110

11. Value of $\left(\cfrac{\sqrt{6}}{\sqrt{3}-\sqrt{2}+\cfrac{\sqrt{6}}{\sqrt{3}-\sqrt{2}+\cfrac{\sqrt{6}}{(\sqrt{3}-\sqrt{2})+\ddots}}}\right)^2$ is

FM0111

JEE-MAINS/ JEE-ADVANCE

JEE-MAINS

1. If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then :-

(1) $B = C$ (2) $A \cap B = \phi$ (3) $A = B$ (4) $A = C$

FM0112

2. In a class of 140 students numbered 1 to 140, all even numbered students opted mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is :

(1) 102 (2) 42 (3) 1 (4) 38

FM0113

3. Two newspapers A and B are published in a city. It is known that 25% of the city populations reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisement is :-

(1) 12.8 (2) 13.5 (3) 13.9 (4) 13

FM0114

4. Let $\mathbb{Z}$ be the set of integers. If $A = \left\{x \in \mathbb{Z} : 2^{(x+2)(x^2-5x+6)} = 1\right\}$ and $B = \{x \in \mathbb{Z} : -3 < 2x - 1 < 9\}$, then the number of subsets of the set A × B, is:

(1) 2^{18} (2) 2^{10} (3) 2^{15} (4) 2^{12}

FM0115

JEE-ADVANCED

1. If X and Y are two sets, then $X \cap (X \cup Y)^c$ equals

(a) X (b) Y (c) ϕ (d) none of these

FM0116

2. The expression $\dfrac{12}{3+\sqrt{5}+2\sqrt{2}}$ is equal to

(a) $1-\sqrt{5}+\sqrt{2}+\sqrt{10}$ (b) $1+\sqrt{5}+\sqrt{2}-\sqrt{10}$

(c) $1+\sqrt{5}-\sqrt{2}+\sqrt{10}$ (d) $1-\sqrt{5}-\sqrt{2}+\sqrt{10}$

FM0117

3. If $x < 0$, $y < 0$, $x + y + \dfrac{x}{y} = \dfrac{1}{2}$ and $(x + y)\dfrac{x}{y} = -\dfrac{1}{2}$, then x = ... and y = ...

FM0118

4. The equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ has

(A) no root (B) one root (C) two equal roots (D) infinitely many roots

FM0119

5. If A = {1, 2, 3, 4, 5}, then the number of proper subsets of A is-

(1) 120 (2) 30 (3) 31 (4) 32

FM0120

6. If A, B, C be three sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$, then -

(1) A = B (2) B = C (3) A = C (4) A = B = C

FM0121

7. Sets A and B have 3 and 6 elements respectively. What can be the minimum number of elements in $A \cup B$?

(1) 3 (2) 6 (3) 9 (4) 18

FM0122

8. Find all real values of x which satisfy $x^2 - 3x + 2 > 0$ and $x^2 - 2x - 4 \le 0$.

FM0123

9. Find the set of all x for which $\frac{2x}{(2x^2+5x+2)} > \frac{1}{(x+1)}$.

FM0124

10. The sum of all real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$ is ...

FM0125

11. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is

(A) 4 (B) 1 (C) 2 (D) 3

FM0126

12. If S is the set of all real x such that $\frac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains

(A) $\left(-\infty, -\frac{3}{2}\right)$ (B) $\left(-\frac{3}{2}, -\frac{1}{4}\right)$ (C) $\left(-\frac{1}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, 3\right)$

(E) None of these

FM0127

13. Let $y = \sqrt{\frac{(x+1)(x-3)}{(x-2)}}$

Find all the real values of x for which y takes real values.

FM0128

ANSWER KEY

Do yourself-1

1. All are true **2.** C 3. D

Do yourself-3

2. C **3.** B **4.** B **5.** B **6.** C **7.** C **8.** D **9.** B
10. D **11.** B **12.** B

Do yourself-4

1. a = 3, b = 4

2. **(i)** $\frac{2}{11}$ **(ii)** $\frac{16}{99}$ **(iii)** $\frac{419}{990}$

4. **(i)** x + y, x – y, xy, x/y are rational numbers.
(ii) x + y, x – y, xy, x/y, all are may or may not be rational.
(iii) x + y, x – y, xy, x/y are irrational numbers.

5. x is irrational number (Non terminating and non-recurring)

Do yourself-5

1. $(x,y) \equiv (1,20); (2,10); (4,5); (5,4); (10,2); (20,1)$

2. **(i)** $(x,y) \equiv (2,2)$ **(ii)** $(x,y) \equiv (5,13); (6,4) ; (7,1)$

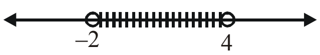
3. $(x,y) \equiv (8,56); (14,14); (56,8)$

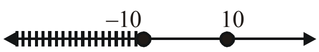
4. 651 **5.** 293 **6.** 4 **7.** 30 **8.** 3

Do yourself-6

(i) 

(ii) 

(iii) 

(iv) 

Do yourself-7

1. B **2.** C **3.** A **4.** D **5.** C

6. **(i)** 1 **(ii)** 0 **(iii)** 1 **(iv)** 1 **(v)** 8

7. 2 **8.** (a) $-\frac{47}{44}$, (b) $-\frac{9}{44}$ **9.** 5 **10.** 26

11. 1 **12.** $x = y = z = \frac{a}{3}$

Do yourself-8

1. **(i)** $(x-y)(x+y)(x^2+y^2)$ **(ii)** $(3a-2x+y)(3a+2x-y)$

(iii) $(2x+3y)(2x-3y-3)$

2. **(i)** $(2x-3y)(4x^2+6xy+9y^2)$ **(ii)** $(2x-5y)[4x^2+10xy+25y^2+1]$

3. **(i)** $(x+8)(x-5)$ **(ii)** $(x-8)(x+5)$ **(iii)** $(x+7)(x-2)$

(iv) $(x-4)(x+1)$ **(v)** $(x-3)(x+1)$ **(vi)** $(3x-4)(x-2)$

(vii) $(4x+7)(3x-5)$ **(viii)** $(3x-2)(x-1)$ **(ix)** $(x-1)(3x-4)$

(x) $(7x-1)(x-1)$ **(xi)** $(x-2)(2x-13)$ **(xii)** $(a-3)(3a+2)$

(xiii) $(2a+1)(7a-3)$

4. **(i)** $(a-b-1)(a+b-3)$ **(ii)** $(x^2-6x+18)(x^2+6x+18)$

(iii) $(x^2-y+1)(x^2+y+1)$ **(iv)** $(2a^2-a-1)(2a^2+a-1)$

(v) $(2x^2-6x+9)(2x^2+6x+9)$ **(vi)** $(x^4-x^2+1)(x^4+x^2+1)$

5. **(i)** $(x-1)(x-2)(x-3)$ **(ii)** $(x+1)(x+3)(2x+1)$

(iii) $(x-1)(x-2)(2x-3)$ **(iv)** $(x^2+2)(x^2+1)\left(x-\sqrt{3}\right)\left(x+\sqrt{3}\right)$

(v) $3(x+y)(y+z)(z+x)$

6. **(i)** $(2a^2-a+1)(4a^4+2a^3-a^2+a+1)$

7. **(i)** $(x^2+5x+1)(x^2+5x+9)$

(ii) $2(2x^2+2x+3)(4x^2+4x-9)$

(iii) $(x^2+5x-22)(x+1)(x+4)$

Do yourself-9

1. (a) 7 (b) 47 (c) 18 **4.** 104 **5.** $-\sqrt{2}$ **6.** 1

8. (a) 0, 4 (b) no solution (c) no solution

Do yourself-10

1. 1 **2.** –20 **3.** 4 **4.** –2 **5.** 3x + 4

6. $\ell = 2, m = -18$ **7.** $\ell = -1,\ m = -3$ **8.** a = 3, b = 1

9. $3x^3 + 4x^2 - 5x - 2$ **10.** 6

Do yourself-11

1. 1 or –8 **2.** 1,32 **3.** $\pm\sqrt{2},\ \pm\sqrt{\frac{2}{3}}$

4. 3 or 4 **5.** $x = 1, x = \frac{3 \pm \sqrt{5}}{2}$ **6.** $x = \pm 1,\ 2,\ -\frac{1}{2}$

7. 0,–1 **8.** 2,4 **9.** ±1

10. –4,4 **11.** –4,1 **12.** –6,1

13. $x = -3 \pm \sqrt{5}$ **14.** $\pm 1,\ 1 \pm \sqrt{2}$

15. (A)→(U); (B)→(U); (C)→(T); (D)→(P,Q,R,S,T,U,V);(E)→(R,T); (F)→(S)

Do yourself-12

1. x = 5, y = 3 **2.** x = 2, y = 1 or x = –1, y = –2

3. x = 4, y = –2 or x = –2, y = 4 **4.** x = –6, y = –2 or x = –4, y = –4

5. x = 3, y = 2 or x = –3, y = –2 **6.** x = 5, y = 1 or x = –5, y = –1

7. $x = \frac{11}{13},\ y = -\frac{24}{5}$ **8.** x = 3, y = 2 or x = –3, y = –2

9. x = 4, y = ±3 **10.** x = –1, y = 3 or $x \in \mathbb{R}$, y = 2

11. x = 2, y = –1or x = –1, y = 2 **12.** $(x,y) \cong (3,1),(-3,-1),(1,3),(-1,-3)$

Do yourself-13

1. (a) F (b) T (c) F (d) T (e) T (f) F

2. (a) $\frac{17}{21}$ (b) $\frac{37}{41}$ **3.** 3^{30} **4.** $(2,\infty)$ **5.** $3 + \sqrt{14}$

Do yourself-14

1. $4\sqrt{2}$ **2.** –12 **3.** 4

Do yourself-15

1. $\frac{20}{27}$ **2.** 21 : 35 : 65 **3.** $\frac{sp}{p+q},\frac{sq}{p+q}$ **4.** $\frac{a^4}{b^4}$

Do yourself-16

1. $(-1,4)-\{1\}$

2. $\left(-\frac{1}{4},\frac{5}{6}\right)$

3. $(-\infty,-2)(-2,-1)\cup(1,\infty)$

4. $\left(-\infty,\frac{1}{2}\right)\cup(2,\infty)-\{4\}$

5. $[-1,2)-\{0\}$

6. $(-1,1]\cup[3,\infty)\cup\{2\}$

7. $\left(-\frac{1}{2},1\right)$

8. $(-2,-1)\cup(0,5)-\{2\}$

9. $\left[-\sqrt{2},\sqrt{2}\right]\cup[3,4)-\{-1\}$

10. $(-2,-1)\cup(1,2)$

11. $\left(-\infty,\frac{-\sqrt{63}}{2}\right)\cup\left(-3,\frac{\sqrt{63}}{2}\right)\cup(4,\infty)$

12. $\left(\frac{3}{4},1\right)\cup(7,\infty)$

13. $\left(-\sqrt{2},0\right)\cup\left(1,\sqrt{2}\right)\cup(2,\infty)$

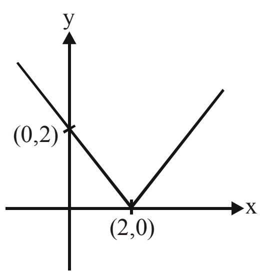
14. $(-\infty,-3]\cup[2,\infty)$

15 $(-2,-1)$

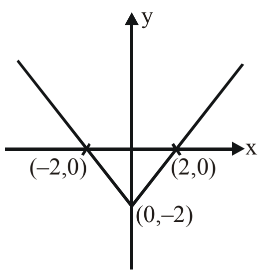
16. $\left(-\frac{1}{4},\frac{1}{3}\right]$

Do yourself-17

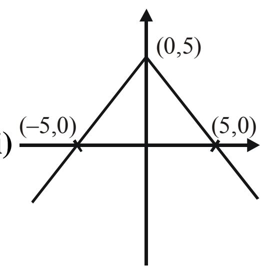
1. **(i)**



(ii)



(iii)



2. $\left\{-\frac{3}{2},-\frac{7}{2}\right\}$ **3.** $\{-1,6\}$ **4.** ϕ **5.** ϕ **6.** $\left\{-\frac{25}{3},\frac{17}{3}\right\}$

7. $\{0,3\}$ **8.** $\left\{-8,-\frac{2}{5}\right\}$ **9.** $\left\{\frac{6}{5},18\right\}$ **10.** ϕ

11.	$\left\{1,\frac{7}{2}\right\}$	**12.**	ϕ	**13.**	$\{1,-2\}$	**14.**	$\left\{0,-\frac{24}{7}\right\}$
15.	$\{-2,0\}$	**16.**	$\left\{-\frac{5}{2},\frac{3}{2}\right\}$	**17.**	$\{-2\}$	**18.**	$[0,\infty)$
19.	$\{-3,-1\}$	**20.**	ϕ	**21.**	$\mathbb{R}$		

EXERCISE (O-1)

1. B	**2.** A	**3.** A	**4.** A	**5.** B	**6.** A	**7.** B
8. A	**9.** A	**10.** D	**11.** D	**12.** A	**13.** A	**14.** A
15. A	**16.** C	**17.** A	**18.** A	**19.** B	**20.** D	**21.** D
22. C	**23.** C	**24.** B	**25.** B	**26.** C	**27.** D	**28.** D
29. A	**30.** A					

EXERCISE (O-2)

1. C	**2.** C	**3.** C	**4.** B	**5.** B	**6.** B,C	**7.** A,C,D
8. A,B	**9.** A,B,C	**10.** A,C	**11.** A,B,C	**12.** A,B,C	**13.** C	**14.** B
15. C	**16.** B	**17.** D	**18.** B	**19.** D	**20.** D	**21.** C
22. 9.00	**23.** 4	**24.** 12.80	**25.** 5.5			

EXERCISE (S-1)

1. 20 **2.** **(a)** 7 **(b)** 47 **(c)** 18 **3.** A **4.** B **5.** 32

6. **(i)** $(x - y - 5)(x - y + 4)$. **(ii)** $(x + 2y - 2)(x + 2y + 4)$.

(iii) $16(3x + 4)(x + 1)$. **(iv)** $(x^{\frac{1}{3}} + 2)(x^{\frac{1}{3}} - 1)$.

(v) $(x - 2)(x^2 + 2x + 4)(x + 1)(x^2 - x + 1)$.

(vi) $(x^2 + 16)(x - 1)(x + 1)$.

(vii) $(x + 1)(x - 2)(x - 3)(x + 2)$. **(viii)** $(x^2 + 5x + 16)(x + 6)(x - 1)$.

(ix) $-2(a^2 + 1)(a^2 + 5)$. **(x)** $(x + y + z - 3)^2$.

(xi) $(2x - 1)(x - 2)(x + 1)^2$. **(xii)** $(x^2 + 2)(x + 1)(x - 1)(x^4 + x^2 + 1)$.

7. 32 **8.** **(a)** 1 **(b)** 3 **(c)** 3

9. $a = 8$ and $b = 5$ **10.** 22

11. (i) $a \neq 4,\ b \in \mathbb{R}$ (ii) $a = 4,\ b \neq 10$ (iii) $a, b \in \phi$ (iv) $a = 4,\ b = 10$

12. 36 **13.** 3 **14.** 610

15. **(i)** $[-3,1]$ **(ii)** $[-7,1]$

(iii) $[-3, 3]$ **(iv)** $(-\infty, 0) \cup (3, +\infty)$

(v) $(-\infty, 3) \cup (4, +\infty)$ **(vi)** $(-\infty, -2) \cup (-2, -1) \cup (1, +\infty)$

(vii) $(-1, 5)$ **(viii)** $[1, 3] \cup (5, +\infty)$

(ix) $\left(-\frac{9}{2},-2\right)\cup(3,\infty)$ **(x)** $(-1,\ 1)\cup(4,\ 6)$

(xi) $(-1,1)-\{0\}$ **(xii)** $\left(-\infty,-\frac{\sqrt{7}}{2}\right)\cup\left(-1,\frac{\sqrt{7}}{2}\right)\cup\left(\frac{4}{3},\infty\right)$

(xiii) $(-\infty,\ -2]\cup(-1,4)$ **(xiv)** $(-\infty,\ -5)\cup(1,\ 2)\cup(6,\ +\infty)$

(xv) $(-\infty,-3)\cup(-2,3)$ **(xvi)** $(-\infty,-2)\cup(-1,3)\cup(4,\infty)$

EXERCISE (S-2)

1. $39\le x\le 63$ **2.** 7 **3.** $x^4-10x^2+1=0$ **4.** 216.50 **5.** 36 **6.** 16 **8.** $\frac{4x}{x^2+4}$

10. x =1 **11. 2**

JEE-MAINS/ JEE-ADVANCE

JEE-MAINS

1. 1 **2.** 4 **3.** 3 **4.** 3

JEE-ADVANCED

1. C **2.** B **3.** $x=-\frac{1}{4},y=-\frac{1}{4}$ **4.** A **5.** 3

6. 2 **7.** 2 **8.** $x\in\left[1-\sqrt{5},1\right)\cup\left(2,1+\sqrt{5}\right]$

9. $(-2,-1)\cup\left(-\frac{2}{3},-\frac{1}{2}\right)$ **10.** 4 **11.** A **12.** A,D **13.** $x\in[-1,\ 2)\cup[3,\ \infty)$

Exercise from new module

EXERCISE (O-1)

Straight Objective Type

1. Which of the following statements is true ?

(A) $3 \subseteq \{1, 3, 5\}$ (B) $3 \in \{1, 3, 5\}$ (C) $\{3\} \in \{1, 3, 5\}$ (D) $\{3, 5\} \in \{1, 3, 5\}$

FM0005

2. Which of the following is a null set ?

(A) $A = \{x : x > 1 \text{ and } x < 1]$ (B) $B = \{x : x + 3 = 3\}$

(C) $C = \{\phi\}$ (D) $D = \{x : x \geq 1 \text{ and } x \leq 1\}$

FM0006

3. If A and B be any two sets, then $(A \cap B)'$ is equal to-

(A) $A' \cap B'$ (B) $A' \cup B'$ (C) $A \cap B$ (D) $A \cup B$

FM0001

4. Let A and B be two sets in the universal set. Then A – B equals-

(A) $A \cap B'$ (B) $A' \cap B$ (C) $A \cap B$ (D) none of these

FM0002

5. If $A \subseteq B$, then $A \cap B$ is equal to-

(A) A (B) B (C) A' (D) B'

FM0003

6. If A and B are any two sets, then $A \cup (A \cap B)$ is equal to-

(A) A (B) B (C) A' (D) B'

FM0004

7. If A and B are not disjoint, then $n(A \cup B)$ is equal to-

(A) $n(A) + n(B)$ (B) $n(A) + n(B) - n(A \cap B)$

(C) $n(A) + n(B) + n(A \cap B)$ (D) $n(A).n(B)$

FM0007

8. In a college of 300 students every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspapers is

(A) at least 30 (B) at most 20 (C) exactly 25 (D) none of these

FM0031

9. Number of integers in between 200 to 2023 (both 200 and 2023 including) are divisible by 2 or 5

(A) 1093 (B) 1094 (C) 1275 (D) 1277

ST0047

10. Number of integers in between 500 to 2023 are divisible by 3 or 5 but not multiple of 2

(A) 356 (B) 355 (C) 710 (D) 711

ST0048

11. Let S = {1,2}, number of elements in the set $\{(A,B) : A \cup B = S\}$

(A) 4 (B) 6 (C) 9 (D) 10

ST0049

12. How many integers between 100 and 999 (all three digits of integer are not distinct) inclusive, have the property that some permutation of its digits is a multiple of 11 between 100 and 999? For example both 121 and 211 have this property

(A) 48 (B) 44 (C) 42 (D) 40

ST0050

13. How many of the following statements are correct

P : In roster form, the order in which the elements are listed is immaterial

Q : While writing a set in roster form, an element is not generally repeated.

R : The collection of five most renowned mathematicians of the world is not a set.

S : A collection of most dangerous animals of the world is a set

(A) 1 (B) 2 (C) 3 (D) 4

ST0051

14. How many of the following statements are sets.

(i) The collection of all the months of a year beginning with the letter J.

(ii) The collection of ten most talented writers of india.

(iii) A team of eleven best-cricket bastsmen of the world.

(iv) The collection of all boys in your class.

(v) The collection of all natural numbers less than 100.

(A) 1 (B) 2 (C) 3 (D) 4

ST0052

15. How many of the following statements are correct ?

(1) Let A = $\{x : 1 < x < 2$, x is a natural number$\}$. Then A is the empty set, because there is no natural number between 1 and 2.

(2) B = $\{x : x^2 - 2 = 0$ and x is rational number$\}$. Then B is the empty set because the equation $x^2 - 2 = 0$ is not satisfied by any rational value of x.

(3) C = $\{x$: x is an even prime number greater than $2\}$. Then C is the empty set, because 2 is the only even prime number

(4) D = $\{x : x^2 = 4$, x is odd$\}$. Then D is the empty set, because the equation $x^2 = 4$ is not satisfied by any odd value of x.

(A) 1 (B) 2 (C) 3 (D) 4

ST0053

16. How many of the following statements are correct ?

(1) Let W be the set of the days of the week. Then W is finite.

(2) Let S be the set of solutions of the equation $x^2 - 16 = 0$. Then S is finite.

(3) Let G be the set of points on a line. Then G is infinite.

(A) 0 (B) 1 (C) 2 (D) 3

ST0054

17. How many of the following statements are incorrect ?

(1) Let A = $\{1,2,3,4\}$ and B = $\{3,1,4,2\}$. Then A = B.

(2) Let A be the set of prime numbers less than 6 and P the set of prime factors of 30. Then A and P are equal, since 2,3 and 5 are the only prime factors of 30 and also these are less than 6.

(3) A set does not change if one or more elements of the set are repeated. For example, the sets A = $\{1,2,3\}$ and B = $\{2,2,1,3,3\}$ are equal, since each element of A is in B and vice-verse. That is why we generally do not repeat any element in describing a set.

(A) 0 (B) 1 (C) 2 (D) 3

ST0055

18. How many of the following statements are correct.

P : All infinite sets cannot be described in the roster form.

Q : Let W be the set of the days of the week. Then W is finite

R : Let A = {x : 1 < x< 2, x is a natural number}. Then A is the empty set,

S : Let G be the set of points on a line. Then G is infinite

(A) 4 (B) 1 (C) 2 (D) 3

ST0056

19. How many of the following statements are correct ?

P. $A \subset B$ if $a \in A \Rightarrow a \in B$

Q. A is a subset of B if a is an element of A implies that a is also an element of B

R. ϕ is a subset of every set

S. Every set A is a subset of itself

(A) 4 (B) 1 (C) 2 (D) 3

ST0057

20. If $x + y = 1$ and $x^2 + y^2 = 2$ then the value of $x^4 + y^4$ equals

(A) 7 (B) 6 (C) $\frac{7}{2}$ (D) $\frac{19}{4}$

FM0032

21. If $x = 2^{1/3} - 2^{-1/3}$ then the value of $2x^3 + 6x$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

FM0033

22. Consider the operation "minus the reciprocal of, defind by "$a \diamond b = a - \frac{1}{b}$".

What is $((1\diamond 2)\diamond 3) - (1\diamond(2\diamond 3))$?

(A) $-\frac{7}{30}$ (B) $-\frac{1}{6}$ (C) 0 (D) $\frac{1}{6}$

ST0058

23. Pranshi has written down one integer two times and another integer three times. The sum of the five numbers is 100, and one of the numbers is 28. The other number is

(A) 6 (B) 8 (C) 10 (D) 12

ST0059

24. If $x,y \in \mathbb{N}$ and $xy - 3x - 2y = 54$, then number of ordered pairs (x,y) is

(A) 7 (B) 8 (C) 12 (D) 24

ST0060

25. How many ordered triples (x,y,z) of positive integers satisfy lcm(x,y) = 72, lcm(x,z) = 600 and lcm(y,z) = 900 ?

(A) 14 (B) 15 (C) 16 (D) 20

ST0061

EXERCISE (O-2)

Multiple Correct Answer Type

1. If U = {1,2,3,4,5,6,7,8}, A ={1,2,3,5,6} and B = {2,3,4,7,8} which of the following are correct

(A) $A \cup B = U$ (B) $B - A = \{1,5,6\}$

(C) $A' \cup B = \{2,3,4,7,8\}$ (D) $(A - B)' = \{2,3,4,7,8\}$

ST0062

2. If A = {1,2,3,4}, B = {3,4,5,6}, C = {5,6,7,8} and D = {7,8,9,10}. Which of the following option(s) is/are true ?

(A) $A \cup C = \{1,2,3,4,5,6,7,8\}$ (B) $B \cup D = \{3,4,5,6,7,8,9,10\}$

(C) $A \cup B \cup D = \{1,2,3,4,5,6,7,8,9,10\}$ (D) $B \cup C \cup D = \{3,4,5,6,7,8,9,10\}$

ST0063

3. Which of the following are correct?

(A) $[2, 10] - \{2, 10\} = (2, 10)$ (B) $[3, 5] - \{5\} = (3,5]$

(C) $(-3, 5] - \{-3\} = (3, 5]$ (D) $[-2, 15] - (2, 15] = [-2, 2]$

ST0064

4. If $A \subseteq B$ then which of the following option(s) is/are correct ?

(A) $A' \subseteq B'$ (B) $B' \subseteq A'$ (C) $A \cap B' = \phi$ (D) $A' \cap B = \phi$

FM0036

5. Which of the following intervals are subsets of [2, 3] ?

(A) $((0, 2.3] \cup (2.3, 3]) \cap (0, 2.3)$

(B) $((2, 2.5] \cup (2.5, 4]) \setminus (0, 2.3)$

(C) $((2, 2.5] \cup (2.5, 4]) \cap (0, 3)$

(D) $((0, 2.3] \cup (2.2, 3]) \setminus (0, 2.3)$

ST0065

6. Which of the following are correct ?

(A) [a,b) is an open interval from a to b, including a but excluding b.

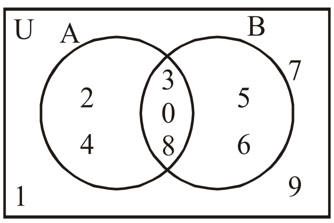
(B) (a,b] is an open interval from a to b, including b but excluding a.

(C) [a,b) is an open interval from a to b, including b but excluding a.

(D) (a,b] is an open interval from a to b, including a but excluding b.

ST0066

7. From the adjoining Venn-diagram, find which of the following are correct :



(A) $A' = \{1,5,6,7,9\}$ (B) $B' = \{1,2,4,7,9\}$

(C) $(A \cap B)' = \{1,2,4,5,6,7,9\}$ (D) $(A - B)' = \{1,2,4,5,6,7,9\}$

ST0067

8. Which of the following option(s) is/are true ?

(A) $A \cup A' = U$ (B) $A \subset B \Rightarrow A \cup B = B$

(C) $A \subset B \Rightarrow A \cap B = A$ (D) $A \cap \phi = \phi$

ST0068

9. Which of the following option(s) is/are true ?

(A) If $A \subset B$ and $A \neq B$, then A is called a proper subset of B

(B) If $A \subset B$, then B is called superset of A.

(C) $A = \{1,2,3\}$ is NOT a proper subset of $B = \{1,2,3,4\}$

(D) If a set A has only one element, we call it a singleton set.

ST0069

10. In which of the following options, $A \neq B$:

(A) $A = \{x : x + 2 = 3\}$, $B = \{x : x \in N$ and is less than $2\}$

(B) $A = \{x : x \in N$ and $3x - 1 < 2\}$, $B = \{x : x \in W$ and $3x - 1 < 2\}$

(C) $A = \{x : x \in N$ and is prime factor of $36\}$, $B = \{1,2,3,4,6,9,12\}$

(D) $A = \{x : \in I$ and $x^2 \leq 4\}$, $B = \{x : x \in R$ and $x^2 - 3x + 2 = 0\}$

ST0070

11. Which of the following are examples of the null set

(A) Set of odd natural numbers divisible by 2

(B) Set of even prime numbers

(C) $\{x : x$ is a natural numbers $x < 5$ and $x > 7\}$

(D) $\{y : y$ is a point common to any two parallel lines$\}$

ST0071

12. Consider the set ϕ, $A = \{1,3\}$, $B = \{1,5,9\}$, $C = \{1,3,5,7,9\}$. Which of the following statements are correct ?

(A) $\phi \subset B$ as ϕ is a subset of every set.

(B) $A \not\subset B$ as $3 \in A$ and $3 \notin B$

(C) $A \subset C$ as $1,3 \in A$ also belongs to C

(D) $B \subset C$ as each element of B is also an element of C.

ST0072

13. Which of the following options are correct ?

(A) The set Q of rational numbers is a subset of the set R of real numbers, and we write $Q \subset R$.

(B) If A is the set of all divisor of 56 and B the set of all prime divisors of 56, then B is a subset of A and we write $B \subset A$.

(C) Let $A = \{1,3,5\}$ and $B = \{x : x$ is an odd natural number less than $6\}$. Then $A \subset B$ and $B \subset A$ and hence $A = B$.

(D) Let $A = \{a,e,i,o,u\}$ and $B = \{a,b,c,d\}$. Then A is not a subset of B, also B is not a subset of A.

ST0073

14. Which of the following options are incorrect

(A) {2,3,4,5} and {3,6} are disjoint sets

(B) {a,e,i,o,u} and {a,b,c,d} are disjoint sets

(C) {2,6,10,14} and {3,7,11,15} are disjoint sets

(D) {2,6,10} and {3,7,11} are disjoint sets

ST0074

15. Consider the following sets.

$S_1 = \{x|x \in \mathbb{R}, 0 < x < 10\}$

$S_2 = \{x|x \in \mathbb{Z}, -10 \le x \le 10\}$

$S_3 = \{x|x \in \mathbb{Q}, 10 \le x \le 15\}$

Choose the set of correct options.

(A) $(S_2 \cap S_3) \cup (S_3 \cap S_1) = \{10\}$

(B) $(S_2 \backslash S_1) \cap (S_3 \backslash S_1) = \phi$

(C) The cardinality of the set $S_1 \cap S_2$ is 10

(D) The cardinality of the set $(S_1 \cap S_2) \cup (S_2 \cap S_3)$ is 10

ST0075

16. Let A, B, C $\subseteq$ X. Which of the following option(s) is/are true ?

(A) $[A\backslash(A \cap B)] \cup [B\backslash(A \cap B)] \cup (A \cap B) = A \cup B$

(B) $[(A \cup B)^c \cup B] \cap [B^c \cup (A \cup B)] = (A\backslash B)^c$

(C) $[(A \cup B) \cap (A \cap C)] \cap B^c = (A\backslash B)]\backslash C^c$

(D) $[A \cap (B^c \cap A^c)^c]^c \cup A = X$

ST0076

17. Let A, B, C, D $\subseteq$ X. Which of the following option(s) is/are true ?

(A) $(C \cup B)^c\backslash(A \cap (B\backslash C)) = B^c \cap C^c$

(B) $A \cup [(A \cup (A \cup B)) \cap (B \cup (A \cap B))] = A \cup B$

(C) $C^c \cup [D^c \cup (D\backslash C^c)^c] = C^c \cup D^c$

(D) $[(A \cup B) \cup B^c] \cap [B^c \cap (A \cup B)] = A\backslash B$

ST0077

18. Let A be a set and let $B = \{(a, S) \in A \times P(A) : a \in S\}$

For which of the following option(s) B is not an empty set :

(A) A = {1} (B) A = {1, 2} (C) A = {1, 2, 3} (D) None of these

ST0078

19. If complex number $-3 + ix^2y$ and $x^2 + y + 4i$ are conjugate of each other. Then real value of x & y can be

(A) $x = 1, y = -4$ (B) $x = -1, y = -4$ (C) $x = 1, y = 4$ (D) $x = -2, y = -3$

FM0038

20. If a mod 3 = 1 and b mod 3 = 2, where $a, b \in \mathbb{N}$, then choose the set of correct options.

(a mod b = c, a, b, c $\in \mathbb{N}$; means a = nb + c where n $\in \mathbb{N}$)

(A) (ab) mod 3 = 0 (B) (ab) mod 3 = 2

(C) (a + b) mod 3 = 0 (D) (a + b) mod 2 = 1

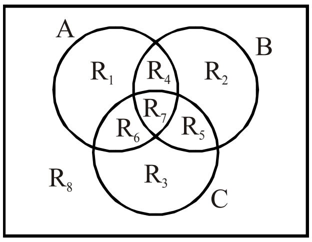
ST0079

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for 1 TO 3

R_i denotes region for i = 1, 2, ..., 8 as shown in figure.



1. Which of the following option(s) is\are true ?

(A) $R_1 = A \cap (B \cup C)'$ (B) $R_2 = B \cap (A' \cap C')$

(C) $R_3 = (A \cup B)'$ (D) $R_8 = (A \cup B \cup C)'$

ST0081

2. Which of the following option(s) is\are true ?

(A) $R_4 = (A \cap B) \cap C'$ (B) $R_5 = (B \cap C)\backslash A$

(C) $R_6 = (A \cap C) \cap (A \cap B \cap C)'$ (D) $R_8 = A' \cap B' \cap C'$

ST0081

3. Which of the following option(s) is/are true ?

(A) $R_7 = A \cap B \cap C$ (B) $R_7 = (A' \cup B' \cup C')'$

(C) $R_3 = (A' \cap B') \cap C$ (D) $R_5 = (B\backslash C') \cap A'$

ST0082

Paragraph for 4 TO 6

Let S be the universal set, A, B & C are sets such that $|(A \cup B \cup C)'| = 1$ & $|S| = 8$.

4. If $|A \cap B \cap C|$ is maximum then which of the following option(s) is\are true ?

(A) $|A| = |B| = |C|$ (B) $A \cap B' = B \cap A' = A \cap C' = A' \cap C$

(C) $A\backslash B = (A \cup B \cup C) - (A \cap B \cap C)$ (D) $A' = S\backslash B$

ST0083

5. If $|A \cap B \cap C| = 6$

(A) If $A = B$, then $|C| = 7$

(B) If $A = B$ and $A\backslash C \neq \phi$, then $|C| = 6$

(C) If $A = B$ and $A\backslash C \neq \phi$, then $|A \cap B| = 7$

(D) If $A = B = C$ then $|A \cup B \cup C| = 7$

ST0084

6. If $|A| = |B| = |C|$, then which of the following option(s) is\are true ?

(A) Sum of minimum and maximum value of $|A \cap B \cap C|$ is 7.

(B) Let X be the set of all possible values of cardinality of $A \cap B \cap C$, then $|X| = 7$.

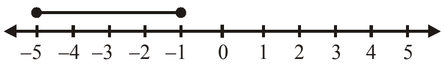
(C) Let X be the set of all possible values of cardinality of $A \cap B \cap C$, then $|X| = 6$.

(D) If $|A| = 4$, then $|A \cap B \cap C| \in \{0, 1, 2\}$.

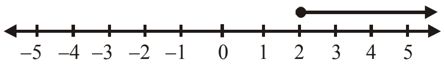
ST0085

Matrix Match Type

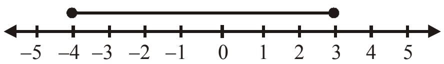
7. **Column-I** **Column-II**

(A) 

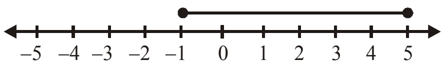
(P) $-1 \le x \le 5;\ x \in R$

(B) 

(Q) $-5 \le x \le -1;\ x \in R$

(C) 

(R) $x \ge 2;\ x \in R$

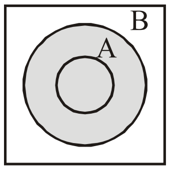
(D) 

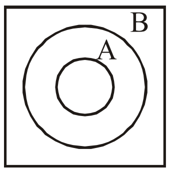
(S) $-4 \le x \le 3;\ x \in R$

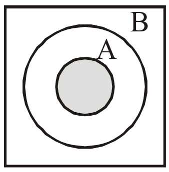
ST0086

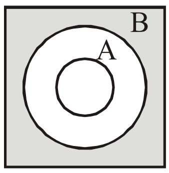
8. Given sets A and B; such that $A \subseteq B$. Match theVenn-diagrams in column-II with the sets I column-I.

Column-I **Column-II**

(A) $A \cup B$ (P) 

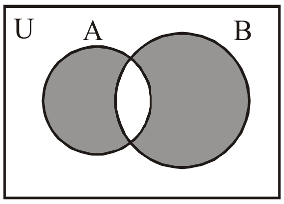
(B) $B' \cap A$ (Q) 

(C) $A \cap B$ (R) 

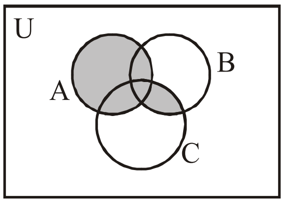
(D) $(A \cup B)'$ (S) 

ST0087

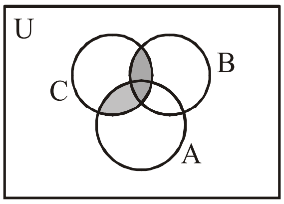
9. **Column-I** **Column-II**

(A) 

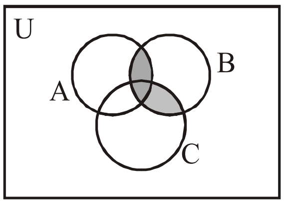
(P) $A \cup (B \cap C)$

(B) 

(Q) $(A - B) \cup (B - A)$

(C) 

(R) $(A \cap C) \cup (B \cap C)$

(D) 

(S) $(A \cup C) \cap B$

ST0088

10. Match the following sets in set builder form in column-I with roster form in column-II

Column-I	Column-II
(A) $\left\{x : x \in I, -\frac{1}{2} < x < \frac{9}{2}\right\}$	(P) $\{1,2,3,4\}$
(B) $\{x : x \in N \text{ and } 4x - 1 \le 15\}$	(Q) $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, ...\right\}$
(C) $\left\{x : x = \frac{n}{n+1}, n \in N\right\}$	(R) $\left\{\frac{1}{2}, \frac{3}{4}, \frac{5}{6}, ...\right\}$
(D) $\left\{x : x = \frac{2n-1}{2n}, n \in N\right\}$	(S) $\{0,1,2,3,4\}$

ST0089

EXERCISE (O-4)

Numerical Grid Type

1. An investigator interviewed 100 students to determine their preferences for the three drinks : milk (M), coffee (C) and tea (T). He reported the following : 10 students had all the three drinks M, C and T; 20 had M and C; 30 had C and T; 25 had M and T; 12 had M only; 5 had C only; and 8 had T only. Using a Venn diagram find how many did not take any of the three drinks.

FM0058

2. A survey shows that 63% of the Americans like cheese whereas 76% like apples. If x% of the Americans like both cheese and apples, number of integral values of x is.

FM0101

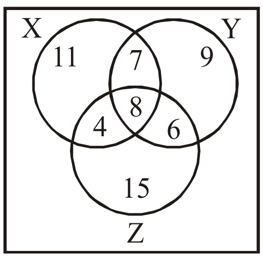
3. How many ordered pairs of integers which satisfy the equation $\frac{1}{m}+\frac{2}{n}=\frac{1}{2}$?

FM0102

4. Find the smallest natural number that leaves the remainders 3 and 4 when divided by 5 and 6 respectively.

ST0090

5. Suppose there are 100 people in a city C. Let X be the set of people who are engineers in the city C, Y be the set of people whose age is above 40 years in the city C, Z be the set of people who has an independent house in the city C. Now, study the following Venn diagram and answer the question given below.



Find the number of engineers who do not have an independent house ?

ST0091

6. Consider a set S = {a | a ∈ ℕ, a ≤ 18}. Let R_1 and R_2 be sets of ordered pairs, defined as $R_1 = \{(x, y) \mid x, y \in S, y = 3x\}$ and $R_2 = \{(x, y) \mid x, y \in S, y = x^2\}$. Find the cardinality of the set $R_1 \backslash (R_1 \cap R_2)$.

ST0092

7. If A = {4, 5, 6}, B = {4, 7, 8} and C = {4, 6, 7}, then find the cardinality of $(A - B) \times (B \cap C)$

ST0093

8. In DDPS – ALLEN, out of 400 students, 300 students opt for at least one subject out of Biology, Chemistry and Mathematics. If 120 students opted for Biology, 150 students opted for Chemistry, 200 students opted for Mathematics and 30 opted for all the three subjects, find the number of students who opted for exactly two subjects.

ST0094

9. Let M be the set of positive integers less than or equal to 2000 such that the difference of any two numbers in M is neither 5 nor 8. Find the maximum number of elements in M.

ST0095

10. In a Zoo, there are 6 Bengal white tigers and 6 Bengal royal tigers. Out of these tigers, 5 are males and 10 are either Bengal royal tigers or males. Find the number of female Bengal white tigers in the Zoo.

ST0096

EXERCISE (JM)

1. If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then :-

(1) $B = C$ (2) $A \cap B = \phi$ (3) $A = B$ (4) $A = C$

FM0112

2. In a class of 140 students numbered 1 to 140, all even numbered students opted mathematics course, those whose number is divisible by 3 opted Physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is :

(1) 102 (2) 42 (3) 1 (4) 38

FM0113

3. Two newspapers A and B are published in a city. It is known that 25% of the city populations reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisement is :-

(1) 12.8 (2) 13.5 (3) 13.9 (4) 13

FM0114

4. Let $\mathbb{Z}$ be the set of integers. If $A = \left\{x \in \mathbb{Z} : 2^{(x+2)(x^2-5x+6)} = 1\right\}$ and $B = \{x \in \mathbb{Z} : -3 < 2x - 1 < 9\}$, then the number of subsets of the set $A \times B$, is:

(1) 2^{18} (2) 2^{10} (3) 2^{15} (4) 2^{12}

FM0115

EXERCISE (JA)

1. If X and Y are two sets, then $X \cap (X \cup Y)^c$ equals

(a) X (b) Y (c) ϕ (d) none of these

FM0116

2. If A = {1, 2, 3, 4, 5}, then the number of proper subsets of A is-

(1) 120 (2) 30 (3) 31 (4) 32

FM0120

3. If A, B, C be three sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$, then -

(1) A = B (2) B = C (3) A = C (4) A = B = C

FM0121

4. Sets A and B have 3 and 6 elements respectively. What can be the minimum number of elements in $A \cup B$?

(1) 3 (2) 6 (3) 9 (4) 18

FM0122

ANSWER KEY

Do yourself-1

1. All are true **2.** C 3. D

4. (A-P, B-R, C-Q, D-S) **5.** (A-P, B-S, C-Q, D-R) **6.** 8 **7.** 2

8. 7 **9.** (A-iii-I ; B-i-II; C-ii-III, D-iv-IV)

10. (C) **11.** (3) **12.** 2 **13.** **3**

Do yourself-2

2. (A-R, B-P, C-Q, D-S) **3.** 2 **4.** 4

5. (A)→(S); (B)→(P); (C)→(Q); (D)→(P) **6.** 2 **7.** 4

Do yourself-3

2. C **3.** B **4.** B **5.** B **6.** C **7.** C **8.** D **9.** B

10. D **11.** B **12.** B **13.** P-30, Q-19, R-60,S-No. **14.** 4 **15.** 2

16. 2 **17.** 8 **18.** B **19.** A, B. D **20.** A

Do yourself-4

1. a = 3, b = 4

2. **(i)** $\frac{2}{11}$ **(ii)** $\frac{16}{99}$ **(iii)** $\frac{419}{990}$

4. **(i)** x + y, x – y, xy, x/y are rational numbers.

(ii) x + y, x – y, xy, x/y, all are may or may not be rational.

(iii) x + y, x – y, xy, x/y are irrational numbers.

6. x is irrational number (Non terminating and non-recurring)

Do yourself-5

1. (x,y) ≡ (1,20); (2,10); (4,5); (5,4); (10,2); (20,1)

2. **(i)** (x,y) ≡ (2,2) **(ii)** (x,y) ≡ (5,13); (6,4) ; (7,1)

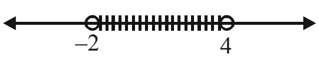
3. (x,y) ≡ (8,56); (14,14); (56,8)

4. 651 **5.** 293 **6.** 4 **7.** 30 **8.** 3

Do yourself-6

(i) 

(ii) 

(iii) 

(iv) 

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	B	A	A	A	B	C	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	C	C	D	D	A	A	A	C
Que.	21	22	23	24	25					
Ans.	C	A	B	C	B					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C,D	A,D	B,C	C,D	A,B	A,B,C	A,B,C,D	A,B,D	B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C,D	A,B,C,D	A,B,C,D	A,B	A,D	A,B,C,D	A,B,C,D	A,B,C	A,B	B,C

EXERCISE # O-3

Que.	1	2	3	4	5	6				
Ans.	A,B,D	A,B,C,D	A,B,C,D	A,B,C,D	B,C,D	A,B,D				
Q.7	A	B	C	D	Q.8	A	B	C	D	
	Q	R	S	P		P	Q	R	S	
Q.9	A	B	C	D	Q.10	A	B	C	D	
	Q	P	R	S		S	P	Q	R	

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	20	25	7	28	18	5	4	110	770	2

EXERCISE # JEE-MAIN

Que.	1	2	3	4
Ans.	1	4	3	3

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4
Ans.	C	3	2	2

FUNDAMENTALS OF ALGEBRA

1. INDICES AND SURDS

1.1 Indices

Definition of Indices :

If 'a' is any non zero real or imaginary number and 'm' is a positive integer, then $a^m = a.\ a.\ a....a$ (m times). Here a is called the base and m is the index, power or exponent.

Law of indices :

(i) $a^0 = 1, (a \neq 0)$

(ii) $a^{-m} = \dfrac{1}{a^m}, (a \neq 0)$

(iii) $a^m.a^n = a^{m+n}$

(iv) $\dfrac{a^m}{a^n} = a^{m-n}, a \neq 0$

(v) $(a^m)^n = a^{mn} = (a^n)^m$

(vi) $\sqrt[q]{a^p} = a^{p/q}, \ q \in \mathbb{N}$ and $q \geq 2$

(vii) If x = y, then $a^x = a^y$, but the converse may not be true. e.g. : $(1)^6 = (1)^8$, but $6 \neq 8$

For $a^x = a^y$ we have following possibilities

- If $a \neq \pm 1, 0$, then x = y
- If a = 1, then x, y may be any real number
- If a = –1, then x, y may be both even or both odd
- If a = 0, then x, y may be any positive real number

But if we have to solve the equations like $\left(f\left(x\right)\right)^{g(x)} = \left(f\left(x\right)\right)^{h(x)}$ (i.e same base, different indices) then we have to solve :

(a) f(x) = 1 (b) f(x) = –1 (c) f(x) = 0 (d) g (x) = h(x)

Verification should be done in (b) and (c) cases

Illustration-1 : Solve $(4-x)^{x^3-4x} = (4-x)^{(x^2-4)(x-1)}$

Solution : **Case-1 :** $4 - x = 1 \Rightarrow x = 3$

Case-2 : $4 - x = -1 \Rightarrow x = 5$

for $x = 5$, $x^3 - 4x$ is odd

and $(x^2 - 4)(x - 1)$ is even

so $x = 5$ does not satisfy the given equation

Case-3 : $4 - x = 0 \Rightarrow x = 4$

For $x = 4$, $x^3 - 4x > 0$ and $(x^2 - 4)(x - 1) > 0$

$\Rightarrow x = 4$ satisfies the given equation

Case-4 : $x^3 - 4x = (x^2 - 4)(x - 1)$

$\Rightarrow x^3 - 4x = x^3 - x^2 - 4x + 4$

$\Rightarrow x^2 - 4 = 0$, $\Rightarrow x = \pm 2$ which satisfies the given equation.

From case-1, case-2, case-3 and case-4, solutions set of the given equation is $\{-2, 2, 3, 4\}$

(viii) If $a^x = b^x$ then consider the following cases :

- If $a \neq \pm b$, then $x = 0$
- If $a = b \neq 0$, then x may have any real value for which a^x is well defined.
- If $a = -b \neq 0$, then x is even.
- If $a = b = 0$, then x can be any positive real.

If we have to solve the equation of the form $[f(x)]^{h(x)} = [g(x)]^{h(x)}$, i.e., same index, different bases, then we have to solve :

(a) $f(x) = g(x)$ (b) $f(x) = -g(x)$ (c) $h(x) = 0$

Verification should be done in (a), (b) and (c) cases.

Illustration-2: Solve $(x^2 - 4)^{2x} = (x^2 + 2x)^{2x}$.

Solution : **Case-1 :** when $x = 0$

Base : $x^2 + 2x = 0$, so $x = 0$ does not satisfy given equation

Case-2 : $x^2 - 4 = x^2 + 2x \neq 0$

$\Rightarrow x \in \phi$

Case-3 : $x^2 - 4 = -x^2 - 2x \neq 0 \Rightarrow 2x^2 + 2x - 4 = 0$

$x^2 + x - 2 = 0 \Rightarrow x = 1$

satisfies the given equation

Case-4 : $x^2 - 4 = x^2 + 2x \Rightarrow x = -2$

$x = -2$ does not satisfy the given equation.

From all above cases, $x \in \{1\}$

Illustration-3: If $a^x = b$, $b^y = c$, $c^z = a$, prove that $xyz = 1$, where a,b,c are distinct numbers.

Solution : We have, $a^{xyz} = (a^x)^{yz}$

$\Rightarrow$ $a^{xyz} = (b)^{yz}$ $[\because a^x = b]$

$\Rightarrow$ $a^{xyz} = (b^y)^z$

$\Rightarrow$ $a^{xyz} = c^z$ $[\because b^y = c]$

$\Rightarrow$ $a^{xyz} = a$ $[\because c^z = a]$

$\therefore$ $a^{xyz} = a^1$

$\Rightarrow$ $xyz = 1$

1.2 Surds

If 'x' is a rational number, which is not the n^{th} power $(n \in \mathbb{N}\backslash\{1\})$ of any rational number, then the number $x^{1/n}$ usually denoted by $\sqrt[n]{x}$ is called surd. The sign '$\sqrt[n]{\ }$' is called the radical sign. The number in the angular part of the sign, i.e., 'n' is called order of the surd. In case of n = 2 the expression $\sqrt[2]{x}$, simply written as $\sqrt{x}$.

Note :

- If $\sqrt[n]{x}$ is a surd then $-\left(\sqrt[n]{x}\right)$ is also a surd.
- Every surd is an irrational number (but every irrational number is not a surd).
- To rationalize the denominator of a fraction of the form $\frac{a}{\sqrt{b}}$, multiply the numerator and denominator of the fraction by $\sqrt{b} \Rightarrow \frac{a}{\sqrt{b}} = \frac{a}{\sqrt{b}}.\frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{\sqrt{b^2}} = \frac{a\sqrt{b}}{b}$.

Eg.

(a) $\sqrt{3}$ is a surd and $\sqrt{3}$ is an irrational number.

(b) $\sqrt[3]{5}$ is surd and $\sqrt[3]{5}$ is an irrational number.

(c) π is an irrational number, but it is not a surd.

1.2.1 Conjugate of a Surd

If two binomial surds (surds containing two terms such as $2+\sqrt{3}$, $2\sqrt{5}-\sqrt{7}$ etc) are such that only the sign connecting the individual terms are different, then they are said to be conjugate of each other. If these surds are quadratic, then their product would always be rational. So in case of a binomial quadratic surd, we use its conjugate as its rationalizing factor.

Ex. Conjugate of $3\sqrt{2}+\sqrt{5}$ is $3\sqrt{2}-\sqrt{5}$ or $-3\sqrt{2}+\sqrt{5}$

Illustration-4 : Rationalize the denominator of $\dfrac{1}{3\sqrt{2}+\sqrt{5}}$

Solution : A conjugate of $3\sqrt{2}+\sqrt{5}$ is $3\sqrt{2}-\sqrt{5}$

Therefore multiplying the conjugate in the numerator and denominator of the given fraction.

$$\frac{3\sqrt{2}-\sqrt{5}}{\left(3\sqrt{2}+\sqrt{5}\right)\left(3\sqrt{2}-\sqrt{5}\right)}$$

$$=\frac{3\sqrt{2}-\sqrt{5}}{\left(3\sqrt{2}\right)^2-\left(\sqrt{5}\right)^2}$$

$$=\frac{3\sqrt{2}-\sqrt{5}}{18-5}$$

$$=\frac{3\sqrt{2}-\sqrt{5}}{13}$$

Illustration-5 : Using the fact that $\left(\sqrt{x}+\sqrt{y}\right)^2 = x+y+\sqrt{xy}$ to find the square root of $7+2\sqrt{10}$. If this number can be expressed in the form $\sqrt{a}+\sqrt{b}$, where a ≤ b, find the value of b – a.

(A) 3 (B) 4 (C) 0 (D) 2

Solution : Let $\sqrt{7+2\sqrt{10}} = \sqrt{a}+\sqrt{b}$, where $a,b \in \mathbb{Q}$

$\Rightarrow 7+2\sqrt{10} = a+b+2\sqrt{ab}$

$\Rightarrow$ a + b = 7 and $\sqrt{ab} = \sqrt{10}$

$(b-a)^2 = (a+b)^2 - 4ab$

$= 49 - 40 = 9$

$\Rightarrow \sqrt{(b-a)^2} = 3 \;\Rightarrow\; b-a = 3$ (since b > a)

OR

$$7+2\sqrt{10}=7+2\cdot\sqrt{2}\cdot\sqrt{5}=\left(\sqrt{2}\right)^2+\left(\sqrt{5}\right)^2+2\cdot\sqrt{2}\cdot\sqrt{5}=\left(\sqrt{2}+\sqrt{5}\right)^2$$

So $\sqrt{7+2\sqrt{10}}=\sqrt{2}+\sqrt{5}\ \therefore\ b-a=3$

Illustration-6 : If $x=\dfrac{1}{2+\sqrt{3}}$, find the value of $x^3-x^2-11x+4$.

Solution : as $x=\dfrac{1}{2+\sqrt{3}}\times\dfrac{2-\sqrt{3}}{2-\sqrt{3}}=\dfrac{2-\sqrt{3}}{(2)^2-(\sqrt{3})^2}$

$x=\dfrac{2-\sqrt{3}}{4-3}=2-\sqrt{3}$

$x-2=-\sqrt{3}$, squaring both sides, we get

$(x-2)^2=\left(-\sqrt{3}\right)^2 \Rightarrow x^2+4-4x=3$

$\Rightarrow x^2-4x+1=0$

Now $x^3-x^2-11x+4$

$=x^3-4x^2+x+3x^2-12x+4$

$=x(x^2-4x+1)+3(x^2-4x+1)+1=x\times 0+3(0)+1=0+0+1=1$

Illustration-7 : If $x=3-2\sqrt{2}$, find $x^2+\dfrac{1}{x^2}$

Solution : We have, $x=3-2\sqrt{2}$.

$\therefore\ \dfrac{1}{x}=\dfrac{1}{3-2\sqrt{2}}=\dfrac{1}{3-2\sqrt{2}}\times\dfrac{3+2\sqrt{2}}{3+2\sqrt{2}}$

$=\dfrac{3+2\sqrt{2}}{(3)^2-(2\sqrt{2})^2}=\dfrac{3+2\sqrt{2}}{9-8}=3+2\sqrt{2}$

Thus, $x^2+\dfrac{1}{x^2}=(3-2\sqrt{2})^2+(3+2\sqrt{2})^2$

$=2((3)^2+(2\sqrt{2})^2)=2(9+8)=34$

Illustration-8 : Rationalise the denominator of $\dfrac{1}{\sqrt{3}-\sqrt{2}-1}$

Solution : $\dfrac{1}{\sqrt{3}-\sqrt{2}-1}=\dfrac{1}{\sqrt{3}-\sqrt{2}-1}\times\dfrac{\sqrt{3}+\sqrt{2}+1}{\sqrt{3}+\sqrt{2}+1}$

$$=\frac{\sqrt{3}+\sqrt{2}+1}{(\sqrt{3}-\sqrt{2}-1)(\sqrt{3}+\sqrt{2}+1)}=\frac{\sqrt{3}+\sqrt{2}+1}{(\sqrt{3})^2-(\sqrt{2}+1)^2}=\frac{\sqrt{3}+\sqrt{2}+1}{-2\sqrt{2}}=-\left(\frac{\sqrt{6}+\sqrt{2}+2}{4}\right)$$

Illustration-9 : Evaluate the following

(i) $\left(\sqrt[3]{64}\right)^{\frac{-1}{2}}$ (ii) $\left(\dfrac{121}{169}\right)^{-3/2}$

Solution : (i) $\left(\sqrt[3]{64}\right)^{\frac{-1}{2}} = \left[(64)^{\frac{1}{3}}\right]^{\frac{-1}{2}} = (64)^{\frac{1}{3}\times\frac{-1}{2}} = (64)^{\frac{-1}{6}}$

$= (2^6)^{\frac{-1}{6}} = 2^{6\times\left(\frac{-1}{6}\right)} = 2^{-1} = \dfrac{1}{2}$

(ii) $\left(\dfrac{11\times 11}{13\times 13}\right)^{-3/2} = \left(\dfrac{11^2}{13^2}\right)^{-3/2} = \left(\dfrac{11}{13}\right)^{2\times\frac{-3}{2}} = \left(\dfrac{11}{13}\right)^{-3} = \left(\dfrac{13}{11}\right)^{3} = \dfrac{2197}{1331}$

Illustration 10 : Simplify $\left[\sqrt[3]{\sqrt[6]{a^9}}\right]^4\left[\sqrt[6]{\sqrt[3]{a^9}}\right]^4$

(A) a^{16} (B) a^{12} (C) a^8 (D) a^4

Solution. $a^{9(1/6)(1/3)4}\cdot a^{9(1/3)(1/6)4} = a^2\cdot a^2 = a^4$.

Illustration 11 : Simplify $a\left(\dfrac{\sqrt{a}+\sqrt{b}}{2b\sqrt{a}}\right)^{-1} + b\left(\dfrac{\sqrt{a}+\sqrt{b}}{2a\sqrt{b}}\right)^{-1}$

Solution. The given expression is equal to

$$a\left(\frac{2b\sqrt{a}}{\sqrt{a}+\sqrt{b}}\right) + b\left(\frac{2a\sqrt{b}}{\sqrt{a}+\sqrt{b}}\right)$$

$$= \frac{2ab}{\sqrt{a}+\sqrt{b}}\left(\sqrt{a}+\sqrt{b}\right) = 2ab$$

Illustration 12 : Evaluate $\sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{7+\sqrt{48}}}}$

Solution. $\sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{7+\sqrt{48}}}}$

$$= \sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{4+3+2\sqrt{12}}}} = \sqrt{3+\sqrt{3}+\sqrt{2+\sqrt{3}+\sqrt{4}+\sqrt{3}}}$$

$$= \sqrt{3+\sqrt{3}+\sqrt{3}+1} = \sqrt{4+2\sqrt{3}} = \sqrt{3}+1 = \sqrt{3}+1$$

Illustration 13 : Find rational numbers a and b, such that $\dfrac{4+3\sqrt{5}}{4-3\sqrt{5}} = a + b\sqrt{5}$

Solution. $\dfrac{4+3\sqrt{5}}{4-3\sqrt{5}} \times \dfrac{4+3\sqrt{5}}{4+3\sqrt{5}} = a + b\sqrt{5}$

$$\frac{61+24\sqrt{5}}{-29} = a + b\sqrt{5}$$

$$a = -\frac{61}{29},\ b = -\frac{24}{29}$$

Do yourself-1 :
Only One Option is Correct for Q.1 to Q.5

1. $\left(\left((625)^{-1/2}\right)^{-1/4}\right)^{2} =$

(A) 4　(B) 5　(C) 2　(D) 3

2. $\left(5\left(8^{1/3}+27^{1/3}\right)^{3}\right)^{1/4} =$

(A) 3　(B) 6　(C) 5　(D) 4

3. $\left\{\sqrt[4]{\left(\dfrac{1}{x}\right)^{-12}}\right\}^{-2/3} =$

(A) $\dfrac{1}{x^2}$　(B) $\dfrac{1}{x^4}$　(C) $\dfrac{1}{x^3}$　(D) $\dfrac{1}{x}$

4. $\dfrac{\sqrt{x^3} \times \sqrt[3]{x^5}}{\sqrt[5]{x^3}} \times \sqrt[30]{x^{77}} =$

(A) $x^{76/15}$　(B) $x^{78/15}$　(C) $x^{79/15}$　(D) $x^{77/15}$

5. If $\sqrt[4]{\sqrt[3]{x^2}} = x^k$, then k =

(A) $\dfrac{2}{6}$　(B) 6　(C) $\dfrac{1}{6}$　(D) 7

6. Simplify :

(i) $\dfrac{\left(5\sqrt{3}+\sqrt{50}\right)\left(5-\sqrt{24}\right)}{\left(\sqrt{75}-5\sqrt{2}\right)}$　(ii) $\dfrac{3\sqrt{2}}{\sqrt{6}+\sqrt{3}} - \dfrac{4\sqrt{3}}{\sqrt{6}+\sqrt{2}} + \dfrac{\sqrt{6}}{\sqrt{3}+\sqrt{2}}$

(iii) $\dfrac{\sqrt{\left[6+2\sqrt{3}+2\sqrt{2}+2\sqrt{6}\right]-1}}{\sqrt{5+2\sqrt{6}}}$ (iv) $\dfrac{2.3^{n+1}-7.3^{n-1}}{3^{n+1}+2\left(\frac{1}{3}\right)^{1-n}}$

(v) $\left(\frac{1}{3}\right)^{-10}.27^{-3}+\left(\frac{1}{5}\right)^{-4}.(25)^{-2}+\left(64^{-\frac{1}{9}}\right)^{-3}$

7. If $\dfrac{\left(2^{n+1}\right)^{m}\left(2^{2n}\right)2^{n}}{\left(2^{m+1}\right)^{n}\left(2^{2m}\right)}=1$, then find the value of $\dfrac{m}{n}$

8. Find rational numbers a and b, such that $\dfrac{2+3\sqrt{5}}{1-3\sqrt{5}}=a+b\sqrt{5}$

9. The square root of $11+\sqrt{112}$ is $a+\sqrt{b}$, $a,b \in \mathbb{N}$ then b – a is

10. $\sqrt{11+\sqrt{21}}=\sqrt{\dfrac{a}{b}}+\sqrt{\dfrac{c}{d}}$, where a, b, c and d are natural numbers with gcd(a, b) = gcd(c, d) = 1. Find a + b + c + d.

11. For $x \neq 0$ find the value of $\left(\dfrac{x^{\ell}}{x^{m}}\right)^{\ell^2+\ell m+m^2}.\left(\dfrac{x^{m}}{x^{n}}\right)^{m^2+mn+n^2}.\left(\dfrac{x^{n}}{x^{\ell}}\right)^{n^2+n\ell+\ell^2}$

12. For $a^x = (x + y + z)^y$, $a^y = (x + y + z)^z$, $a^z = (x + y + z)^x$, then find the values of x,y and z. Where $a > 0$ and $a \neq 1$.

2. FACTORIZATIONS

2.1 $a^2 - b^2 = (a - b)(a + b)$

Illustration-14 : $(3x - y)^2 - (2x - 3y)^2$

Solution : Use $a^2 - b^2 = (a - b)(a + b)$

$(3x - y)^2 - (2x - 3y)^2 = (3x - y + 2x - 3y)(3x - y - 2x + 3y) = (5x - 4y)(x + 2y)$

2.2 Factorising the Quadratic expression

Illustration-15 : $x^2 + 6x - 187$

Solution : $x^2 + 6x - 187 = x^2 + 17x - 11x - 187$

$= x(x + 17) - 11(x + 17)$

$= (x + 17)(x - 11)$

2.3 Factorisation by converting the given expression into a perfect square.

Illustration-16 : $9x^4 - 10x^2 + 1$

Solution : $9x^4 - 10x^2 + 1 = (3x^2)^2 - 2\cdot3x^2 + 1 - 4x^2$

$= (3x^2 - 1)^2 - (2x)^2$

$= (3x^2 - 1 - 2x)(3x^2 - 1 + 2x)$

$= (x - 1)(3x + 1)(x + 1)(3x - 1)$

2.4 $a^3 \pm b^3 \equiv (a \pm b)(a^2 \mp ab + b^2)$

Illustration-17 : $a^6 - b^6$

Solution : $a^6 - b^6 = (a^2)^3 - (b^2)^3$

$= (a^2 - b^2)\ (a^4 + a^2b^2 + b^4)$

$= (a - b)\ (a + b)\ (a^2 - ab + b^2)\ (a^2 + ab + b^2)$

2.5 Using Factor Theorem :

Illustration-18 : $x^3 - 13x - 12$

Solution : As $x = -1$ makes given expression 0, $x + 1$ is a factor

$$
\begin{array}{r|l|l}
x+1 & x^3 - 13x - 12 & x^2 - x - 12 \\
 & \underline{x^3 + x^2} & \\
 & -x^2 - 13x - 12 & \\
 & \underline{-x^2 - x} & \\
 & -12x - 12 & \\
 & \underline{-12x - 12} & \\
 & 0 &
\end{array}
$$

$\therefore\ x^3 - 13x - 12 = (x + 1)\ (x^2 - x - 12)$

$= (x + 1)\ (x - 4)\ (x + 3)$

2.6 $a^3 + b^3 + c^3 - 3abc = (a + b + c)\ (a^2 + b^2 + c^2 - ab - bc - ac)$

Illustration-19 : $8x^3 + y^3 + 27z^3 - 18xyz$

Solution : $8x^3 + y^3 + 27z^3 - 18xyz = (2x)^3 + (y)^3 + (3z)^3 - 3(2x)\ (y)\ (3z)$

$= (2x + y + 3z)\ (4x^2 + y^2 + 9z^2 - 2xy - 6xz - 3yz)$

2.7 Cyclic Expression and its Factorization :

An expression is said to be cyclic with regard to the variables $x_1, x_2, ..., x_n$ arranged in this order, when it is unchanged by changing x_1 into x_2, x_2 into x_3, x_3 into x_4, ..., x_n into x_1.

For three variables

$E(x, y, z)$ is cyclic if $E(x, y, z) = E(y, z, x) = E(z, x, y)$

Ex. 1. $x + y + z$ is a cyclic expression

2. $x^2 + y^2 + z^2 + xy + yz + zx$ is a cyclic expression

3. $E(x, y, z) = x(y - z) + y(z - x) + z(x - y)$ is a cyclic expression because

$\Rightarrow E(y, z, x) = y(z - x) + z(x - y) + x(y - z)$

and $E(z, x, y) = z(x - y) + x(y - z) + y(z - x)$

$\Rightarrow E(x, y, z) = E(y, z, x) = E(z, x, y)$

Theorem :

If $E(x, y, z)$ is a cyclic expression and $x - y$ is a factor of $E(x, y, z)$ then $y - z$ and $z - x$ are also factors of $E(x, y, z)$.

Illustration-20 : Factorize $x^2(y - z) + y^2(z - x) + z^2(x - y)$

Solution : $x^2(y - z) + x(z^2 - y^2) + yz(y - z)$

$= (y - z)\,(x^2 - x(z + y) + yz)$

$= (y - z)\,(x^2 - xz - xy + yz)$

$= (y - z)\,(x(x - z) - y\,(x - y))$

$= (y - z)\,(x - z)\,(x - y)$

$= -(x - y)\,(y - z)\,(z - x)$

OR

Let $E(x, y, z) = x^2(y - z) + y^2(z - x) + z^2(x - y)$

when $y = x$, $E(x, y, z) = x^2(x - z) + x^2(z - x) + z^2(x - x) = x^3 - x^2z + x^2z - x^3 = 0$

$\Rightarrow x - y$ is factor of $E(x, y, z)$

So, $y - z$ and $z - x$ are also factors of $E(x, y, z)$

$E(x, y, z) = A(x - y)(y - z)(z - x)$ ...(i)

Since degree of $E(x, y, z)$ is 3 so A is constant. We can find A by substituting values of x, y & z in(i)

Let $x = 0, y = 1, z = 2$

$E(0, 1, 2) = A(-1)(-1)(2)$

$\Rightarrow 1(2 - 0) + 2^2(0 - 1) = 2A \Rightarrow A = -1$

$E(x, y, z) = -(x - y)(y - z)(z - x)$

2.8 Important Algebraic Identities

- $xy + ay + bx + ab = (x + a)(y + b)$
- $x^2 + 2xy + y^2 = (x + y)^2$
- $x^2 + y^2 + z^2 + 2xy + 2yz + 2zx = (x + y + z)^2$
- $x^2 - y^2 = (x - y)(x + y)$
- $x^4 + x^2 + 1 = (x^2 + 1)^2 - x^2 = (x^2 + x + 1)(x^2 - x + 1)$
- $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \ldots + y^{n-1}),\ n \in \mathbb{N}$
- $x^{2n+1} + y^{2n+1} = (x + y)(x^{2n} - x^{2n-1}\,y + x^{2n-2}\,y^2 - \ldots + y^{2n}),\ n \in \mathbb{N}$

- $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

$$= \frac{1}{2}(x+y+z)\left\{(x-y)^2 + (y-z)^2 + (z-x)^2\right\}$$

- $x^3 + y^3 + z^3 = 3xyz$ if $x + y + z = 0$ or $x = y = z$
- $x^3 + y^3 + z^3 + 3(x + y)(y + z)(z + x) = (x + y + z)^3$

Problem Solving Strategies :

- When facing a problem with gigantic numbers, try replacing them with smaller numbers and look for a pattern. You can often prove your pattern works and solve the problem by substituting variable expressions for the numbers.

Illustration-21 : Compute $\sqrt{2022 \times 2020 \times 2018 \times 2016 + 16}$ without using calculator.

Solution : Let x = 2019

$$\sqrt{2022 \times 2020 \times 2018 \times 2016 + 16}$$

$$= \sqrt{(x+3)(x+1)(x-1)(x-3) + 16}$$

$$= \sqrt{(x^2 - 9)(x^2 - 1) + 16}$$

$$= \sqrt{x^4 - 10x^2 + 25} = \sqrt{(x^2 - 5)^2} = x^2 - 5$$

$= 2019^2 - 5 = (2000)^2 + (19)^2 + 2 \times 2000 \times 19 - 5$

$= 4000000 + 361 + 76000 - 5$

= 4,076, 356

- Focusing on what makes a problem tricky helps identify what strategies might be best to solve the problem.
- If you have a problem that involves expressions of the form a + b and a – b, where a and/or b involve square roots, consider finding a way to multiply the expressions to get rid of the square roots.

Illustration-22 : Suppose $x = z - \sqrt{z^2 - 5}$ and $5y = z + \sqrt{z^2 - 5}$.

Find x when y = 2/3.

Solution : x = $z - \sqrt{z^2 - 5}$ and 5y = $z + \sqrt{z^2 - 5}$, multiplying both equations

we get $5xy = z^2 - (z^2 - 5) = 5$

$\Rightarrow xy = 1 \Rightarrow x = \frac{3}{2}$.

- When making a substitution, take some time to look for the substitution that simplifies your work the most.
- If we have the product of two variables added to linear terms with both variables, such as mn + 3m + 5n, then there is a constant we can add that will allow us to factor. For example, adding 15 to mn + 3m + 5n gives us mn + 3m + 5n + 15 = (m + 5)(n + 3).

Illustration-23 : Find all integral solutions of x and y when xy – y – 2x = 3.

Solution : $xy - y - 2x = 3$

$\Rightarrow xy - y - 2x + 2 = 5$

$\Rightarrow y(x - 1) - 2(x - 1) = 5$

$\Rightarrow (x - 1)(y - 2) = 5$

$x-1$	$y-2$	(x,y)
5	1	(6,3)
1	5	(2,7)
–5	–1	(–4,1)
–1	–5	(0,–3)

All possible (x, y) and (6, 3), (2, 7), (0, –3) and (–4, 1)

- Guessing has a long and glorious history in mathematics and science. It is often a very important first step in many discoveries. Don't be afraid to guess! But make sure you test your guesses – a guess itself is only a first step.

Illustration-24 : Find all pairs of positive integers m and n such that m^2 is 105 greater than n^2.

Solution : Turning the words into math is easy :

$m^2 = n^2 + 105.$

$\Rightarrow (m - n)(m + n) = 105.$

$(m - n)(m + n) = 1.105 = 3.35 = 5.21 = 7.15$

Because m and n are positive, we know that m – n is smaller than m + n, so we only have these four cases to consider :

$m - n = 1$	$m - n = 3$	$m - n = 5$	$m - n = 7$
$m + n = 105$	$m + n = 35$	$m + n = 21$	$m + n = 15$

Each of these systems of equations gives us a solution (m, n). Adding the equations in the first case gives us $2m = 106$, so $m = 53$. Substitution then gives $n = 52$. Similarly, we can work through each of the other three cases to find the four solutions $(m, n) = (53, 52); (19, 16); (13, 8); (11, 4)$.

Illustration-25 : The number 7,999,999,999 has two prime factors. Find them.

Solution : Let $7{,}999{,}999{,}999 = 8{,}000{,}000{,}000 - 1$

$= (2000)^3 - 1^3 = (2000 - 1)(2000^2 + 2000 + 1)$

$= (1999)(4002001)$

According to question, 7,999,999,999 has two prime factors, they must be 1999 and 4002001.

Illustration-26 : Factor $x^4 + 4y^4$.

Solution : $x^4 + 4y^4 = (x^2)^2 + (2y^2)^2 - 2(x^2)(2y^2) + (2xy)^2$

$= (x^2 + 2y^2)^2 - (2xy)^2$

$= (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)$

Do yourself-2 :

1. Factorize following expressions

(i) $x^4 - y^4$ **(ii)** $9a^2 - (2x - y)^2$ **(iii)** $4x^2 - 9y^2 - 6x - 9y$

2. Factorize following expressions

(i) $8x^3 - 27y^3$ **(ii)** $8x^3 - 125y^3 + 2x - 5y$

3. Factorize following expressions

(i) $x^2 + 3x - 40$ **(ii)** $x^2 - 3x - 40$ **(iii)** $x^2 + 5x - 14$

(iv) $x^2 - 3x - 4$ **(v)** $x^2 - 2x - 3$ **(vi)** $3x^2 - 10x + 8$

(vii) $12x^2 + x - 35$ **(viii)** $3x^2 - 5x + 2$ **(ix)** $3x^2 - 7x + 4$

(x) $7x^2 - 8x + 1$ **(xi)** $2x^2 - 17x + 26$ **(xii)** $3a^2 - 7a - 6$

(xiii) $14a^2 + a - 3$

4. Factorize following expressions

(i) $a^2 - 4a + 3 + 2b - b^2$ **(ii)** $x^4 + 324$ **(iii)** $x^4 - y^2 + 2x^2 + 1$

(iv) $4a^4 - 5a^2 + 1$ **(v)** $4x^4 + 81$ **(vi)** $1 + x^4 + x^8$

5. Factorize following expressions

(i) $x^3 - 6x^2 + 11x - 6$ **(ii)** $2x^3 + 9x^2 + 10x + 3$

(iii) $2x^3 - 9x^2 + 13x - 6$ **(iv)** $x^6 - 7x^2 - 6$

(v) $(x + y + z)^3 - x^3 - y^3 - z^3$

6. (i) Factorize the expressions $8a^6 + 5a^3 + 1$

(ii) Show that $(x - y)^3 + (y - z)^3 + (z - x)^3 = 3(x - y)(y - z)(z - x)$.

7. Factorize following expressions

(i) $(x + 1)(x + 2)(x + 3)(x + 4) - 15$

(ii) $4x(2x + 3)(2x - 1)(x + 1) - 54$

(iii) $(x - 3)(x + 2)(x + 3)(x + 8) + 56$

8. Factorize the following expressions :

(i) $(x - y)(x - y - 1) - 20$ **(ii)** $(x + 2y)(x + 2y + 2) - 8$

(iii) $3(4x + 5)^2 - 2(4x + 5) - 1$ **(iv)** $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2$

(v) $x^6 - 7x^3 - 8$ **(vi)** $x^4 + 15x^2 - 16$

(vii) $(x^2 - x - 3)(x^2 - x - 5) - 3$ **(viii)** $(x^2 + 5x + 6)(x^2 + 5x + 4) - 120$

(ix) $(a^2 + 1)^2 + (a^2 + 5)^2 - 4(a^2 + 3)^2$

(x) $(x - 1)^2 + (y - 1)^2 + (z - 1)^2 + 2(x - 1)(y - 1) + 2(y - 1)(z - 1) + 2(z - 1)(x - 1)$

(xi) $2x^4 - x^3 - 6x^2 - x + 2$ **(xii)** $(x^4 + x^2 - 4)(x^4 + x^2 + 3) + 10$

2.9 Miscellaneous Algebraic Manipulations

Illustration-27 : Suppose $x + \frac{1}{x} = 5$ find **(i)** $x^2 + \frac{1}{x^2}$ **(ii)** $x^4 - \frac{1}{x^4}$

Solution : (i) Given : $x + \frac{1}{x} = 5$

By squaring both sides $\left(x + \frac{1}{x}\right)^2 = (5)^2 \Rightarrow x^2 + \frac{1}{x^2} + 2 = 25 \Rightarrow x^2 + \frac{1}{x^2} = 23$

(ii) $\left(x-\frac{1}{x}\right)^2=\left(x+\frac{1}{x}\right)^2-4=21 \Rightarrow x-\frac{1}{x}=\pm\sqrt{21}$

$x^2-\frac{1}{x^2}=\pm5\sqrt{21}$

from (i) we have $x^2+\frac{1}{x^2}=23$

So $x^4-\frac{1}{x^4}=\pm23\times5\sqrt{21}=\pm115\sqrt{21}$

Illustration-28 : Simplify $\sqrt{6+\sqrt{11}}+\sqrt{6-\sqrt{11}}$

Solution : Let $x=\sqrt{6+\sqrt{11}}+\sqrt{6-\sqrt{11}}$

By squaring both sides

$x^2=\left(\sqrt{6+\sqrt{11}}\right)^2+\left(\sqrt{6-\sqrt{11}}\right)^2+2\sqrt{36-11}$

= 12 + 10 = 22

Since x is positive, so $x=\sqrt{22}$.

OR

$$x=\sqrt{6+\sqrt{11}}+\sqrt{6-\sqrt{11}}=\frac{1}{\sqrt{2}}\left(\sqrt{12+2\sqrt{11}}+\sqrt{12-2\sqrt{11}}\right)$$

$$=\frac{1}{\sqrt{2}}\left(\sqrt{\left(\sqrt{11}+1\right)^2}+\sqrt{\left(\sqrt{11}-1\right)^2}\right)=\frac{1}{\sqrt{2}}\left(\sqrt{11}+1+\sqrt{11}-1\right)$$

$x=\sqrt{22}$

Illustration-29 : If $\left(a+\frac{1}{a}\right)^2=3$, then $a^3+\frac{1}{a^3}$ equals :

(A) $6\sqrt{3}$ (B) $3\sqrt{3}$ (C) 0 (D) $6\sqrt{3}$

Solution : $a+\frac{1}{a}=\pm\sqrt{3}$

$$a^3+\frac{1}{a^3}=\left(a+\frac{1}{a}\right)^3-3\left(a+\frac{1}{a}\right)=\pm3\sqrt{3}\mp3\sqrt{3}=0$$

Illustration-30 : If $x = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$ and $y = \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$, then find $x^3 + y^3$.

Solution : $x = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} = \left(\sqrt{3}-\sqrt{2}\right)^2 = 5-2\sqrt{6}$

$y = 5+2\sqrt{6}$

$x^3 + y^3 = (x + y)(x^2 - xy + y^2)$

$= (x + y)[(x + y)^2 - 3xy] = 10 \times [100 - 3] = 970$

Do yourself-3 :

1. Suppose $a+\frac{1}{a}=3$:

(a) Find $a^2 + \frac{1}{a^2}$ **(b)** Find $a^4 + \frac{1}{a^4}$ **(c)** Find $a^3 + \frac{1}{a^3}$

2. If $x+\frac{1}{x}=2$, then prove that : $x^2+\frac{1}{x^2}=x^4+\frac{1}{x^4}=x^8+\frac{1}{x^8}$.

3. If $2x + 3y + 4z = 0$, then prove that $8x^3 + 27y^3 + 64z^3 = 72xyz$.

4. I'm thinking of two numbers. The sum of my numbers is 14 and the product of my numbers is 46. What is the sum of the squares of my numbers ?

5. Simplify $\sqrt{7-\sqrt{13}}-\sqrt{7+\sqrt{13}}$.

6. Simplify the expression $\sqrt[3]{2+\sqrt{5}}+\sqrt[3]{2-\sqrt{5}}$.

7. If x, y, z are all different real numbers, then prove that

$$\frac{1}{(x-y)^2}+\frac{1}{(y-z)^2}+\frac{1}{(z-x)^2}=\left(\frac{1}{x-y}+\frac{1}{y-z}+\frac{1}{z-x}\right)^2.$$

8. Solve for x :

(a) $\sqrt{2x+1}=\sqrt{x}+1$ **(b)** $\sqrt{x^2-1}=x-3$ **(c)** $\sqrt{x+1}+2\sqrt{2x-3}=-3$

3. POLYNOMIAL IN ONE VARIABLE

An algebraic expression of the form $p(x) = a_nx^n + a_{n-1}x^{n-1} + ... + a_1x + a_0$ is called a polynomial function in 'x' where $a_i(i = 0, 1, 2, ..., n)$ is a constant which belongs to the set of real numbers and sometimes to the set of complex numbers and n is a whole number.

- a_i is the coefficient of x^i for i = 1, 2, 3, ..., n and a_0 is constant term of p(x).
- If $a_n \neq 0$, then a_nx^n is called leading term and a_n is called **leading co-efficient.**
- If $a_n = 1$, then polynomial is called **monic polynomial.**

- If $a_n \neq 0$, then **degree of the polynomial** is n.
- $f(x) = a_0$ is called **constant polynomial.** Its degree is 0, if $a_0 \neq 0$. If $a_0 = 0$ the polynomial $f(x)$ is called **ZERO polynomial.** Its degree is defined as $-\infty$ to preserve following two properties listed below. Some people prefer not to define degree of zero polynomial.

If $f(x)$ is a polynomial of degree n and g(x) is a polynomial of degree m then

1. $f(x) \pm g(x)$ is a polynomial of degree $\leq \max\{n, m\}$
2. $f(x) \times g(x)$ is a polynomial of degree m + n.

3.1 Types of Polynomials (w.r.t. Degree)

Degree of the polynomial in one variable is the largest exponent of the variable. For example, the degree of the polynomial $3x^7 - 4x^6 + x + 9$ is 7 and the degree of the polynomial $5x^6 - 4x^2 - 6$ is 6.

Polynomials classified by degree

Degree	Name	General form	Example
undefined or $-\infty$	Zero polynomial	0	0
0	(Non-zero) constant polynomial	a; $(a \neq 0)$	4
1	Linear polynomial	$ax + b$; $(a \neq 0)$	$x + 1$
2	Quadratic polynomial	$ax^2 + bx + c$; $(a \neq 0)$	$x^2 + 1$
3	Cubic polynomial	$ax^3 + bx^2 + cx + d$; $(a \neq 0)$	$x^3 + 1$

Usually, a polynomial of degree n, for n greater than 3, is called a polynomial of degree n, although the phrases quartic polynomial for degree 4 and quintic polynomial for degree 5 are sometimes used.

Note :

Polynomials having only one term are called monomials. E.g. 2, 2x, $7y^5$, $12t^7$ etc. Polynomials having exactly two dissimilar terms are called binomials. E.g. $p(x) = 2x + 1$, $r(y) = 2y^7 + 5y^6$ etc. Polynomials having exactly three distinct terms are called trinomials. E.g. $p(x) = 2x^2 + x + 6$, $q(y) = 9y^6 + 4y^2 + 1$ etc.

3.2 Division in Polynomials

Consider two polynomials P(x) & d(x) with d(x) being not identically zero and degree of d(x) $\leq$ degree of P(x) then there exists unique polynomials Q(x) and r(x) such that

$P(x) = Q(x) . d(x) + r(x)$

Here P(x) is called as dividend,

d(x) is called as divisor,

Q(x) is called as quotient,

& r(x) is called as remainder with degree of r(x) < degree of d(x)

Note : If d(x) is a divisor of P(x) then kd(x) will also be a divisor of P(x); $k \in \mathbb{R} - \{0\}$ and d(–x) will be a divisor of P(–x).

3.3 Remainder Theorem

Statement : Let p(x) be a polynomial of degree ≥ 1 and 'a' is any real number. If p(x) is divided by (x – a), then the remainder is p(a).

Illustration-31 : Let p(x) be $x^3 - 7x^2 + 6x + 4$. Divide p(x) with (x – 6) and find the remainder

Solution : Put x = 6 in p(x) i.e. p(6) will be the remainder.

$\therefore$ required remainder be

$p(6) = (6)^3 - 7\cdot6^2 + 6\cdot6 + 4 = 216 - 252 + 36 + 4 = 256 - 252 = 4$

$$
\begin{array}{r|l|l}
x-6 & x^3 - 7x^2 + 6x + 4 & x^2 - x \\
 & \underset{-\quad +}{x^3 - 6x^2} & \\
\hline
 & -x^2 + 6x + 4 & \\
 & \underset{+\quad -}{-x^2 + 6x} & \\
\hline
 & \text{Remainder} = 4 &
\end{array}
$$

Thus, p(a) is remainder on dividing p(x) by (x – a).

Remark : (i) p(–a) is remainder on dividing p(x) by (x + a)

$[\because x + a = 0 \Rightarrow x = -a]$

(ii) $p\left(\frac{b}{a}\right)$ is remainder on dividing p(x) by (ax – b)

$[\because ax - b = 0 \Rightarrow x = b/a]$

(iii) $p\left(\frac{-b}{a}\right)$ is remainder on dividing p(x) by (ax + b)

$[\because ax + b = 0 \Rightarrow x = -b/a]$

(iv) $p\left(\frac{b}{a}\right)$ is remainder on dividing p(x) by (b – ax)

$[\because b - ax = 0 \Rightarrow x = b/a]$

Illustration-32 : Find the remainder when

$x^3 - ax^2 + 6x - a$ is divided by x – a

Solution : Let $p(x) = x^3 - ax^2 + 6x - a$

$p(a) = a^3 - a(a)^2 + 6(a) - a$

$= a^3 - a^3 + 6a - a = 5a$

So, by the Remainder theorem, remainder = 5a

3.4 Factor Theorem

Statement : Let f(x) be a polynomial of degree ≥ 1 and a be any real constant such that f(a) = 0, then (x – a) is a factor of f(x). Conversely, if (x – a) is a factor of f(x), then f(a) = 0.

Proof : By Remainder theorem, if f(x) is divided by (x – a), the remainder will be f(a). Let q(x) be the quotient. Then, we can write,

$f(x) = (x - a) \times q(x) + f(a)$ ($\because$ Dividend = Divisor × Quotient + Remainder)

If $f(a) = 0$, then $f(x) = (x - a) \times q(x)$

Thus, $(x - a)$ is a factor of $f(x)$.

Converse Let $(x - a)$ is a factor of $f(x)$.

Then we have a polynomial $q(x)$ such that $f(x) = (x - a) \times q(x)$

Replacing x by a, we get $f(a) = 0$.

Hence, proved.

Illustration-33 : Use the factor theorem to determine whether $(x - 1)$ is a factor of $f(x) = 2\sqrt{2}x^3 + 5\sqrt{2}x^2 - 7\sqrt{2}$

Solution : By using factor theorem, $(x - 1)$ is a factor of $f(x)$, only when $f(1) = 0$

$f(1) = 2\sqrt{2}(1)^3 + 5\sqrt{2}(1)^2 - 7\sqrt{2} = 2\sqrt{2} + 5\sqrt{2} - 7\sqrt{2} = 0$

Hence, $(x - 1)$ is a factor of $f(x)$.

3.5 Fundamental Theorem of Algebra

Every polynomial function of degree ≥ 1 has atleast one zero in the complex numbers.

In other words, if we have

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0,\ a_i \in \text{complex number } \forall\ i = 0, 1, 2, \ldots, n$$

with $n \geq 1$, then there exists atleast one $h \in \mathbb{C}$, such that

$$a_n h^n + a_{n-1}h^{n-1} + \ldots + a_1 h + a_0 = 0.$$

From this, it is easy to deduce that a polynomial function of degree 'n' (≥ 1) has exactly n zeroes.

i.e., $f(x) = a(x - r_1)(x - r_2)\ldots(x - r_n)$

Illustration-34: Find the constants a, b, c such that $(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

Solution : **Method : 1**

$(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

By comparing coefficient of x^4 from both sides

$$2a = 2 \Rightarrow a = 1$$

By comparing coefficient of x^3 from both sides

$$2b + 3a = 11 \Rightarrow b = 4$$

By comparing coefficient of x^2 from both sides

$$2c + 3b + 7a = 9 \Rightarrow c = -5$$

By comparing coefficient of x from both sides

$$3c + 7b = 13 \qquad \text{(b \& c satisfy)}$$

By comparing constant

$$7c = -35 \quad \Rightarrow c = -5$$

So $\quad a = 1, b = 4, c = -5$

Method : 2

Given that $(2x^2 + 3x + 7)(ax^2 + bx + c) = 2x^4 + 11x^3 + 9x^2 + 13x - 35$

$$\Rightarrow \qquad ax^2 + bx + c = \frac{2x^4 + 11x^3 + 9x^2 + 13x - 35}{2x^2 + 3x + 7}$$

$$= x^2 + 4x - 5$$

By comparing

$$a = 1,\ b = 4 \text{ and } c = -5.$$

Illustration 35 : Show that $(x - 3)$ is a factor of the polynomial $x^3 - 3x^2 + 4x - 12$.

Solution. Let $p(x) = x^3 - 3x^2 + 4x - 12$ be the given polynomial. By factor theorem, $(x - a)$ is a factor of a polynomial $p(x)$ iff $p(a) = 0$. Therefore, in order to prove that $x - 3$ is a factor of $p(x)$, it is sufficient to show that $p(3) = 0$.

Now, $p(x) = x^3 - 3x^2 + 4x - 12$

$\Rightarrow p(3) = 3^3 - 3 \times 3^2 + 4 \times 3 - 12 = 27 - 27 + 12 - 12 = 0$

Hence, $(x - 3)$ is a factor of $p(x) = x^3 - 3x^2 + 4x - 12$.

Illustration 36 : Without actual division prove that $2x^4 - 6x^3 + 3x^2 + 3x - 2$ is exactly divisible by $x^2 - 3x + 2$.

Solution. Let $f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$ and $g(x) = x^2 - 3x + 2$ be the given polynomials.

Then $g(x) = x^2 - 3x + 2 = x^2 - 2x - x + 2 = x(x - 2) - 1(x - 2) = (x - 1)(x - 2)$

In order to prove that $f(x)$ is exactly divisible by $g(x)$, it is sufficient to prove that $x - 1$ and $x - 2$ are factors of $f(x)$. For this it is sufficient to prove that $f(1) = 0$ and $f(2) = 0$.

Now, $f(x) = 2x^4 - 6x^3 + 3x^2 + 3x - 2$

$\Rightarrow \quad f(1) = 2 \times 1^4 - 6 \times 1^3 + 3 \times 1^2 + 3 \times 1 - 2 \Rightarrow f(1) = 0$

and, $f(2) = 2 \times 2^4 - 6 \times 2^3 \times 2^2 + 3 \times 2 - 2 \Rightarrow f(2) = 0$

Hence, $(x - 1)$ and $(x - 2)$ are factors of $f(x)$.

$\Rightarrow \ g(x) = (x - 1)(x - 2)$ is a factors of $f(x)$.

Hence, $f(x)$ is exactly divisible by $g(x)$.

Illustration 37 : The polynomials $P(x) = kx^3 + 3x^2 - 3$ and $Q(x) = 2x^3 - 5x + k$, when divided by $(x - 4)$ leave the same remainder. The value of k is

(A) 2 (B) 1 (C) 0 (D) –1

Solution : $P(4) = 64k + 48 - 3 = 64k + 45$

$Q(4) = 128 - 20 + k = k + 108$

given $P(4) = Q(4)$

$64k + 45 = k + 108$

$\Rightarrow 63k = 63$ $\Rightarrow k = 1$ $\Rightarrow$ Option **(B)** is correct

Illustration-38 : Let P(x) be a polynomial such that when P(x) is divided by x – 19, the remainder is 99, and when P(x) is divided by x – 99, the remainder is 19. What is the remainder when P(x) is divided by (x – 19)(x – 99) ?

Solution : Because we are dividing by a quadratic, the degree of the remainder is not greater than 1. So, the remainder is ax + b, for some constants a and b. Therefore, we have

$$P(x) = (x - 19)(x - 99)Q(x) + ax + b,$$

where Q(x) is the quotient when P(x) is divided by (x – 19)(x – 99). We eliminate the Q(x) term by letting x = 19 or by letting x = 99. Doing each in turn gives us the system of equations

$$P(19) = 19a + b = 99,$$

$$P(99) = 99a + b = 19.$$

Solving this system of equations gives us a = –1 and b = 118.

So, the remainder is –x + 118.

Illustration-39 : The polynomial $f(x) = x^4 + ax^3 + bx^2 + cx + d$ has roots 1, 3, 5 and 7. Determine all the coefficients of $f(x)$.

Solution : By applying factor theorem

$f(x)= (x - 1)(x - 3)(x - 5)(x - 7)$

$= (x^2 - 4x + 3)(x^2 - 12x + 35)$

$= x^2(x^2 - 12x + 35) - 4x(x^2 - 12x + 35) + 3(x^2 - 12x + 35)$

$= x^4 - 16x^3 + 86x^2 - 176x + 105$

So a = –16, b = 86, c = –176 and c = 105

Illustration-40 : If $f(x)$ is monic polynomial of degree 6 such that $f(0) = 0$, $f(1) = -1$, $f(2) = -2$, $f(3) = -3$, $f(4) = -4$ and $f(5) = -5$, then find $f(x)$.

Solution : According to question

$f(0) = 0, f(1) = -1, f(2) = -2, ..., f(5) = -5$

$\Rightarrow f(x) + x = 0$ has the roots x = 0, 1, 2, ..., 5

$\Rightarrow f(x) + x = x(x - 1)(x - 2)(x - 3)(x - 4)(x - 5)$ (By factor theorem)

$\Rightarrow f(x) = x(x - 1)(x - 2)(x - 3)(x - 4)(x - 5) - x.$

Illustration-41 : If $P(x) = 2x^3 + ax^2 + bx + c$, where a, b, c $\in \mathbb{Z}$. If $P(\sqrt{3}) = 10 - 2\sqrt{3}$.

Find (i) $P(-\sqrt{3})$ (ii) $3a + c$

Solution : $P(x) = 2x^3 + ax^2 + bx + c$

$P(\sqrt{3}) = 10 - 2\sqrt{3}$

$\Rightarrow 6\sqrt{3} + 3a + b\sqrt{3} + c = 10 - 2\sqrt{3}$

$\Rightarrow (3a + c) + \sqrt{3}(6 + b) = 10 - 2\sqrt{3}$

$\Rightarrow 3a + c = 10$ and $6 + b = -2$ $\quad$ (a, b, c $\in$ I)

$3a + c = 10$ and $b = -8$

$P(-\sqrt{3}) = -6\sqrt{3} + 3a - \sqrt{3}b + c$

$= -6\sqrt{3} + (3a + c) + 8\sqrt{3}$

$P(-\sqrt{3}) = 2\sqrt{3} + 10$

Illustration-42 : Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where a, b, c, d are constants. If $P(1) = 10$, $P(2) = 20$, $P(3) = 30$, then compute $P(4) + P(0)$.

Solution : $P(1) = 10$, $P(2) = 20$ and $P(3) = 30$

we can write these information as

$P(x) = 10x$ for $x = 1, 2, 3$

$\Rightarrow P(x) - 10x = 0$ has roots $x = 1, 2$ and 3

By factor theorem

$P(x) - 10x = (x - \alpha)(x - 1)(x - 2)(x - 3)$

$\Rightarrow P(x) = (x - \alpha)(x - 1)(x - 2)(x - 3) + 10x$

$P(4) + P(0) = (4 - \alpha)(3)(2)(1) + 40 + (-\alpha)(-1)(-2)(-3) + 0$

$= 24 - 6\alpha + 40 + 6\alpha$

$= 64$

Illustration-43 : Let a, b and c are roots of $2x^3 + x^2 + x + 1 = 0$

find (i) $a + b + c$ (ii) abc

Solution : By factor theorem

$2x^3 + x^2 + x + 1 = 2(x - a)(x - b)(x - c)$

$\Rightarrow 2x^3 + x^2 + x + 1 = 2(x^3 - (a + b + c)x^2 + (ab + bc + ca)x - abc)$

$\Rightarrow 2x^3 + x^2 + x + 1 = 2x^3 - 2(a + b + c)x^2 + 2(ab + bc + ca)x - 2abc$

By comparing coefficient of x^2 and constant term, we have

$-2(a + b + c) = 1$ and $-2abc = 1$

$\Rightarrow a + b + c = \frac{-1}{2}$ and $abc = \frac{-1}{2}$.

Do yourself-4 :

1. Determine the remainder when the polynomial $P(x) = x^4 - 3x^2 + 2x + 1$ is divided by $(x - 1)$

2. Find the remainder when $f(x) = 3x^3 + 6x^2 - 4x - 5$ is divided by $(x + 3)$.

3. Determine the value of k for which $x^3 - 6x + k$ may be divisible by $(x - 2)$.

4. Find the value of a, if $(x - a)$ is a factor of $x^3 - a^2x + x + 2$.

5. Find remainder when $f(x) = x^5 - x^3 + 3x^2 + 3x + 1$ is divided by $(x^2 - 1)$.

6. Find the value of ℓ and m if $8x^3 + \ell x^2 - 27x + m$ is divisible by $2x^2 - x - 6$.

7. Find ℓ and m if $2x^3 - (2\ell + 1)x^2 + (\ell + m)x + m$ may be exactly divisible by $2x^2 - x - 3$.

8. f(x) when divided by $x^2 - 3x + 2$ leaves the remainder $ax + b$. If $f(1) = 4$ and $f(2) = 7$, determine a and b.

9. A polynomial in x of the third degree which will vanish when $x = 1$ & $x = -2$ and will have the values 4 & 28 when $x = -1$ and $x = 2$ respectively. Find the polynomial

10. If $f(x)$ is polynomial of degree 4 such that $f(1) = 1$, $f(2) = 2$, $f(3) = 3$, $f(4) = 4$ & $f(0) = 1$ find $f(5)$.

11. Suppose a and b are constants such that
$(x^3 + bx^2 - 7x + 9)(x^2 + ax + 5) = x^5 + 13x^4 + 38x^3 - 22x^2 + 37x + 45\ \forall\ x \in \mathbb{R}$. Find a and b.

12. If P(x) is a cubic polynomial such that $P(1) = 1$; $P(2) = 2$; $P(3) = 3$ with leading coefficient 3 then find the value of P(4).

4. EQUATIONS REDUCIABLE TO QUADRATIC EQUATIONS

There are certain equations which can be reduced to $ax^2 + bx + c = 0$ by some proper substitution.

4.1 $a(f(x))^2 + b(f(x)) + c = 0$, where $f(x)$ is expression of x

Method of solving : Put $f(x) = y$

Illustration-44 : **(a)** Solve $2^{x+3} + 2^{-x} - 6 = 0$ **(b)** Solve $\left(\frac{x}{x+1}\right)^2 + 6 - 5\left(\frac{x}{x+1}\right) = 0$

Solution : **(a)** Put $2^x = y$

$$8y + \frac{1}{y} - 6 = 0$$

$$8y^2 - 6y + 1 = 0$$

$$(4y - 1)(2y - 1) = 0$$

$$y = \frac{1}{4}, \frac{1}{2}$$

$\therefore\ 2^x = 2^{-2}$ and $2^x = 2^{-1}$ $\therefore\ x = -2, x = -1$

(b) Put $\dfrac{x}{x+1} = y$

$y^2 - 5y + 6 = 0$

$(y-2)(y-3) = 0 \Rightarrow \dfrac{x}{x+1} = 2$ and $\dfrac{x}{x+1} = 3$

4.2 $(x-a)^4 + (x-b)^4 = c,$

Method of Solving : Put $\dfrac{x-a+x-b}{2} = t \Rightarrow x = \dfrac{(a+b)}{2} + t$

Illustration-45 : $(x-1)^4 + (x-7)^4 = 272$

Solution : Put $x = \dfrac{7+1}{2} + t$

$\Rightarrow x = t + 4$

$\Rightarrow (t+3)^4 + (t-3)^4 = 272$

$\Rightarrow 2(t^4 + 6.9t^2 + 81) = 272$

$\Rightarrow t^4 + 54t^2 + 81 = 136$

$\Rightarrow t^4 + 54t^2 - 55 = 0$

$\Rightarrow (t^2-1)(t^2+55) = 0$

$\Rightarrow t^2 = 1$

$\Rightarrow t = \pm 1$

$\Rightarrow x = 5,3$

4.3 $ma^{2x} + n(ab)^x + rb^{2x} = 0 \qquad (a \,\&\, b > 0)$

Method of Solving : Divide the equation by b^{2x} and put $\left(\dfrac{a}{b}\right)^x = t$ for $t > 0$

Illustration-46 : $3^{2x+2} + 5.6^x - 4^{x+1} = 0$

Solution : $3^{2x+2} + 5.6^x - 4^{x+1} = 0$

$\Rightarrow 9 \times (3)^{2x} + 5 \times (2 \times 3)^x - 4(2)^{2x} = 0$

$\Rightarrow 9\left(\dfrac{3}{2}\right)^{2x} + 5\left(\dfrac{3}{2}\right)^x - 4 = 0 \qquad \text{...(1)}$

Let $\left(\frac{3}{2}\right)^x = t$ for $t > 0$

Equation 1 becomes

$9t^2 + 5t - 4 = 0$

$\Rightarrow t = \frac{-5 \pm \sqrt{25+144}}{18} = -1$ or $\frac{4}{9}$

$t = -1$ (rejected)

$t = \frac{4}{9} \Rightarrow \left(\frac{3}{2}\right)^x = \left(\frac{2}{3}\right)^2 = \left(\frac{3}{2}\right)^{-2}$

$\Rightarrow x = -2$

Solution of the given equation is $x = -2$

4.4 $m \cdot a^{f(x)} + n \cdot b^{f(x)} + r = 0$, where $ab = 1$, a & $b > 0$ and $f(x)$ is expression of x

Method of Solving : put $a^{f(x)} = t$, then $b^{f(x)} = \frac{1}{t}$

Illustration-47 : Solve $\left(\sqrt{5+2\sqrt{6}}\right)^x + \left(\sqrt{5-2\sqrt{6}}\right)^x = 10$

Solution : Let $a = \sqrt{5+2\sqrt{6}}$ and $b = \sqrt{5-2\sqrt{6}}$

$ab = 1$

Let $a^x = t$

Given equation become

$t + \frac{1}{t} = 10 \Rightarrow t^2 - 10t + 1 = 0$

$\Rightarrow t = \frac{10 \pm \sqrt{96}}{2} = 5 \pm 2\sqrt{6}$

$$\Rightarrow \left(5+2\sqrt{6}\right)^{x/2} = \left(5+2\sqrt{6}\right) \Rightarrow \frac{x}{2} = 1 \Rightarrow x = 2$$

or $\left(5+2\sqrt{6}\right)^{x/2} = \left(5-2\sqrt{6}\right) = \left(5+2\sqrt{6}\right)^{-1}$

$$\Rightarrow \frac{x}{2} = -1 \Rightarrow x = -2$$

Solutions of the given equation is x = 2 or –2.

4.5 (x + a)(x + b)(x + c)(x + d) + e = 0 when b + c = a + d

Illustration-48 : Solve x(x + 1) (x + 2) (x + 3) – 8 = 0

Solution : x(x + 1) (x + 2) (x + 3) – 8 = 0

$\Rightarrow$ x(x + 3) (x + 1) (x + 2) – 8 = 0

$\Rightarrow (x^2 + 3x)(x^2 + 3x + 2) - 8 = 0$

$\Rightarrow (x^2 + 3x)^2 + 2(x^2 + 3x) - 8 = 0$

$\Rightarrow (x^2 + 3x)^2 + 4(x^2 + 3x) - 2(x^2 + 3x) - 8 = 0$

$\Rightarrow (x^2 + 3x)(x^2 + 3x + 4) - 2(x^2 + 3x + 4) = 0$

$\Rightarrow (x^2 + 3x - 2)(x^2 + 3x + 4) = 0$

$\Rightarrow (x^2 + 3x - 2) = 0$ or $(x^2 + 3x + 4) = 0$

$$\Rightarrow x = \frac{-3 \pm \sqrt{17}}{2} \text{ or } x = \frac{-3 \pm \sqrt{7}\,i}{2}$$

4.6 (x + a)(x + b)(x + c)(x + d) + ex² = 0, where ad = bc.

Method of solving : Divide given equation by x^2 and put $x + \frac{ad}{x} = t$

Illustration-49 : $(x + 1)(x + 2)(x + 3)(x + 6) = 3x^2$

Solution : $(x + 1)(x + 6)(x + 2)(x + 3) - 3x^2 = 0$

$\Rightarrow (x^2 + 7x + 6)(x^2 + 5x + 6) - 3x^2 = 0$

$$\Rightarrow \left(x + \frac{6}{x} + 7\right)\left(x + \frac{6}{x} + 5\right) - 3 = 0$$

Let $x + \frac{6}{x} = t$

$(t + 7)(t + 5) - 3 = 0$

$\Rightarrow t^2 + 12t + 32 = 0$

$t = -8$ or -4

when $x + \frac{6}{x} = -8 \Rightarrow x^2 + 8x + 6 = 0 \Rightarrow x = -4 \pm \sqrt{10}$

$x + \frac{6}{x} = -4 \Rightarrow x^2 + 4x + 6 = 0 \Rightarrow x = -2 \pm \sqrt{2}\,i$

Illustration-50 : Solve $(x^2 - 3x)(x^2 - 3x + 2) + 1 = 0$

Solution : Let $x^2 - 3x = y$

$\Rightarrow y(y + 2) + 1 = 0$

$\Rightarrow y^2 + 2y + 1 = 0$

$\Rightarrow (y + 1)^2 = 0$

$\Rightarrow y = -1$

Putting $y = 1$ in $x^2 - 3x = y$

we have $x^2 - 3x = -1$

$\Rightarrow x^2 - 3x + 1 = 0$

$\Rightarrow x = \frac{3 \pm \sqrt{5}}{2}$

4.7 $ax^4 + bx^3 + cx^2 + dx + e = 0$, where $a = e$ & $b = \pm d$

Method of solving : Divide given equation by x^2 and put $x + \frac{1}{x} = t$ or $x - \frac{1}{x} = t$ whichever is applicable.

Illustration-51 : Solve $x^4 - 2x^3 + 3x^2 - 2x + 1 = 0$

Solution : $x^4 - 2x^3 + 3x^2 - 2x + 1 = 0$

By dividing x^2 both sides we have

$x^2 - 2x + 3 - \frac{2}{x} + \frac{1}{x^2} = 0$

$$\Rightarrow x^2+\frac{1}{x^2}-2\left(x+\frac{1}{x}\right)+3=0$$

Let $x+\frac{1}{x}=t$

Above equation become

$t^2 - 2 - 2t + 3 = 0$

$\Rightarrow t^2 - 2t + 1 = 0 \Rightarrow (t - 1)^2 = 0$

$\Rightarrow t = 1 \quad \Rightarrow x + \frac{1}{x} = 1 \Rightarrow x^2 - x + 1 = 0 \Rightarrow x=\frac{1\pm\sqrt{3}\,i}{2}$

Roots are $\frac{1\pm\sqrt{3}\,i}{2}$

4.8 By Guessing Rational Roots of Polynomial.

Illustration-52 : Solve : $x^4 + x^3 - 2x^2 - x + 1 = 0$

Solution : Let $P(x) = x^4 + x^3 - 2x^2 - x + 1$

$P(1) = 0$ and $P(-1) = 0$

$\Rightarrow (x - 1)(x + 1)$ factor of $P(x)$

We can find other factor of $P(x)$ by dividing $x^2 - 1$ from $P(x)$.

$P(x) = (x^2 - 1)(x^2 + x - 1) = 0$

$\Rightarrow x = \pm 1$ or $x^2 + x - 1 = 0$

$\Rightarrow x = \pm 1$ or $x=\frac{-1\pm\sqrt{5}}{2}$

solution of $P(x) = 0$ are $x = \pm 1$ or $\frac{-1\pm\sqrt{5}}{2}$.

Do yourself-5 :

Solve the following equations for x :

1. $x^{2/3} + x^{1/3} - 2 = 0$

2. $x^{\frac{2}{5}}-3x^{\frac{1}{5}}+2=0$

3. $3x^4 - 8x^2 + 4 = 0$

4. $4^x - 3.2^{x+3} + 128 = 0$

5. $x^2+\frac{1}{x^2}-5\left(x+\frac{1}{x}\right)+8=0$

6. $2\left(x^2+\frac{1}{x^2}\right)-3\left(x-\frac{1}{x}\right)-4=0$

7. $2^{2x+1} - 7 \times 10^x + 5^{2x+1} = 0$

8. $(x - 1)^4 + (x - 5)^4 = 82$

9. $\left(\sqrt{3}+\sqrt{2}\right)^{x}+\left(\sqrt{3}-\sqrt{2}\right)^{x}-2\sqrt{3}=0$　　**10.** $\left(3+2\sqrt{2}\right)^{x/2}+\left(3-2\sqrt{2}\right)^{x/2}-34=0$

11. $x(x+1)(x+2)(x+3)=24$　　**12.** $(x+1)(x+2)(x+3)(x+4)=120$

13. $(x+1)(x+2)^2(x+4)-2x^2=0$　　**14.** $x^4-2x^3-2x^2+2x+1=0$

15. Match the values of x given in **Column-II** satisfying the exponential equation in Column-I (Do not verify). Remember that for a > 0, the term a^x is always greater than zero $\forall\ x \in R$.

Column-I	**Column-II**
(A) $5^x-24=\dfrac{25}{5^x}$	**(P)** –3
(B) $(2^{x+1})(5^x)=200$	**(Q)** –2
(C) $4^{2/x}-5(4^{1/x})+4=0$	**(R)** –1
(D) $\dfrac{2^{x-1}.4^{x+1}}{8^{x-1}}=16$	**(S)** 0
(E) $4^{x^2+2}-9\left(2^{x^2+2}\right)+8=0$	**(T)** 1
(F) $5^{2x}-7^x-5^{2x}(35)+7^x(35)=0$	**(U)** 2
	(V) 3
	(W) None

5. SYSTEM OF EQUATIONS

Observe the following Illustrations :

Illustrations-53 : If x – y = 2 and xy = 24, find the value of $\dfrac{1}{x}+\dfrac{1}{y}$.

Solution :
$(x+y)^2=(x-y)^2+4xy=4+4(24)$
$\Rightarrow (x+y)^2=100$
$\Rightarrow x+y=10, -10$

$$\therefore\ \frac{x+y}{xy}=\frac{10}{24}=\frac{5}{12};\frac{x+y}{xy}=-\frac{10}{24}=-\frac{5}{12}$$

Illustrations-54 : If 2x – 3y – z = 0 and x + 3y – 14z = 0, then find $\dfrac{x^2+3xy}{y^2+z^2}$.

Solution :
$$\frac{2x}{z}-\frac{3y}{z}=1\ \ \&\ \ \frac{x}{z}+\frac{3y}{z}=14$$

Solving $\dfrac{x}{z}=5;\dfrac{y}{z}=3$

$$\Rightarrow \frac{\left(\frac{x}{z}\right)^2+\frac{3x}{z}.\frac{y}{z}}{\left(\frac{y}{z}\right)^2+1}=\frac{25+3(5)(3)}{(3)^2+1}=\frac{70}{10}=7$$

Illustration-55 : Given $a + b = 20$ and $a^3 + b^3 = 800$, find $a^2 + b^2$.

Solution : $a + b = 20$

$\Rightarrow a^2 + b^2 + 2ab = 400$ .... (1)

$a^3 + b^3 = 800$

$\Rightarrow (a + b)(a^2 - ab + b^2) = 800$

$\Rightarrow a^2 + b^2 - ab = 40$ .... (2)

By adding twice of second equation with first equation.

$3(a^2 + b^2) = 480$

$\Rightarrow a^2 + b^2 = 160.$

Illustrations-56 : $x(y + z) = 29$, $y(z + x) = 26$; $z(x + y) = 51$ find x,y,z.

Solution : $xy + zx = 29$ ...(1)

$yz + xy = 26$ ...(2)

$xz + yz = 51$ ...(3)

$\Rightarrow 2[xy + yz + zx] = 106$

$\Rightarrow xy + yz + zx = 53$

Now : $xy = 2$, $zx = 27$; $yz = 24$

$\Rightarrow x^2y^2z^2 = 24 \times 2 \times 27 = (36)^2$

$\Rightarrow xyz = \pm 36 \therefore (x,y,z) \equiv \left(\frac{3}{2},\frac{4}{3},18\right)$ or $\left(-\frac{3}{2},-\frac{4}{3},-18\right)$

Illustrations-57 : If $x^3 + y^3 = 35$; and $x^2y + xy^2 = 30$, then find (x,y).

Solution : $\frac{x^3+y^3}{x^2y+xy^2} = \frac{(x+y)(x^2-xy+y^2)}{xy(x+y)} = \frac{35}{30} = \frac{7}{6}$

$\Rightarrow 6x^2 - 13xy + 6y^2 = 0 \Rightarrow (3x - 2y)(2x - 3y) = 0 \Rightarrow 3x = 2y$ or $2x = 3y$

Case -I : $3x = 2y$, $x^3 + \left(\frac{3x}{2}\right)^3 = 35 \Rightarrow 35x^3 = 8 \times 35 \Rightarrow x = 2, y = 3$

Case -II : $2x = 3y$, $x^3 + \left(\frac{2x}{3}\right)^3 = 35 \Rightarrow 35x^3 = 27 \times 35 \Rightarrow x = 3, y = 2$

Illustrations-58 : If x,y and z are real numbers such that $x^2 + y^2 + 2z^2 - 4x + 2z - 2yz + 5 = 0$, find the value of $(x - y - z)$.

Solution : $x^2 + y^2 + 2z^2 - 4x + 2z - 2yz + 5 = 0$

$\Rightarrow x^2 - 4x + 4 + y^2 + z^2 - 2yz + z^2 + 2z + 1 = 0$

$\Rightarrow (x - 2)^2 + (y - z)^2 + (z + 1)^2 = 0$

$\Rightarrow x = 2,\ y = z,\ z = -1$

$\therefore\ x - y - z = 2 + 1 + 1 = 4$

Illustrations-59 : $xy = 12$, $yz = 15$, $zx = 20$ find $x + y + z$; $x,y,z \in \mathbb{R}^+$

Solution : $(xyz)^2 = (3 \times 4 \times 5)^2$

$\Rightarrow xy\,z = 3 \times 4 \times 5$

$\Rightarrow z = 5,\ y = 3,\ x = 4$

$\Rightarrow x + y + z = 5 + 3 + 4 = 12$

Do yourself-6 :

Solve the following systems of equations :

1. $\begin{cases} x + 2y = 7 \\ 3x + 5y = 18 \end{cases}$

2. $\begin{cases} 3x - 2y = 13 \\ 4x + 5y = 25 \end{cases}$

3. $\begin{cases} x^2 - y^2 = 16, \\ x + y = 8 \end{cases}$

4. $\begin{cases} x - y = 1, \\ x^3 - y^3 = 7 \end{cases}$

5. $x + y = 2$ and $x^3 + y^3 = 56$; $x, y \in \mathbb{R}$, then find x and y.

6. $\begin{cases} x^2 + y^2 + 6x + 2y = 0, \\ x + y + 8 = 0 \end{cases}$

7. $\begin{cases} \dfrac{x}{y} - \dfrac{y}{x} = \dfrac{5}{6}, \\ x^2 - y^2 = 5 \end{cases}$

8. $\begin{cases} \dfrac{x+y}{x-y} + \dfrac{x-y}{x+y} = \dfrac{13}{6}, \\ xy = 5 \end{cases}$

9. $\begin{cases} \dfrac{1}{x+1} + \dfrac{1}{y} = \dfrac{1}{3}, \\ \dfrac{1}{(x+1)^2} - \dfrac{1}{y^2} = \dfrac{1}{4} \end{cases}$

10. $\begin{cases} x^2 + y^2 = 25 - 2xy \\ y(x + y) = 10 \end{cases}$

11. $\begin{cases} \dfrac{1}{y-1} - \dfrac{1}{y+1} = \dfrac{1}{x}, \\ y^2 - x - 5 = 0 \end{cases}$

12. $\begin{cases} 2xy + y^2 - 4x - 3y + 2 = 0, \\ xy + 3y^2 - 2x - 14y + 16 = 0 \end{cases}$

13. $\begin{cases} x^3 + y^3 = 7, \\ xy(x + y) = -2 \end{cases}$

14. $\begin{cases} x^4 + y^4 = 82, \\ xy = 3 \end{cases}$

6. INEQUALITIES

6.1 Basic Rules :

- If a > b and b > c, then a > c.
- If x > y, then x + c > y + c for any real number c. Additionally, if a > b, then x + a > y + b.
- If x > y and a > 0, then xa > ya.
- If we multiply or divide an inequality by a negative number, we must reverse the sign. For example, if x > y and a < 0, then xa < ya.
- If x > y > 0 and a > b > 0, then xa > yb.
- If x > y and x and y have the same sign (positive or negative), then $\frac{1}{x} < \frac{1}{y}$.
- If x > y ≥ 0, then for any positive real number a, we have $x^a > y^a$.

  In particular, if 0 < a < b, then $\sqrt[n]{a} < \sqrt[n]{b}$ for all positive integral values of n > 1.

  E.g. $\sqrt[4]{2} < \sqrt[4]{7}$, $\sqrt[3]{3} < \sqrt[3]{5}$, $\sqrt[5]{10} < \sqrt[5]{13}$ etc.

  If two simple surds of different orders viz. $\sqrt[n]{a}$ and $\sqrt[m]{b}$ have to be compared, they have to be expressed as surds of the same order i.e. LCM of n and m .

  Ex. compare $\sqrt[4]{6}$ and $\sqrt[3]{5}$,

  we express both as the surds of 12th order.

  $\therefore\ \sqrt[4]{6} = \sqrt[12]{6^3}$ and $\sqrt[3]{5} = \sqrt[12]{5^4}$. As $6^3 < 5^4 \Rightarrow \sqrt[4]{6} < \sqrt[3]{5}$

Illustration-60 : Solve :

(i) $7 - 3x \geq 8 + 2x$ (ii) $\frac{8-x}{7} \geq 4$ (iii) $\frac{1}{x} \geq 5$ (iv) $\frac{4}{x+1} \leq 2$

Solution : (i) $7 - 3x \geq 8 + 2x \Rightarrow 7 - 8 \geq 2x + 3x$

$$\Rightarrow \frac{-1}{5} \geq x \Rightarrow x \in \left(-\infty, -\frac{1}{5}\right]$$

(ii) $\frac{8-x}{7} \geq 4 \Rightarrow 8 - x \geq 28 \Rightarrow -x \geq 20$

$\Rightarrow x \leq -20 \Rightarrow x \in (-\infty, -20]$

(iii) $\frac{1}{x} \geq 5$

$$\Rightarrow \infty > \frac{1}{x} \geq 5 \qquad \Rightarrow 0 < x \leq \frac{1}{5} \qquad \Rightarrow x \in \left(0, \frac{1}{5}\right]$$

(iv) $\frac{4}{x+1} \le 2$

$\Rightarrow -\infty < \frac{4}{x+1} < 0$ or $0 < \frac{4}{x+1} \le 2$ $\qquad$ ($\frac{4}{x+1} = 0$ is not possible)

$\Rightarrow 0 > \frac{x+1}{4} > -\infty$ or $\infty > \frac{x+1}{4} \ge \frac{1}{2}$

$\Rightarrow x + 1 < 0$ or $2 \le x + 1$

$\Rightarrow x < -1$ or $1 \le x$ $\quad \Rightarrow x \in (-\infty, -1) \cup [1, \infty)$

Do yourself : 7

1. Determine which of the following statements is true or false. If it is false, provide an example that shows the statement is false. Assume *a, b, c, x* and *y* are real numbers.

(a) If $a \le b$ and $b \le c$, then $a < c$. $\qquad$ (b) If $a \ge b \ge a$, then $a = b$.

(c) If $a > b$, then $ac > bc$. $\qquad$ (d) If $a > b$ and $c \le 0$, then $ac \le bc$.

(e) If $x + a \ge y + a$, then $x \ge y$. $\qquad$ (f) If $x + a \ge y + b$, then $x \ge y$ and $a \ge b$.

2. Which fraction is larger ?

(a) $\frac{13}{17}$ or $\frac{17}{21}$ $\qquad$ (b) $\frac{31}{35}$ or $\frac{37}{41}$

3. Which of the following numbers is largest : $2^{36}, 3^{30}, 5^{18}, 6^{12}, 7^{8}, 8^{4}$? (No calculators!)

4. What values of x satisfy the inequality, $7 - 3x < x - 1 \le 2x + 9$?

5. Which of the following is greater $5+\sqrt{3}, 3+\sqrt{14}$? (Without calculating the values of $\sqrt{3}, \sqrt{14}$)

6.2 Trivial and Sum of squares (SOS) Inequality

The square of any real number is non-negative. So if x is real, then $x^2 \ge 0$. This is known as Trivial inequality. Equality holds only if $x = 0$.

Sum of squares (SOS) of real numbers is non negative. That is $\Sigma x_i^2 \ge 0$. This is know as SOS inequality.

Equality holds if $x_i = 0 \ \forall \ i$

Ex. $x, y, z \in \mathbb{R}$ and $x^2 + y^2 + z^2 = 0 \Rightarrow x = y = z = 0$.

Note :

- $f(x) = [g(x)]^{2n}$ where $n \in \mathbb{N} \Rightarrow f(x) \ge 0$
- $f(x) = [g(x)]^{1/2n}$, $n \in \mathbb{N}$, $g(x) \ge 0 \Rightarrow f(x) \ge 0$

Illustration-61 : If a, b, c $\in \mathbb{R}$ and $a^2 + b^2 + c^2 - ab - bc - ca = 0$, prove that $a = b = c$.

Solution : $a^2 + b^2 + c^2 - ab - bc - ca = 0$

$\Rightarrow 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = 0$

$\Rightarrow (a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ac + a^2) = 0$

$\Rightarrow (a - b)^2 + (b - c)^2 + (c - a)^2 = 0$

$\Rightarrow a - b = b - c = c - a = 0$

$\Rightarrow a = b = c.$

Illustration-62 : For $x,y \in \mathbb{R}$, find the all possible values (range) of expression $4x^2 + 9y^2 - 12x + 6y$.

Solution : If $E(x,y) = 4x^2 + 9y^2 - 12x + 6y$

$= (2x)^2 - 2(2x) \times 3 + (3)^2 + (3y)^2 + 2(3y) + (1)^2 - 10$

$= (2x - 3)^2 + (3y + 1)^2 - 10$

By sum of square (SOS), $(2x - 3)^2 + (3y + 1)^2 \geq 0$

$\Rightarrow E(x,y) = (2x - 3)^2 + (3y + 1)^2 - 10 \geq 0 - 10$

So Range of $E(x,y) = [-10, \infty)$

Illustration-63 : Find all ordered pairs of real numbers (x,y) such that $(4x^2 + 4x + 3)(y^2 - 6y + 13) = 8$

Solution : Rewriting the equation

$((2x + 1)^2 + 2)\,((y - 3)^2 + 4) = 8$

from the SOS, $(2x + 1)^2 \geq 0$ and $(y - 3)^2 \geq 0$,

so we have

$(2x + 1)^2 + 2 \geq 2$ and $(y - 3)^2 + 4 \geq 4$

$\Rightarrow ((2x + 1)^2 + 2)\,((y - 3)^2 + 4) \geq 8$

Equality holds only if

$(2x + 1)^2 = (y - 3)^2 = 0 \Rightarrow (x,y) = \left(-\frac{1}{2}, 3\right)$

Illustration-64 : If $b^2 - 4ac < 0$ and $a > 0$, then show that $ax^2 + bx + c > 0 \ \forall\ x \in \mathbb{R}$

Solution : $ax^2 + bx + c = a\left[x^2 + \frac{b}{a}x\right] + c$

$$= a\left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2}\right] + c$$

$$= a\left[x + \frac{b}{2a}\right]^2 - \left(\frac{b^2 - 4ac}{4a}\right) > 0 \ \forall\ x \in \mathbb{R}.$$ Hence proved

6.3 Mean :

In any collection of data a specific value between two extremes (minimum/maximum) is called a mean of the data.

For Two Variables :

Let x, y be two **positive** real numbers with $x \le y$.

- The **Arithmetic Mean** (**A**) of x and y is $\frac{x+y}{2}$

Observe $x \le \frac{x+y}{2} \le y$

- The **Geometric Mean** (**G**) of x and y is $\sqrt{xy}$

Observe $x \le \sqrt{xy} \le y$

We have $x \le G \le A \le y$

Equality holds only when x = y.

Proof For two positive numbers x and y, the Trivial inequality gives us $\left(\sqrt{x}-\sqrt{y}\right)^2 \ge 0$

$$\Rightarrow x+y-2\sqrt{xy} \ge 0$$

$$\Rightarrow \frac{x+y}{2} \ge \sqrt{xy} \Rightarrow A \ge G$$

Note :

- $x+\frac{1}{x} \ge 2 \ \forall\ x>0$ and $x+\frac{1}{x} \le -2 \ \forall\ x<0$

Illustration : 65 Find all possible values (range) of the expression $x+\frac{4}{x}$, when $x \in \mathbb{R}-\{0\}$.

Sol. When $x>0$, $\frac{x+\frac{4}{x}}{2} \ge \sqrt{x \times \frac{4}{x}} = 2$ (By $A \ge G$)

$\Rightarrow x+\frac{4}{x} \ge 4$,

So, when $x>0$, range = $[4,\infty)$ ...(1)

for $x<0$, $\frac{x+\frac{4}{x}}{2} \le -\sqrt{x \times \frac{4}{x}}$

$\Rightarrow x+\frac{4}{x} \le -4$ ...(2)

From (1) and (2), range = $\left(-\infty,-4\right] \cup \left[4,\infty\right)$

Do yourself : 8

1. Find the minimum value of $\frac{x^4+8}{x^2}$.

2. For x < 0, find the maximum value of $\frac{3x^2+12}{x}$.

3. If a, b $\in \mathbb{R}^+$, find minimum possible value of $(a+b)\left(\frac{1}{a}+\frac{1}{b}\right)$.

7. RATIO AND PROPORTION

If a and b be two quantities of the same kind, then their ratio is a : b; which may be denoted by the fraction $\frac{a}{b}$ (This may be an integer or fraction)

In the ratio a : b, a is the first term (Antecedent) and b is the second term (Consequent)

A ratio may represented in a number of ways e.g. $\frac{a}{b}=\frac{ma}{mb}=\frac{na}{nb}=...$ where m, n, ... are non-zero numbers.

Let a, b, c, d be positive integers now to compare two ratios a : b and c : d we use following :

- (a : b) > (c : d) if ad > bc
- (a : b) = (c : d) if ad = bc
- (a : b) < (c : d) if ad < bc

To compare two or more ratio, reduce them to common denominator.

Note :

- If a > b > 0 and x > 0, then $\frac{a}{b}>\frac{a+x}{b+x}$, e.g. $\frac{41}{40}>\frac{45}{44}$
- If 0 < a < b and x > 0, then $\frac{a}{b}<\frac{a+x}{b+x}$

Illustration-66 : What term must be added to each term of the ratio 5 : 37 to make it equal to 1 : 3 ?

Solution : Let x be added to each term of the ratio 5 : 37.

Then $\frac{x+5}{x+37}=\frac{1}{3}$ $\Rightarrow$ 3x + 15 = x + 37 i.e. x = 11

Illustration-67 : If x : y = 3 : 4; find the ratio of 7x – 4y : 3x + y

Solution : $\frac{x}{y}=\frac{3}{4} \Rightarrow x=\frac{3}{4}y$

Now $\frac{7x-4y}{3x+y}=\frac{7.\frac{3}{4}y-4y}{3.\frac{3}{4}y+y}$ (putting the value of x)

$=\frac{5y}{13y}=\frac{5}{13}$ i.e. 5 : 13

7.1 Proportion :

When two ratios are equal, then the four quantities compositing them are said to be proportional.

so, if $\frac{a}{b}=\frac{c}{d}$, then it is written as a : b = c : d or a : b :: c : d

Where 'a' and 'd' are known as extremes and 'b and c' are known as means.

(i) An important property of proportion : Product of extremes = product of means.

(ii) If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}$ then each is equal to $\frac{a+c+e}{b+d+f}$

(iii) If a : b = c : d, then b : a = d : c (Invertando)

i.e. $\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{b}{a}=\frac{d}{c}$

(iv) If a : b = c : d, then a : c = b : d (Alternando)

i.e. $\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a}{c}=\frac{b}{d}$

(v) If a : b = c : d, then $\frac{a+b}{b}=\frac{c+d}{d}$ (Componendo)

i.e. $\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a}{b}+1=\frac{c}{d}+1$

(vi) If a : b = c : d, then $\frac{a-b}{b}=\frac{c-d}{d}$ (Dividendo)

i.e. $\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a}{b}-1=\frac{c}{d}-1$

(vii) If $a : b = c : d$, then $\dfrac{a+b}{a-b} = \dfrac{c+d}{c-d}$ (Componendo and dividendo)

i.e. $\dfrac{a}{b} = \dfrac{c}{d} \Rightarrow \dfrac{a}{b} + 1 = \dfrac{c}{d} + 1 \Rightarrow \dfrac{a+b}{b} = \dfrac{c+d}{d}$ ... (1)

$\dfrac{a}{b} - 1 = \dfrac{c}{d} - 1 \Rightarrow \dfrac{a-b}{b} = \dfrac{c-d}{d}$ ... (2)

Dividing equation (1) by (2) we obtain

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Illustration-68 : If $\dfrac{x}{a} = \dfrac{y}{b} = \dfrac{z}{c}$ show that $\dfrac{x^3+a^3}{x^2+a^2} + \dfrac{y^3+b^3}{y^2+b^2} + \dfrac{z^3+c^3}{z^2+c^2} = \dfrac{(x+y+z)^3+(a+b+c)^3}{(x+y+z)^2+(a+b+c)^2}$

Solution : $\dfrac{x}{a} = \dfrac{y}{b} = \dfrac{z}{c} = k$ (constant)

x = ak; y = bk; z = ck

substituting these values of x, y, z in the given expression, we obtain

$$\text{L.H.S.} = \frac{a^3k^3+a^3}{a^2k^2+a^2} + \frac{b^3k^3+b^3}{b^2k^2+b^2} + \frac{c^3k^3+c^3}{c^2k^2+c^2}$$

$$= \frac{a^3(k^3+1)}{a^2(k^2+1)} + \frac{b^3(k^3+1)}{b^2(k^2+1)} + \frac{c^3(k^3+1)}{c^2(k^2+1)} = \frac{(k^3+1)}{(k^2+1)}.(a+b+c)$$

$$\text{Now R.H.S} = \frac{(ak+bk+ck)^3+(a+b+c)^3}{(ak+bk+ck)^2+(a+b+c)^2}$$

$$= \frac{k^3(a+b+c)^3+(a+b+c)^3}{k^2(a+b+c)^2+(a+b+c)^2}$$

$$= \frac{(k^3+1)(a+b+c)^3}{(k^2+1)(a+b+c)^2} = \frac{(k^3+1)}{(k^2+1)}.(a+b+c) = \text{L.H.S}$$

Illustration-69 : If $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}$, prove that

$(ab + bc + cd + de)^2 = (a^2 + b^2 + c^2 + d^2)\ (b^2 + c^2 + d^2 + e^2)$

Solution : $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}$, then we have

$$\frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e}=\frac{\sqrt{\left(a^2+b^2+c^2+d^2\right)}}{\sqrt{\left(b^2+c^2+d^2+e^2\right)}}=k \quad \text{(say)}$$

i.e. $a = bk$ $\quad \therefore\ ab = b^2k$

$b = ck$ $\quad \therefore\ bc = c^2k$

$c = dk$ $\quad \therefore\ cd = d^2k$

$d = ek$ $\quad \therefore\ de = e^2k$

so, $(a^2 + b^2 + c^2 + d^2) = k^2\ (b^2 + c^2 + d^2 + e^2)$ ... (i)

Now L.H.S. $= (ab + bc + cd + de)^2$

$= (kb^2 + kc^2 + kd^2 + ke^2)^2$

$= k^2(b^2 + c^2 + d^2 + e^2)^2$

$= k^2(b^2 + c^2 + d^2 + e^2)\ (b^2 + c^2 + d^2 + e^2)$

$= (a^2 + b^2 + c^2 + d^2)\ (b^2 + c^2 + d^2 + e^2) =$ R.H.S (by use of (i))

Illustration-70 : Solve the equation $\frac{3x^4+x^2-2x-3}{3x^4-x^2+2x+3}=\frac{5x^4+2x^2-7x+3}{5x^4-2x^2+7x-3}$

Solution : $\frac{3x^4+x^2-2x-3}{3x^4-x^2+2x+3}=\frac{5x^4+2x^2-7x+3}{5x^4-2x^2+7x-3}$

Applying componendo and dividendo, we have

$$\frac{3x^4}{x^2-2x-3}=\frac{5x^4}{2x^2-7x+3}$$

or $3x^4\ (2x^2 - 7x + 3) - 5x^4(x^2 - 2x - 3) = 0$

or $x^4\ [6x^2 - 21x + 9 - 5x^2 + 10x + 15] = 0$

or $x^4\ (x^2 - 11x + 24) = 0$

$\therefore\ x = 0$ or $x^2 - 11x + 24 = 0$

$\therefore\ x = 0$ or $(x - 8)\ (x - 3) = 0$

$\therefore\ x = 0, 8, 3$

Do yourself : 9

1. If $\frac{a}{b}=\frac{2}{3}$ and $\frac{b}{c}=\frac{4}{5}$, then find value of $\frac{a+b}{b+c}$

2. If $\frac{a}{b}=\frac{3}{5}$ and $\frac{b}{c}=\frac{7}{13}$, then find the value of a : b : c

3. If sum of two numbers is s and their quotient is $\frac{p}{q}$. Find number.

4. If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}$, then find the value of $\frac{2a^4b^2+3a^2c^2-5e^4f}{2b^6+3b^2d^2-5f^5}$ in terms of a and b.

5. If x : a = y : b = z : c, then show that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

8. SIGN-SCHEME (WAVY CURVE) METHOD

Given $f(x)$ and g(x) are polynomials.

To solve the inequalities of the type $\frac{f(x)}{g(x)} * 0$, where '*' can be >, ≥, < or ≤, we take the following steps :

(i) Find all the roots of f(x) = 0 and g(x) = 0

(ii) Write all these roots on the real line in increasing order of values.

(iii) Check the sign of the expression $\frac{f(x)}{g(x)}$ at some x greater than the largest root. If it is positive, put + sign in rightmost interval. In case of negative, put -ve sign in rightmost interval and while moving from right to left change sign in accordance with step (iv).

(iv) If a root occurs even number of times, then sign of expression will be same on both sides of the root and if a root occurs odd number of times, then sign of the expression will be different on both sides of the root.

(v) Write the answer according to need of the question.

Note :

- We don't give equality sign on '±∞' in the solution as they are two improper points of number line.
- We can't take zeroes of denominator in the final answer as at these points expression is not defined (because division by '0' is not defined).
- In case of ≥ 0 or ≤ 0, zeroes of numerator will be part of the answer provided they are not appearing in denominator also.
- Do not cross multiply the terms in the inequalities

Illustration-71 : Find the solution of $-x^2 + 6x + 7 \geq 0$

Solution : $-x^2 + 6x + 7 \geq 0$

$\Rightarrow x^2 - 6x - 7 \leq 0$

$\Rightarrow (x + 1)(x - 7) \leq 0$

$\Rightarrow x \in [-1,7]$

Illustration-72 : Find the solution of $2x^2 - 7x + 3 \geq 0$

Solution : $2x^2 - 7x + 3 \geq 0$

$\Rightarrow (2x - 1)(x - 3) \geq 0$

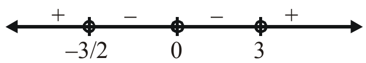
$$x \in \left(-\infty, \frac{1}{2}\right] \cup [3, \infty)$$

Illustration-73 : Solve $2x^4 > 3x^3 + 9x^2$

Solution : $2x^4 - 3x^3 - 9x^2 > 0$

$x^2(2x^2 - 3x - 9) > 0$

$x^2(2x + 3)(x - 3) > 0$



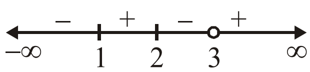
$$x \in \left(-\infty, -\frac{3}{2}\right) \cup (3, \infty)$$

Illustration-74 : Find the solution of the inequalities $\dfrac{(x-1)(x-2)}{(x-3)} \geq 0$

Solution : $x - 1 = 0,\ x - 2 = 0,\ x - 3 = 0$

$\Rightarrow x = 1, 2, 3$

Since $x - 3 \neq 0,\ x \neq 3$

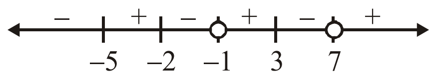


so, $x \in [1, 2] \cup (3, \infty)$

Illustration-75 : If $f(x) = \dfrac{x^3 + 4x^2 - 11x - 30}{x^2 - 6x - 7}$ then find x such that

(i) $f(x) > 0$ **(ii)** $f(x) < 0$.

Solution : Given $f(x) = \dfrac{(x-3)(x+2)(x+5)}{(x+1)(x-7)}$



(i) $f(x) > 0 \Rightarrow x \in (-5, -2) \cup (-1, 3) \cup (7, \infty)$

(ii) $f(x) < 0 \Rightarrow x \in (-\infty, -5) \cup (-2, -1) \cup (3, 7)$

Illustration 76: Solve for real x : $\dfrac{1}{x+1} + \dfrac{2}{x+2} > \dfrac{3}{x+3}$

Solution : $\dfrac{3x+4}{(x+1)(x+2)} > \dfrac{3}{x+3}$

$\Rightarrow \dfrac{3x+4}{(x+1)(x+2)} - \dfrac{3}{x+3} > 0$ $\Rightarrow \dfrac{4x+6}{(x+1)(x+2)(x+3)} > 0$

so, $x \in (-\infty, -3) \cup \left(-2, -\dfrac{3}{2}\right) \cup (-1, \infty)$

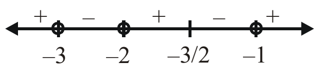
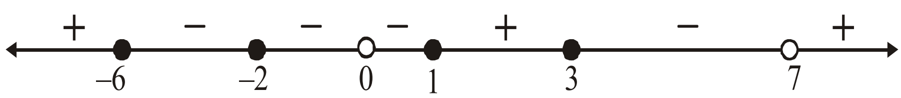


Illustration 77: Let $f(x) = \dfrac{(x-1)^3(x+2)^4(x-3)^5(x+6)}{x^2(x-7)^3}$. Solve the following inequality

(i) $f(x) > 0$ (ii) $f(x) \geq 0$ (iii) $f(x) < 0$ (iv) $f(x) \leq 0$

Solution : We mark on the number line zeroes of numerator of expression : 1, –2, 3 and –6 (with black circles) and the zeroes of denominator 0 and 7 (with white circles), isolate the double points : –2 and 0 and draw the wavy curve :



From graph, we get

(i) $x \in (-\infty, -6) \cup (1, 3) \cup (7, \infty)$

(ii) $x \in (-\infty, -6] \cup \{-2\} \cup [1, 3] \cup (7, \infty)$

(iii) $x \in (-6, -2) \cup (-2, 0) \cup (0, 1) \cup (3, 7)$

(iv) $x \in [-6, 0) \cup (0, 1] \cup [3, 7)$

Do yourself-10 :

Solve following inequalities over the set of real numbers :

1. $(x-1)^2(x+1)^3(x-4)<0$

2. $\frac{6x-5}{4x+1}<0$

3. $\frac{(x-1)(x+2)^2}{-1-x}<0$

4. $\frac{(2x-1)(x-1)^2(x-2)^3}{(x-4)^4}>0$

5. $\frac{(x-1)^2(x+1)^3}{x^4(x-2)}\le 0$

6. $\frac{(x-2)^2(1-x)(x-3)^3(x-4)^2}{(x+1)}\le 0$

7. $\frac{x^2+4x+4}{2x^2-x-1}<0$

8. $\frac{x^3(x-2)(5-x)}{(x^2-4)(x+1)}>0$

9. $\frac{(2-x^2)(x-3)^3}{(x+1)(x^2-3x-4)}\ge 0$

10. $x^4-5x^2+4<0$

11. $\frac{15-4x}{x^2-x-12}<4$

12. $\frac{x^2+1}{4x-3}>2$

13. $\frac{1}{x-2}+\frac{1}{x-1}>\frac{1}{x}$

14. $(x-2)(x+3)\ge 0$

15 $\frac{x}{x+1}>2$

16. $\frac{3x-1}{4x+1}\le 0$

17. **Solve following Inequalities over the set of real numbers -**

(i) $x^2+2x-3\le 0$ **(ii)** $x^2+6x-7\le 0$

(iii) $x^4-2x^2-63\le 0$ **(iv)** $\frac{x+1}{(x-1)^2}<1$

(v) $\frac{x^2-7x+12}{2x^2+4x+5}>0$ **(vi)** $\frac{(x-1)(x+2)^2}{-1-x}<0$

(vii) $\frac{x^4+x^2+1}{x^2-4x-5}<0$ **(viii)** $\frac{x+7}{x-5}+\frac{3x+1}{2}\ge 0$

(ix) $\frac{1}{x+2}<\frac{3}{x-3}$ **(x)** $\frac{14x}{x+1}-\frac{9x-30}{x-4}<0$

(xi) $\frac{x^2+2}{x^2-1}<-2$ **(xii)** $\frac{5-4x}{3x^2-x-4}<4$

(xiii) $\frac{(x+2)(x^2-2x+1)}{4+3x-x^2}\ge 0$ **(xiv)** $\frac{x^4-3x^3+2x^2}{x^2-x-30}>0$

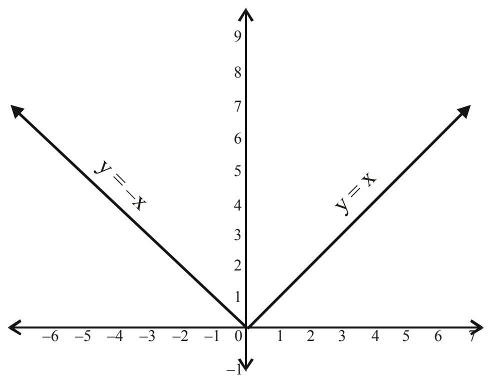
(xv) $\frac{2x}{x^2-9}\le\frac{1}{x+2}$ **(xvi)** $\frac{20}{(x-3)(x-4)}+\frac{10}{x-4}+1>0$

9. MODULUS AND MODULUS EQUATIONS

For any real number x, modulus or absolute value of x is denoted by |x| and is defined as

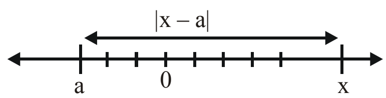
$$|x| = \begin{cases} x, & \text{if} \quad x \geq 0 \\ -x, & \text{if} \quad x < 0 \end{cases}$$

- Graph of y = |x|



Note :

- $|x| = |-x| \geq 0$
- Geometrically |x| is distance of real number x from zero along the real number line
- More generally |x – a| is distance between 'x' and 'a' on the number line.



- $|x| = \sqrt{x^2}$
- $|xy| = |x|\,|y|$

Illustration-78 : Sketch the graph of following equation and also find all possible values (Range) of y

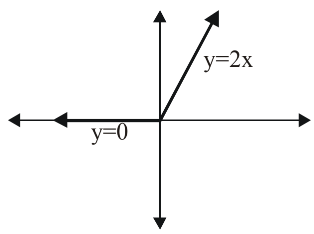
(i) $y = |x| + x$ (ii) $y = |x - 2| + x + 1$

(iii) $y = |x - 3| - |x|$ (iv) $y = 2|x - 2| - |x + 1|$

Soluiton :

(i) $y = |x| + x = \begin{cases} x + x & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$

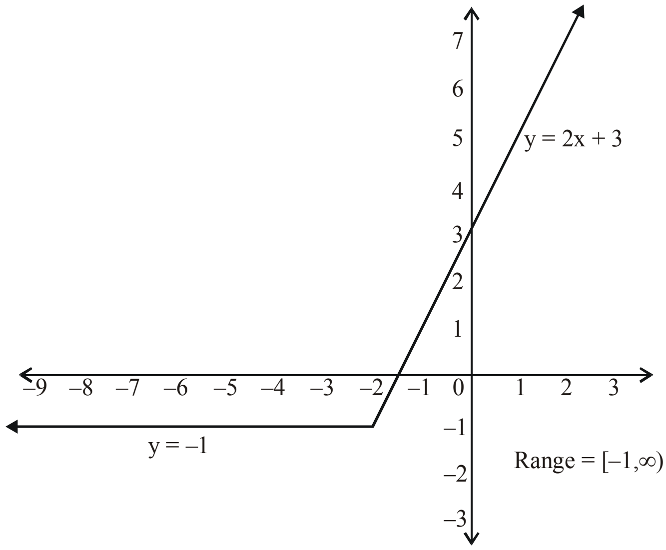
$$= \begin{cases} 2x & , \quad x \geq 0 \\ 0 & , \quad x < 0 \end{cases}$$



From graph we can find all possible values (range) of y which is $[0,\infty)$

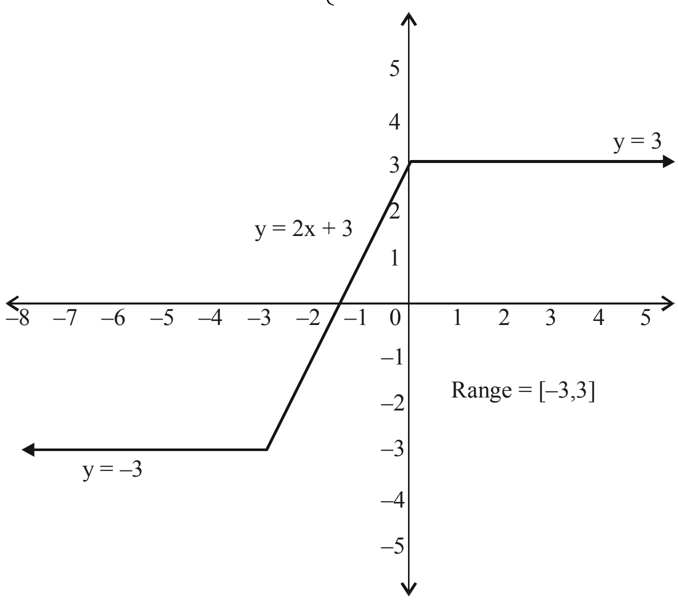
(ii) $y = |x + 2| + x + 1 = \begin{cases} x+2+x+1 & , \quad x \geq -2 \\ -x-2+x+1 & , \quad x < -2 \end{cases}$

$$= \begin{cases} 2x+3 & , \quad x \geq -2 \\ -1 & , \quad x < -2 \end{cases}$$

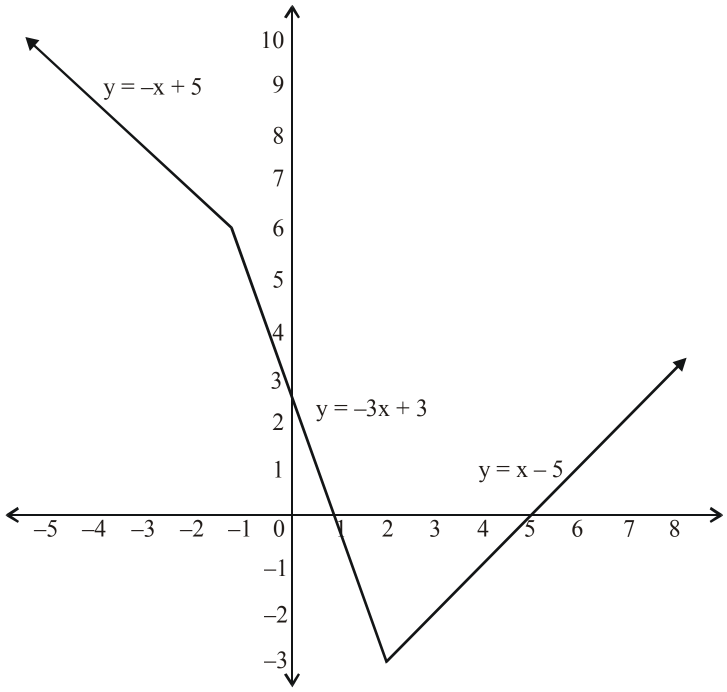


(iii) $y = |x + 3| - |x| = \begin{cases} x+3-x & , \quad x \geq 0 \\ x+3-(-x) & , \quad x \in [-3,0) \\ -x-3-(-x) & , \quad x < -3 \end{cases}$

$$= \begin{cases} 3 & , \quad x \geq 0 \\ 2x+3 & , \quad x \in [-3,0) \\ -3 & , \quad x < -3 \end{cases}$$



(iv) $y = 2|x-2| - |x+1| = \begin{cases} 2(x-2)-(x+1) & , \quad x \geq 2 \\ 2(-x+2)-(x+1) & , \quad x \in [-1,2] \\ 2(-x+2)+(x+1) & , \quad x < -1 \end{cases}$



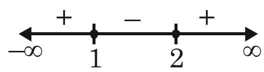
Range = $[-3,\infty)$

Illustration-79 : If $|x-1||x-2| = -(x^2-3x+2)$, then find the interval in which x lies ?

Solution : $|(x-1)(x-2)| = -(x-2)(x-1)$

$\Rightarrow (x-1)(x-2) \leq 0$

$\Rightarrow 1 \leq x \leq 2$



Do yourself-11

(1) Sketch the graph of following

(i) $y = |x-2|$ (ii) $y = |x| - 2$ (iii) $y = 5 - |x|$

Solve for x

(2) $|2x+5| = 2$ **(3)** $|2x-5| = 7$ **(4)** $|x-3| = -1$

(5) $|2x-3| + 4 = 2$ **(6)** $\left|\frac{3x+4}{3}\right| = 7$ **(7)** $|x^2-3x+2| = 2$

(8) $|2x-3| = |3x+5|$ **(9)** $2|x+3| = 3|x-4|$ **(10)** $|x^2+x+1| = |x^2+x+2|$

(11) $|x^2-4x+3| = |x^2-5x+4|$ **(12)** $|x-6| + |x-3| = 1$ **(13)** $|2x-1| + |2x+3| = 6$

(14) $|3x+5| + |4x+7| = 12$ **(15)** $|x| + |x+1| + |x+2| = 3$

(16) $|2x-3| + |2x+1| + |2x+5| = 12$ **(17)** $|x| - 2|x+1| + 3|x+2| = 0$

(18) $|x| - x = 0$ **(19)** $|x^2+3x+2| + x + 1 = 0$

(20) $x^2 + 3|x| + 2 = 0$ **(21)** $|x^2+1| - x^2 - 1 = 0$

Useful Mathematical Symbols for Reference

Symbol	How it is read
$\forall$	For all...
$\exists$	There exists...
$\wedge$	and
$\vee$	or
$<$	is less than
$>$	is greater than
$\leq$	is less than or equal to
$\geq$	is greater than or equal to
$\nless$	is not less than
$\ngtr$	is not greater than
$\in$	belongs to
$\notin$	does not belong to
$\|$, :, s.t.	such that
$\Rightarrow$	implies (If... then...)
$\Leftrightarrow$	implies and implied by (if and only if / iff)
!	factorial
$\sqrt{\ }$	The square root of
$\sqrt[n]{\ }$	n^{th} root, $n \in N$, $n \geq 2$
Σ	The summation of
Π	The product of

Greek Letters (Capital, Small) and there pronounciations

Symbol (Capital, Small)	How it is read	Symbol (Capital, Small)	How it is read
A,α	alpha	N,ν	nu
B,β	beta	Ξ,ξ	xi
Γ,γ	gamma	O,o	omicron
Δ,δ	delta	Π,π	pi
E,ε	epsilon	P,ρ	rho
Z,ζ	zeta	Σ,σ	sigma
H,η	eta	T,τ	tau
Θ,θ	theta	Y,υ	upsilon
I,ι	iota	Φ,ϕ	phi
K,κ	kappa	X,χ	chi
Λ,λ	lambda	Ψ,ψ	psi
M,μ	mu	Ω,ω	omega

EXERCISE (O-1)

Straight Objective Type

1. $\sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6+...}}}}$ =

(A) 3 (B) 2 (C) 1 (D) ±3

FM0008

2. If x = $8-\sqrt{60}$, then $\frac{1}{2}\left[\sqrt{x}+\frac{2}{\sqrt{x}}\right]$ =

(A) $\sqrt{5}$ (B) $\sqrt{3}$ (C) $2\sqrt{5}$ (D) $2\sqrt{3}$

FM0009

3. If $\frac{4+3\sqrt{5}}{4-3\sqrt{5}}$ = a+b$\sqrt{5}$, a, b are rational numbers, then (a, b) =

(A) $\left(\frac{61}{29},\frac{-24}{29}\right)$ (B) $\left(\frac{-61}{29},\frac{24}{29}\right)$ (C) $\left(\frac{61}{29},\frac{24}{29}\right)$ (D) $\left(\frac{-61}{29},\frac{-24}{29}\right)$

FM0010

4. The square root $5+2\sqrt{6}$ is :

(A) $\sqrt{3}+2$ (B) $\sqrt{3}-\sqrt{2}$ (C) $\sqrt{2}-\sqrt{3}$ (D) $\sqrt{3}+\sqrt{2}$

FM0011

5. If $\frac{4}{2+\sqrt{3}+\sqrt{7}}$ = $\sqrt{a}+\sqrt{b}-\sqrt{c}$, then which of the following can be true -

(A) a = 1, b = 4/3, c = 7/3 (B) a = 1, b = 2/3, c = 7/9
(C) a = 2/3, b = 1, c = 7/3 (D) a = 7/9, b = 4/3, c = 1

FM0012

6. The numerical value of $\left(x^{1/(a-b)}\right)^{1/(a-c)} \times \left(x^{1/(b-c)}\right)^{1/(b-a)} \times \left(x^{1/(c-a)}\right)^{1/(c-b)}$ is (a, b, c are distinct real numbers)

(A) 1 (B) 8 (C) 0 (D) None

FM0013

7. $(1^3 + 2^3 + 3^3 + 4^3)^{-3/2}$ =

(A) 10^{-3} (B) 10^{-2} (C) 10^{-4} (D) 10^{-1}

FM0014

8. $\left(7^{\left(-\frac{1}{2}\right)}\times 5^2\right)^2 \div \sqrt{25^3}$ =

(A) $\frac{5}{7}$ (B) $\frac{7}{5}$ (C) 35 (D) $-\frac{5}{7}$

FM0015

9. $(2d^2e^{-1})^3 \times \left(\frac{d^3}{e}\right)^{-2} =$

(A) $8e^{-2}$ (B) $8e^{-3}$ (C) $8e^{-1}$ (D) $8e^{-4}$

FM0016

10. If $x^y = y^x$ and $x = 2y$, then the values of x and y are $(x, y > 0)$

(A) $x = 4, y = 2$ (B) $x = 3, y = 2$ (C) $x = 1, y = 1$ (D) None of these

FM0017

11. If $(a^m)^n = a^{m^n}$, then express 'm' in the terms of n is $(a > 0, a \neq 1, m > 1, n > 1)$

(A) $n^{\left(\frac{1}{n-1}\right)}$ (B) $n^{\left(\frac{1}{n+1}\right)}$ (C) $n^{\left(\frac{1}{n}\right)}$ (D) None

FM0018

12. If $a = x + \frac{1}{x}$, then $x^3 + x^{-3} =$

(A) $a^3 + 3a$ (B) $a^3 - 3a$ (C) $a^3 + 3$ (D) $a^3 - 3$

FM0019

13. If $\left(\sqrt[3]{4}\right)^{2x+\frac{1}{2}} = \frac{1}{32}$, then x =

(A) –2 (B) 4 (C) –6 (D) –4

FM0020

14. If $(5+2\sqrt{6})^{x^2-3} + (5-2\sqrt{6})^{x^2-3} = 10$, then all possible values of x are

(A) –2, 2 (B) $\sqrt{2}$, $-\sqrt{2}$ (C) 2, $+\sqrt{2}$ (D) 2, –2, $\sqrt{2}$, $-\sqrt{2}$

FM0021

15. If $3^{2x^2} - 2.3^{x^2+x+6} + 3^{2(x+6)} = 0$ then the value of x is

(A) – 2 (B) 3 (C) Both (A) and (B) (D) None of these

FM0022

16. How many integers in between 100 to 1500 (both inclusive) are multiples of 5 or 11 ?

(A) 408 (B) 26 (C) 382 (D) 380

FM0023

17. Square root of $4+\sqrt{15}$ is equal to

(A) $\frac{\sqrt{3}+\sqrt{5}}{2}$ (B) $\sqrt{\frac{3}{2}}+\sqrt{\frac{5}{2}}$ (C) $\sqrt{\frac{5}{2}}-\sqrt{\frac{3}{2}}$ (D) None of these

FM0024

18. If $A = \{(x, y) \mid xy = 8 \text{ and } x, y \in \mathbb{Z}\}$, then $n(A) =$

(A) 4 (B) 8 (C) 12 (D) 16

FM0025

19. The set of all real numbers x that satisfy $\frac{x-2}{x+2} > \frac{2x-3}{4x-1}$

(A) $(-\infty,-2)\cup\left(\frac{1}{4},\frac{3}{2}\right)\cup(2,\infty)$ (B) $\left(-2,\frac{1}{4}\right)\cup\left(\frac{3}{2},2\right)$

(C) $(-\infty,-2)\cup\left(\frac{1}{4},1\right)\cup(4,\infty)$ (D) $\left(-2,\frac{1}{4}\right)\cup(1,4)$

FM0026

20. The set of all real numbers x that satisfy $\frac{x^2-4}{x^2-5x+6} \le 0$

(A) $[-2, 3]$ (B) $[-2, 3)$

(C) $(-\infty, -2] \cup (3, \infty)$ (D) None of these

FM0027

21. If $\frac{(x+3)^2(x-1)^9(x+1)^5}{(x-3)(x-5)^4(x-6)^5} \le 0$, then number of possible integral values of x is -

(A) 6 (B) 3 (C) 4 (D) 5

FM0028

22. If $\left(\sqrt{2}+1\right)^x + \left(\sqrt{2}-1\right)^x - 2\sqrt{2} = 0$, then sum of all possible values of x is

(A) 0 (B) 1 (C) 2 (D) 3

FM0029

23. If $x^2 + y^2 + 4z^2 - 6x - 2y - 4z + 11 = 0$, then xyz is equal to

(A) 3/2 (B) 4 (C) 6 (D) 3

FM0030

24. If $\frac{\sqrt{3}+4\sqrt{2}}{4\sqrt{2}-\sqrt{3}} = \frac{a+b\sqrt{6}}{c}$, then the value of a + b + c (where a, b, c $\in$ N and are relatively prime)

(A) 70 (B) 72 (C) 50 (D) 40

FM0034

25. If $x^2 + 4y^2 + z^2 - 2xy - 2yz - zx = 0$ then x : y : z equals

(A) 1 : 2 : 1 (B) 2 : 1 : 2 (C) 1 : 2 : 3 (D) 1 : 1 : 2

FM0035

EXERCISE (O-2)

More than one correct

1. If $x = \sqrt{7\sqrt{7\sqrt{7\sqrt{7...}}}}$ where x, y > 0

$y = \sqrt{20 + \sqrt{20 + \sqrt{20 + ...}}}$ then which of the following is/are correct.

(A) $x + y = 12$ (B) $x - y = 3$ (C) $x^2 + y^2 = 74$ (D) $x^2 - y^2 = 24$

FM0037

2. If $8x^3 + 27x^2 - 9x - 50$ is divided by $(x + 2)$ then remainder is λ then

(A) $\frac{3\lambda + 4}{10}$ is equal to 4

(B) If $\lambda = \frac{p}{q}$ then (p + q) is divisible by 13 (where p & q are coprime)

(C) λ is a natural number

(D) $(\lambda - 2)$ is divisible by 3

FM0039

3. If $f(x) = ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \mathbb{R}$, $a \neq 0$ then which of the following option(s) is(are) correct ?

(A) if $a + b + c + d = 0$ then $x - 1$ is factor of $f(x)$.

(B) if $a + b = c + d$ then $x + 1$ is factor of $f(x)$.

(C) if $a + c = b + d$ then $x + 1$ is factor of $f(x)$.

(D) none of these

FM0040

4. If x = a and x = b are the two roots of the equation $9^x - 4 \times 3^{x+1} + 27 = 0$ then

(A) $a + b = 3$ (B) $(a - b)^2 = 1$ (C) $\frac{a}{b} + \frac{b}{a} = \frac{5}{2}$ (D) $a + b = 4$

FM0041

5. The real values of 'x' satisfying the equation $\left(4+\sqrt{15}\right)^{x^2-x-1} + \left(4-\sqrt{15}\right)^{x^2-x-1} = 8$ is/are :

(A) 0 (B) 1 (C) –1 (D) –2

FM0042

6. The value of $(4 + 1)(4^2 + 1)(4^4 + 1)(4^8 + 1) + \frac{1}{3} = \frac{2^\lambda}{3}$ where $\lambda \in \mathbb{N}$ then the factor(s) of 'λ' is/are

FM0075

(A) 4 (B) 8 (C) 16 (D) 32

7. If $x = \frac{6mn}{m+n}$, then value of $\left(\frac{x+3m}{x-3m} + \frac{x+3n}{x-3n}\right)$, (where $m \neq n \neq 0$) is less than or equal to

(A) 1 (B) 2 (C) 3 (D) 4

FM0128

8. If $x=\sqrt{6+\sqrt{6+\sqrt{6+...\infty}}}$ and $y=\sqrt{6\sqrt{6\sqrt{6...\infty}}}$, then

(A) x = 2y (B) x + y = 9 (C) $x^2 - y = 3$ (D) x is a prime number

FM0129

9. Value of $\left(\frac{x^{\frac{2}{3}}y^{-\frac{1}{2}}}{yx^{-\frac{2}{3}}} \div \sqrt{\frac{xy^{-2}}{yx^{-1}}}\right)^6$ is not equal to

(A) x^2 (B) x (C) x^3 (D) x^4

FM0130

10. If $\frac{1}{2-\sqrt{3}}-\frac{1}{\sqrt{3}+\sqrt{2}}+\frac{5}{3-\sqrt{2}}=a+b\sqrt{2}$; where $a,b \in Q$ then

(Note : Q denotes set of rational numbers)

(A) $a+b=\frac{41}{7}$ (B) $a+b=\frac{39}{7}$ (C) $\frac{a}{b}=5$ (D) $\frac{a}{b}=\frac{29}{12}$

FM0131

11. If $\left(1+\frac{1}{3}\right)\left(1+\frac{1}{3^2}\right)\left(1+\frac{1}{3^4}\right)...\left(1+\frac{1}{3^{512}}\right)=\frac{1}{18}\left(\frac{3^a-1}{3^{1021}}\right)$, then

(A) sum of the digits of number 'a' is 5. (B) sum of the digits of number 'a' is 7.

(C) last digit of number 'a' is 3. (D) 'a' is divisible by 8.

FM0132

12. If x(y + z) = 16; y(z + x) = 66; z(x + y) = 70 and x,y,z > 0, then (x + y + z) is less than

(A) 16 (B) 17 (C) 26 (D) 27

FM0133

13. If $x^2 - 1$ is a factor of $8x^3 + qx^2 + px + 2$, then (p +4q) is less than

(A) –8 (B) –2 (C) –10 (D) –16

FM0134

14. Which of the following option(s) is(are) true

(A) sum of all roots of the equation $|9 - x| = 5$ is 4

(B) sum of all roots of the equation $|9 - x| = 5$ is 18

(C) sum of all roots of the eqaution $|x - 9| = 5$ is 18

(D) sum of all roots of the eqaution $|x - 9| = 5$ is 14

FM0135

15. If $f(x)$ is monic polynomial of degree 3 such that $f(0) = 0$, $f(1) = 1$, $f(2) = 2$ then which of the following option(s) is/are true ?

(A) $f(4) = 4$ (B) $|f(-1)| = 1$ (C) $f(-1) = -7$ (D) $f(-2) = -26$

FM0136

EXERCISE (O-3)

Paragraph for question no. 1 to 3

If $(5+2\sqrt{6})^{x^2-8}+(5-2\sqrt{6})^{x^2-8}=10, x \in \mathbb{R}$

On the basis of above information, answer the following questions :

1. Number of solution(s) of the given equation is/are-

(A) 1 (B) 2 (C) 4 (D) infinite

FM0043

2. Sum of positive solutions is

(A) 3 (B) $3+\sqrt{7}$ (C) $2+\sqrt{5}$ (D) 2

FM0043

3. If $x \in (-3,5]$, then number of possible values of x, is-

(A) 1 (B) 2 (C) 3 (D) 4

FM0043

Paragraph for question no. 4 to 6

A polynomial p(x) when divided by (x – 1), (x + 1) and (x + 2) gives remainder 5, 7 and 2 respectively. If p(x) is divided by $(x^2 - 1)$ and (x – 1) (x + 2) gives remainder as another polynomial R(x) and r(x) respectively. Then

4. The value of R(50) is :

(A) 34 (B) –44 (C) 44 (D) 104

FM0044

5. The value of r(100) is :

(A) 34 (B) 44 (C) 54 (D) 104

FM0044

6. The value of r(10) is :

(A) 4 (B) 14 (C) 24 (D) 35

FM0137

Matching list type

7. Match the list

	List-I		**List-II**	
(P)	The units digit of $2^7 \times 3^{10}$ is	(1)	1	**FM0045**
(Q)	The number of prime factors of $6^{100} \times 15^{200}$ is	(2)	2	**FM0046**
(R)	Number of solutions of xy = 35 in natural numbers with x > y > 1 is	(3)	3	**FM0047**
(S)	Remainder when $x^{100} - 3x^{99} + 2x^{98} + 3x - 2$ is divided by x – 2 is equal to	(4)	4	**FM0048**

Codes :

(A) P-1, Q-2, R-3, S-4 (B) P-2, Q-3, R-1, S-4

(C) P-2, Q-3, R-2, S-3 (D) P-1, Q-4, R-1, S-2

8. Match the list

	List-I		List-II
(P)	If $x=\sqrt{5}-2$ then value of $2x^3 + 11x^2 + 10x + 4$ is equal to	(1)	8
(Q)	If $x=\sqrt{5}+2$ then value of $2x^3 - 7x^2 - 6x + 7$	(2)	7
(R)	If $\left(\left(\sqrt{11+2\sqrt{30}}\right)+\left(\sqrt{10+4\sqrt{6}}\right)\right)^2 = p+\sqrt{q}+\sqrt{r}+\sqrt{s}$ where p, q, r, s are integers & q, r, s are not perfect squares the p + q + r + s is equal to	(3)	161
(S)	If $\left(\left(\sqrt{11+2\sqrt{30}}\right)-\left(\sqrt{10+4\sqrt{6}}\right)\right)^2 = \alpha-\sqrt{\beta}$ where α, β are integers & β is not a perfect square then $\alpha^2 + \beta$ is equal to	(4)	977

FM0049 **FM0050** **FM0051** **FM0052**

Codes :

(A) P-1, Q-2, R-3, S-4 (B) P-2, Q-3, R-1, S-4

(C) P-2, Q-1, R-3, S-4 (D) P-2, Q-1, R-4, S-3

Answer the question question 9 and 10 by appropriately matching the lists based on the information given

	List-I		List-II
(I)	If $2\sqrt{x+5} = x+2$, then integral value of x	(P)	3
(II)	The number of real solution of the equation $x^2 + 5\|x\| + 4 = 0$, is	(Q)	–4
(III)	The real solution of the equation $(x-2)^2 + \|x-2\| - 2 = 0$, is	(R)	–1
(IV)	The number of real solution(s) $\|x+2\| = 2(3-x)$, is	(S)	0
		(T)	4
		(U)	1

9. Which of the following is only **CORRECT** option ?

(A) I $\rightarrow$ R, U (B) II $\rightarrow$ S, T (C) I $\rightarrow$ Q, S (D) II $\rightarrow$ S

FM0053

10. Which of the following is only **INCORRECT** option ?

(A) III $\rightarrow$ P, U (B) IV $\rightarrow$ U (C) III $\rightarrow$ Q, T (D) III $\rightarrow$ P

FM0053

EXERCISE (O-4)

Numerical Grid Type

1. If $a-\frac{1}{a}=\frac{1}{2}$ then $\left(4a^2+\frac{4}{a^2}\right)$ is equal to

FM0054

2. If polynomial Ax^3+4x^2+Bx+5 leaves same remainder, when divided by $x-1$ and $x+2$ respectively then value of 3A + B is equal to

FM0055

3. If $9^x + 6^x = 2.4^x$ then the value of $\frac{x^3+64}{5}$

FM0056

4. If $a + b + c = 6$ & $a^2 + b^2 + c^2 = 14$ and $a^3 + b^3 + c^3 = 36$ then the value of $3\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$

FM0057

5. If $x + y + z = 12$ & $x^2+y^2+z^2=96$ and $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=36$. Find the value of $\frac{x^3+y^3+z^3}{4}$

FM0104

6. If $x=\sqrt{4-2\sqrt{3}}$ and $y=\sqrt{9-4\sqrt{5}}$ then the value of $\left(\sqrt{5}x-\sqrt{3}y\right)^2$ is equal to $\left(a-b\sqrt{c}\right)$ where a, b, c are coprime numbers then a + b + c is equal to (where 'c' is an odd integer)

FM0105

7. If x, y, z $\in$ R and $121x^2 + 4y^2 + 9z^2 - 22x + 4y + 6z + 3 = 0$ then value of $x^{-1} - y^{-1} - z^{-1}$ is equal to

FM0106

8. If $\frac{p}{q}=\frac{x+2}{x-2}$, then find the value of $\frac{p^2-q^2}{p^2+q^2}\times\frac{x^2+4}{x}$. $(x\neq 0)$

FM0108

9. $\frac{3x+\sqrt{9x^2-5}}{3x-\sqrt{9x^2-5}}=5$ find 'x'

FM0110

10. Value of $\left(\frac{\sqrt{6}}{\sqrt{3}-\sqrt{2}+\frac{\sqrt{6}}{\sqrt{3}-\sqrt{2}+\frac{\sqrt{6}}{\left(\sqrt{3}-\sqrt{2}\right)+\ddots}}}\right)^2$ is

FM0111

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\Sets, FOM & Q.Eq\Eng\04_Fundamental of Algebra.p65

JEE-MAINS/ JEE-ADVANCE

1. The expression $\frac{12}{3+\sqrt{5}+2\sqrt{2}}$ is equal to

(a) $1-\sqrt{5}+\sqrt{2}+\sqrt{10}$ (b) $1+\sqrt{5}+\sqrt{2}-\sqrt{10}$

(c) $1+\sqrt{5}-\sqrt{2}+\sqrt{10}$ (d) $1-\sqrt{5}-\sqrt{2}+\sqrt{10}$

FM0117

2. If x < 0, y < 0, x + y + $\frac{x}{y}=\frac{1}{2}$ and (x + y)$\frac{x}{y}=-\frac{1}{2}$, then x = ... and y = ...

FM0118

3. The equation $x-\frac{2}{x-1}=1-\frac{2}{x-1}$ has

(A) no root (B) one root (C) two equal roots (D) infinitely many roots

FM0119

4. Find all real values of x which satisfy $x^2 - 3x + 2 > 0$ and $x^2 - 2x - 4 \le 0$.

FM0123

5. Find the set of all x for which $\frac{2x}{\left(2x^2+5x+2\right)}>\frac{1}{(x+1)}$.

FM0124

6. The sum of all real roots of the equation $|x-2|^2 + |x-2| - 2 = 0$ is ...

FM0125

7. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is

(A) 4 (B) 1 (C) 2 (D) 3

FM0126

8. If S is the set of all real x such that $\frac{2x-1}{2x^3+3x^2+x}$ is positive, then S contains

(A) $\left(-\infty,-\frac{3}{2}\right)$ (B) $\left(-\frac{3}{2},-\frac{1}{4}\right)$ (C) $\left(-\frac{1}{4},\frac{1}{2}\right)$ (D) $\left(\frac{1}{2},3\right)$

(E) None of these

FM0127

9. Let $y=\sqrt{\frac{(x+1)(x-3)}{(x-2)}}$

Find all the real values of x for which y takes real values.

FM0128

ANSWER KEY

Do yourself-1

1.	B	**2.**	C	**3.**	A	**4.**	D	**5.**	C

6. **(i)** 1 **(ii)** 0 **(iii)** 1 **(iv)** 1 **(v)** 8

7. 2 **8.** (a) $-\frac{47}{44}$, (b) $-\frac{9}{44}$ **9.** 5 **10.** 26

11. 1 **12.** $x = y = z = \frac{a}{3}$ and if a is integral multiple of 6 then $x = y = z = \pm a/3$

Do yourself-2

1. **(i)** $(x - y)(x + y)(x^2 + y^2)$ **(ii)** $(3a - 2x + y)(3a + 2x - y)$

(iii) $(2x + 3y)(2x - 3y - 3)$

2. **(i)** $(2x - 3y)(4x^2 + 6xy + 9y^2)$ **(ii)** $(2x - 5y)[4x^2 + 10xy + 25y^2 + 1]$

3. **(i)** $(x + 8)(x - 5)$ **(ii)** $(x - 8)(x + 5)$ **(iii)** $(x + 7)(x - 2)$

(iv) $(x - 4)(x + 1)$ **(v)** $(x - 3)(x + 1)$ **(vi)** $(3x - 4)(x - 2)$

(vii) $(4x + 7)(3x - 5)$ **(viii)** $(3x - 2)(x - 1)$ **(ix)** $(x - 1)(3x - 4)$

(x) $(7x - 1)(x - 1)$ **(xi)** $(x - 2)(2x - 13)$ **(xii)** $(a - 3)(3a + 2)$

(xiii) $(2a + 1)(7a - 3)$

4. **(i)** $(a - b - 1)(a + b - 3)$ **(ii)** $(x^2 - 6x + 18)(x^2 + 6x + 18)$

(iii) $(x^2 - y + 1)(x^2 + y + 1)$ **(iv)** $(2a^2 - a - 1)(2a^2 + a - 1)$

(v) $(2x^2 - 6x + 9)(2x^2 + 6x + 9)$ **(vi)** $(x^4 - x^2 + 1)(x^4 + x^2 + 1)$

5. **(i)** $(x - 1)(x - 2)(x - 3)$ **(ii)** $(x + 1)(x + 3)(2x + 1)$

(iii) $(x - 1)(x - 2)(2x - 3)$ **(iv)** $(x^2 + 2)(x^2 + 1)\left(x - \sqrt{3}\right)\left(x + \sqrt{3}\right)$

(v) $3(x + y)(y + z)(z + x)$

6. **(i)** $(2a^2 - a + 1)(4a^4 + 2a^3 - a^2 + a + 1)$

7. **(i)** $(x^2 + 5x + 1)(x^2 + 5x + 9)$

(ii) $2(2x^2 + 2x + 3)(4x^2 + 4x - 9)$

(iii) $(x^2 + 5x - 22)(x + 1)(x + 4)$

8. **(i)** $(x - y - 5)(x - y + 4)$. **(ii)** $(x + 2y - 2)(x + 2y + 4)$.

(iii) $16(3x + 4)(x + 1)$. **(iv)** $(x^{\frac{1}{3}} + 2)(x^{\frac{1}{3}} - 1)$.

(v) $(x - 2)(x^2 + 2x + 4)(x + 1)(x^2 - x + 1)$.

(vi) $(x^2 + 16)(x - 1)(x + 1)$.

(vii) $(x + 1)(x - 2)(x - 3)(x + 2)$. **(viii)** $(x^2 + 5x + 16)(x + 6)(x - 1)$.

(ix) $-2(a^2 + 1)(a^2 + 5)$. **(x)** $(x + y + z - 3)^2$.

(xi) $(2x - 1)(x - 2)(x + 1)^2$. **(xii)** $(x^2 + 2)(x + 1)(x - 1)(x^4 + x^2 + 1)$.

Do yourself-3

1. (a) 7 (b) 47 (c) 18 **4.** 104 **5.** $-\sqrt{2}$ **6.** 1

8. (a) 0, 4 (b) no solution (c) no solution

Do yourself-4

1. 1 **2.** –20 **3.** 4 **4.** –2 **5.** 3x + 4

6. $\ell = 2, m = -18$ **7.** $\ell = -1, m = -3$ **8.** a = 3, b = 1

9. $3x^3 + 4x^2 - 5x - 2$ **10.** 6 **11.** a = 8 and b = 5 **12.** 22

Do yourself-5

1. 1 or –8 **2.** 1,32 **3.** $\pm\sqrt{2}, \pm\sqrt{\frac{2}{3}}$

4. 3 or 4 **5.** $x = 1, x = \frac{3 \pm \sqrt{5}}{2}$ **6.** $x = \pm 1, 2, -\frac{1}{2}$

7. 0,–1 **8.** 2,4 **9.** ±1

10. –4,4 **11.** –4,1 **12.** –6,1

13. $x = -3 \pm \sqrt{5}$ **14.** $\pm 1, 1 \pm \sqrt{2}$

15. (A)→(U); (B)→(U); (C)→(T); (D)→(P,Q,R,S,T,U,V);(E)→(R,T); (F)→(S)

Do yourself-6

1. x = 1 , y = 3 **2.** x = 5, y = 1

3. x = 5, y = 3 **4.** x = 2, y = 1 or x = –1, y = –2

5. x = 4, y = –2 or x = –2, y = 4 **6.** x = –6, y = –2 or x = –4, y = –4

7. x = 3, y = 2 or x = –3, y = –2 **8.** x = 5, y = 1 or x = –5, y = –1

9. $x = \frac{11}{13}, y = -\frac{24}{5}$ **10.** x = 3, y = 2 or x = –3, y = –2

11. x = 4, y = ±3 **12.** x = –1, y = 3 or $x \in \mathbb{R}$, y = 2

13. x = 2, y = –1or x = –1, y = 2 **14.** $(x, y) \cong (3,1), (-3,-1), (1,3), (-1,-3)$

Do yourself-7

1. (a) F (b) T (c) F (d) T (e) T (f) F

2. (a) $\frac{17}{21}$ (b) $\frac{37}{41}$ **3.** 3^{30} **4.** $(2,\infty)$ **5.** $3+\sqrt{14}$

Do yourself-8

1. $4\sqrt{2}$ **2.** -12 **3.** 4

Do yourself-9

1. $\frac{20}{27}$ **2.** 21 : 35 : 65 **3.** $\frac{sp}{p+q},\frac{sq}{p+q}$ **4.** $\frac{a^4}{b^4}$

Do yourself-10

1. $(-1,4)-\{1\}$

2. $\left(-\frac{1}{4},\frac{5}{6}\right)$

3. $(-\infty,-2)(-2,-1)\cup(1,\infty)$

4. $\left(-\infty,\frac{1}{2}\right)\cup(2,\infty)-\{4\}$

5. $[-1,2)-\{0\}$

6. $(-1,1]\cup[3,\infty)\cup\{2\}$

7. $\left(-\frac{1}{2},1\right)$

8. $(-2,-1)\cup(0,5)-\{2\}$

9. $\left[-\sqrt{2},\sqrt{2}\right]\cup[3,4)-\{-1\}$

10. $(-2,-1)\cup(1,2)$

11. $\left(-\infty,\frac{-\sqrt{63}}{2}\right)\cup\left(-3,\frac{\sqrt{63}}{2}\right)\cup(4,\infty)$

12. $\left(\frac{3}{4},1\right)\cup(7,\infty)$

13. $\left(-\sqrt{2},0\right)\cup\left(1,\sqrt{2}\right)\cup(2,\infty)$

14. $(-\infty,-3]\cup[2,\infty)$

15 $(-2,-1)$

16. $\left(-\frac{1}{4},\frac{1}{3}\right]$

17. **(i)** $[-3,1]$ **(ii)** $[-7,1]$

(iii) $[-3, 3]$ **(iv)** $(-\infty, 0)\cup(3, +\infty)$

(v) $(-\infty, 3)\cup(4, +\infty)$ **(vi)** $(-\infty, -2)\cup(-2, -1)\cup(1, +\infty)$

(vii) $(-1, 5)$ **(viii)** $[1, 3]\cup(5, +\infty)$

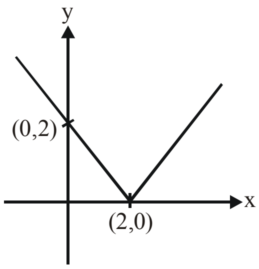
(ix) $\left(-\frac{9}{2},-2\right)\cup(3,\infty)$ **(x)** $(-1, 1)\cup(4, 6)$

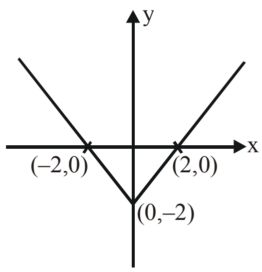
(xi) $(-1,1)-\{0\}$ **(xii)** $\left(-\infty,-\frac{\sqrt{7}}{2}\right)\cup\left(-1,\frac{\sqrt{7}}{2}\right)\cup\left(\frac{4}{3},\infty\right)$

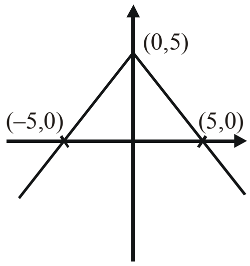
(xiii) $(-\infty,-2]\cup(-1,4)$ **(xiv)** $(-\infty,-5)\cup(1,2)\cup(6,+\infty)$

(xv) $(-\infty,-3)\cup(-2,3)$ **(xvi)** $(-\infty,-2)\cup(-1,3)\cup(4,\infty)$

Do yourself-11

1. (i)


(ii)


(iii)


2. $\left\{-\frac{3}{2},-\frac{7}{2}\right\}$ **3.** $\{-1,6\}$ **4.** ϕ **5.** ϕ **6.** $\left\{-\frac{25}{3},\frac{17}{3}\right\}$

7. $\{0,3\}$ 8. $\left\{-8,-\frac{2}{5}\right\}$ 9. $\left\{\frac{6}{5},18\right\}$ 10. ϕ

11. $\left\{1,\frac{7}{2}\right\}$ **12.** ϕ **13.** $\{1,-2\}$ **14.** $\left\{0,-\frac{24}{7}\right\}$

15. $\{-2,0\}$ **16.** $\left\{-\frac{5}{2},\frac{3}{2}\right\}$ **17.** $\{-2\}$ **18.** $[0,\infty)$

19. $\{-3,-1\}$ **20.** ϕ **21.** $\mathbb{R}$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	A	D	D	A	A	A	A	C	A
Que.	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Ans.	A	B	D	D	C	C	B	B	C	D
Que.	**21**	**22**	**23**	**24**	**25**					
Ans.	D	A	A	B	B					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C	A,C	A,B,C	A,B,C	A,B,C,D	B,C,D	B,C,D	B,C,D	A,D
Que.	**11**	**12**	**13**	**14**	**15**					
Ans.	B,D	C,D	A,B,C	B,C	C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	C	B	D	B	B	D	D	C

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	9.00	4	12.80	5.5	216.50	36	16	4	x = 1	2

EXERCISE # JEE-MAIN/ADVANCED

1. B **2.** $x = -\frac{1}{4}, y = -\frac{1}{4}$ **3.** A **4.** $x \in \left[1-\sqrt{5},1\right) \cup \left(2,1+\sqrt{5}\right]$

5. $(-2,-1) \cup \left(-\frac{2}{3},-\frac{1}{2}\right)$ **6.** 4 **7.** A

8. A,D **9.** $x \in [-1, 2) \cup [3, \infty)$